

UNITED STATES SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

Form 10-Q

(Mark One)



QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the quarterly period ended September 30, 2022

Or



TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the transition period from to

Commission File Number 001-36198

INTERCONTINENTAL EXCHANGE, INC.

(Exact name of registrant as specified in its charter)

Delaware

(State or other jurisdiction of
incorporation or organization)

46-2286804

(IRS Employer
Identification Number)

5660 New Northside Drive,
Atlanta, Georgia

(Address of principal executive offices)

30328

(Zip Code)

(770) 857-4700

Registrant's telephone number, including area code

Securities registered pursuant to Section 12(b) of the Act:

Title of Each Class	Trading Symbol(s)	Name of Each Exchange on Which Registered
Common Stock, \$0.01 par value per share	ICE	New York Stock Exchange

Indicate by check mark whether the registrant: (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports); and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company or an emerging growth company. See definitions of "large accelerated filer," "accelerated filer," "smaller reporting company" and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
		Emerging growth company	☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

As of November 1, 2022, the number of shares of the registrant's Common Stock outstanding was 558,551,625 shares.

INTERCONTINENTAL EXCHANGE, INC.
Form 10-Q
Quarterly Period Ended September 30, 2022
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PART I. Financial Statements

Item 1. Consolidated Financial Statements

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Balance Sheets
(In millions, except per share amounts)

	As of September 30, 2022 (Unaudited)	As of December 31, 2021
Assets:		
Current assets:		
Cash and cash equivalents	\$ 1,183	\$ 607
Short-term restricted cash and cash equivalents	6,032	1,035
Cash and cash equivalent margin deposits and guaranty funds	156,789	145,936
Invested deposits, delivery contracts receivable and unsettled variation margin	7,902	4,493
Customer accounts receivable, net of allowance for doubtful accounts of \$25 and \$24 at September 30, 2022 and December 31, 2021, respectively	1,248	1,208
Prepaid expenses and other current assets	558	1,021
Total current assets	173,712	154,300
Property and equipment, net	1,720	1,699
Other non-current assets:		
Goodwill	21,075	21,123
Other intangible assets, net	13,210	13,736
Long-term restricted cash and cash equivalents	405	398
Other non-current assets	1,170	2,246
Total other non-current assets	35,860	37,503
Total assets	\$ 211,292	\$ 193,502
Liabilities and Equity:		
Current liabilities:		
Accounts payable and accrued liabilities	\$ 788	\$ 703
Section 31 fees payable	58	57
Accrued salaries and benefits	274	354
Deferred revenue	315	194
Short-term debt	7	1,521
Margin deposits and guaranty funds	156,789	145,936
Invested deposits, delivery contracts payable and unsettled variation margin	7,902	4,493
Other current liabilities	188	153
Total current liabilities	166,321	153,411
Non-current liabilities:		
Non-current deferred tax liability, net	3,565	4,100
Long-term debt	18,113	12,397
Accrued employee benefits	189	200
Non-current operating lease liability	267	252
Other non-current liabilities	417	394
Total non-current liabilities	22,551	17,343
Total liabilities	188,872	170,754
Commitments and contingencies		

Equity:

Intercontinental Exchange, Inc. stockholders' equity:		
Preferred stock, \$0.01 par value; 100 shares authorized; none issued or outstanding	—	—
Common stock, \$0.01 par value; 1,500 shares authorized; 634 and 631 issued at September 30, 2022 and December 31, 2021, respectively, and 559 and 561 shares outstanding at September 30, 2022 and December 31, 2021, respectively	6	6
Treasury stock, at cost; 75 and 70 shares at September 30, 2022 and December 31, 2021, respectively	(6,224)	(5,520)
Additional paid-in capital	14,269	14,069
Retained earnings	14,731	14,350
Accumulated other comprehensive loss	(403)	(196)
Total Intercontinental Exchange, Inc. stockholders' equity	22,379	22,709
Non-controlling interest in consolidated subsidiaries	41	39
Total equity	22,420	22,748
Total liabilities and equity	\$ 211,292	\$ 193,502

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Income
(In millions, except per share amounts)
(Unaudited)

	Nine Months Ended September 30,		Three Months Ended September 30,	
	2022	2021	2022	2021
Revenues:				
Exchanges	\$ 4,824	\$ 4,376	\$ 1,577	\$ 1,434
Fixed income and data services	1,555	1,403	534	477
Mortgage technology	880	1,061	276	366
Total revenues	7,259	6,840	2,387	2,277
Transaction-based expenses:				
Section 31 fees	332	204	158	38
Cash liquidity payments, routing and clearing	1,403	1,330	418	437
Total revenues, less transaction-based expenses	5,524	5,306	1,811	1,802
Operating expenses:				
Compensation and benefits	1,058	1,093	344	374
Professional services	101	124	32	43
Acquisition-related transaction and integration costs	81	42	19	14
Technology and communication	513	495	169	168
Rent and occupancy	63	61	22	20
Selling, general and administrative	166	163	54	52
Depreciation and amortization	768	759	258	253
Total operating expenses	2,750	2,737	898	924
Operating income	2,774	2,569	913	878
Other income/(expense):				
Interest income	42	—	33	—
Interest expense	(440)	(321)	(176)	(108)
Other income/(expense), net	(1,132)	1,341	(1,097)	54
Other income/(expense), net	(1,530)	1,020	(1,240)	(54)
Income/(loss) before income tax expense/(benefit)	1,244	3,589	(327)	824
Income tax expense/(benefit)	186	1,049	(152)	187
Net income/(loss)	\$ 1,058	\$ 2,540	\$ (175)	\$ 637
Net income attributable to non-controlling interest	(37)	(9)	(16)	(4)
Net income/(loss) attributable to Intercontinental Exchange, Inc.	\$ 1,021	\$ 2,531	\$ (191)	\$ 633
Earnings/(loss) per share attributable to Intercontinental Exchange, Inc. common stockholders:				
Basic	\$ 1.83	\$ 4.50	\$ (0.34)	\$ 1.12
Diluted	\$ 1.82	\$ 4.48	\$ (0.34)	\$ 1.12
Weighted average common shares outstanding:				
Basic	559	563	558	563
Diluted	561	565	560	566

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Comprehensive Income
(In millions)
(Unaudited)

	Nine Months Ended September 30,		Three Months Ended September 30,	
	2022	2021	2022	2021
Net income/(loss)	\$ 1,058	\$ 2,540	\$ (175)	\$ 637
Other comprehensive income/(loss):				
Foreign currency translation adjustments, net of tax benefit of \$1 for the nine months ended September 30, 2022 and \$1 for both the nine and three months ended September 30, 2021	(207)	(15)	(98)	(32)
Change in equity method investment, net of tax expense of \$1 for the nine months ended September 30, 2021	—	1	—	—
Other comprehensive loss	(207)	(14)	(98)	(32)
Comprehensive income/(loss)	\$ 851	\$ 2,526	\$ (273)	\$ 605
Comprehensive income attributable to non-controlling interest	(37)	(9)	(16)	(4)
Comprehensive income/(loss) attributable to Intercontinental Exchange, Inc.	\$ 814	\$ 2,517	\$ (289)	\$ 601

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Changes in Equity and Redeemable Non-Controlling Interest
(In millions)
(Unaudited)

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Income/(Loss)		
	Shares	Value	Shares	Value					
Balance, as of December 31, 2021	631	\$ 6	(70)	\$ (5,520)	\$ 14,069	\$ 14,350	\$ (196)	\$ 39	\$ 22,748
Other comprehensive loss	—	—	—	—	—	—	(207)	—	(207)
Exercise of common stock options	—	—	—	—	22	—	—	—	22
Repurchases of common stock	—	—	(4)	(632)	—	—	—	—	(632)
Payments relating to treasury shares	—	—	(1)	(72)	—	—	—	—	(72)
Stock-based compensation	—	—	—	—	129	—	—	—	129
Issuance under the employee stock purchase plan	1	—	—	—	49	—	—	—	49
Issuance of restricted stock	2	—	—	—	—	—	—	—	—
Distributions of profits	—	—	—	—	—	—	—	(35)	(35)
Dividends paid to stockholders	—	—	—	—	—	(640)	—	—	(640)
Net income/(loss) attributable to non-controlling interest	—	—	—	—	—	(37)	—	37	—
Net income	—	—	—	—	—	1,058	—	—	1,058
Balance, as of September 30, 2022	634	\$ 6	(75)	\$ (6,224)	\$ 14,269	\$ 14,731	\$ (403)	\$ 41	\$ 22,420

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Income/(Loss)		
	Shares	Value	Shares	Value					
Balance, as of June 30, 2022	633	\$ 6	(75)	\$ (6,223)	\$ 14,201	\$ 15,135	\$ (305)	\$ 47	\$ 22,861
Other comprehensive loss	—	—	—	—	—	—	(98)	—	(98)
Exercise of common stock options	—	—	—	—	2	—	—	—	2
Payments relating to treasury shares	—	—	—	(1)	—	—	—	—	(1)
Stock-based compensation	—	—	—	—	41	—	—	—	41
Issuance under the employee stock purchase plan	1	—	—	—	25	—	—	—	25
Distributions of profits	—	—	—	—	—	—	—	(22)	(22)
Dividends paid to stockholders	—	—	—	—	—	(213)	—	—	(213)
Net income/(loss) attributable to non-controlling interest	—	—	—	—	—	(16)	—	16	—
Net loss	—	—	—	—	—	(175)	—	—	(175)
Balance, as of September 30, 2022	634	\$ 6	(75)	\$ (6,224)	\$ 14,269	\$ 14,731	\$ (403)	\$ 41	\$ 22,420

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Changes in Equity and Redeemable Non-Controlling Interest — (Continued)
(In millions)
(Unaudited)

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity	Redeemable Non-Controlling Interest
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Income/(Loss)			
	Shares	Value	Shares	Value						
Balance, as of December 31, 2020	629	\$ 6	(68)	\$ (5,200)	\$ 13,845	\$ 11,039	\$ (192)	\$ 36	\$ 19,534	\$ 93
Other comprehensive income	—	—	—	—	—	—	(14)	—	(14)	—
Exercise of common stock options	—	—	—	—	10	—	—	—	10	—
Payments relating to treasury shares	—	—	(1)	(69)	—	—	—	—	(69)	—
Stock-based compensation	—	—	—	—	122	—	—	—	122	3
Issuance under the employee stock purchase plan	—	—	—	—	42	—	—	—	42	—
Issuance of restricted stock	2	—	—	—	—	—	—	—	—	—
Distributions of profits	—	—	—	—	—	—	—	(21)	(21)	—
Dividends paid to stockholders	—	—	—	—	—	(561)	—	—	(561)	—
Net income/(loss) attributable to non-controlling interest	—	—	—	—	—	(9)	—	18	9	(9)
Net income	—	—	—	—	—	2,540	—	—	2,540	—
Balance, as of September 30, 2021	631	\$ 6	(69)	\$ (5,269)	\$ 14,019	\$ 13,009	\$ (206)	\$ 33	\$ 21,592	\$ 87

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity	Redeemable Non-Controlling Interest
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Income/(Loss)			
	Shares	Value	Shares	Value						
Balance, as of June 30, 2021	631	\$ 6	(68)	\$ (5,267)	\$ 13,952	\$ 12,563	\$ (174)	\$ 36	\$ 21,116	\$ 89
Other comprehensive income	—	—	—	—	—	—	(32)	—	(32)	—
Exercise of common stock options	—	—	—	—	2	—	—	—	2	—
Payments relating to treasury shares	—	—	(1)	(2)	—	—	—	—	(2)	—
Stock-based compensation	—	—	—	—	41	—	—	—	41	1
Issuance under the employee stock purchase plan	—	—	—	—	24	—	—	—	24	—
Distributions of profits	—	—	—	—	—	—	—	(10)	(10)	—
Dividends paid to stockholders	—	—	—	—	—	(187)	—	—	(187)	—
Net income/(loss) attributable to non-controlling interest	—	—	—	—	—	(4)	—	7	3	(3)
Net income	—	—	—	—	—	637	—	—	637	—
Balance, as of September 30, 2021	631	\$ 6	(69)	\$ (5,269)	\$ 14,019	\$ 13,009	\$ (206)	\$ 33	\$ 21,592	\$ 87

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Cash Flows
(In millions)
(Unaudited)

	Nine Months Ended September 30,	
	2022	2021
Operating activities:		
Net income	\$ 1,058	\$ 2,540
Adjustments to reconcile net income to net cash provided by operating activities:		
Depreciation and amortization	768	759
Stock-based compensation	116	112
Deferred taxes	(515)	134
Gain on sale of Euroclear investment	(41)	—
Gain on sale of Coinbase investment	—	(1,227)
Gain on equity investments	—	(34)
Net losses/(income) from unconsolidated investees	1,112	(42)
Impairment of Bakkt investment	40	—
Other	31	36
Changes in assets and liabilities:		
Customer accounts receivable	(72)	(109)
Other current and non-current assets	(170)	(63)
Section 31 fees payable	1	(193)
Deferred revenue	130	163
Other current and non-current liabilities	4	54
Total adjustments	1,404	(410)
Net cash provided by operating activities	2,462	2,130
Investing activities:		
Capital expenditures	(125)	(117)
Capitalized software development costs	(200)	(211)
Purchases of invested margin deposits	(6,935)	(2,828)
Proceeds from sales of invested margin deposits	4,285	3,281
Cash paid for acquisitions, net of cash acquired	(57)	(10)
Purchases of equity and equity method investments	(69)	(23)
Proceeds from sale of Euroclear investment	741	—
Proceeds from sale of Coinbase investment	—	1,237
Other	(1)	(4)
Net cash provided by/(used in) investing activities	(2,361)	1,325
Financing activities:		
Proceeds from debt facilities, net	7,891	—
Repayments of debt facilities	(2,702)	(1,248)
Redemption of commercial paper, net	(1,012)	(1,081)
Repurchases of common stock	(632)	—
Dividends to stockholders	(640)	(561)
Change in cash and cash equivalent margin deposits and guaranty funds	13,503	24,094
Payments relating to treasury shares received for restricted stock tax payments and stock option exercises	(72)	(69)
Other	37	31
Net cash provided by financing activities	16,373	21,166
Effect of exchange rate changes on cash and cash equivalents, restricted cash and cash equivalents, and cash and cash equivalent margin deposits and guaranty funds	(41)	(4)
Net increase in cash and cash equivalents, restricted cash and cash equivalents, and cash and cash equivalent margin deposits and guaranty funds	16,433	24,617
Cash and cash equivalents, restricted cash and cash equivalents, and cash and cash equivalent margin deposits and guaranty funds at beginning of period	147,976	83,619
Cash and cash equivalents, restricted cash and cash equivalents, and cash and cash equivalent margin deposits and guaranty funds at end of period	\$ 164,409	\$ 108,236

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Cash Flows (Continued)
(In millions)
(Unaudited)

	As of September 30, 2022	As of September 30, 2021
Supplemental cash flow disclosure:		
Cash paid for income taxes	\$ 710	\$ 809
Cash paid for interest	\$ 368	\$ 329

Reconciliation of the components of cash and cash equivalents, restricted cash and cash equivalents, and cash and cash equivalent margin deposits and guaranty funds to the balance sheet:

Cash and cash equivalents	\$	1,183	\$	618
Short-term restricted cash and cash equivalents		6,032		1,045
Long-term restricted cash and cash equivalents		405		398
Cash and cash equivalent margin deposits and guaranty funds		156,789		106,175
Total	\$	164,409	\$	108,236

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Notes to Consolidated Financial Statements
(Unaudited)

1. Description of Business

Nature of Business and Organization

We are a provider of market infrastructure, data services and technology solutions to a broad range of customers including financial institutions, corporations and government entities. Our products, which span major asset classes including futures, equities, fixed income and residential mortgages in the United States, or U.S., provide our customers with access to mission critical tools that are designed to increase asset class transparency and workflow efficiency.

- In our Exchanges segment, we operate regulated marketplaces for the listing, trading and clearing of a broad array of derivatives contracts and financial securities.
- In our Fixed Income and Data Services segment, we provide fixed income pricing, reference data, indices, analytics and execution services as well as global credit default swap, or CDS, clearing and multi-asset class data delivery solutions.
- In our Mortgage Technology segment, we provide an end-to-end technology platform that offers customers comprehensive, digital workflow tools that aim to address the inefficiencies that exist in the U.S. residential mortgage market, from application through closing and the secondary market.

We operate marketplaces and provide technology and data services in the U.S., United Kingdom, or U.K., European Union, or EU, Canada, Asia Pacific and the Middle East.

2. Summary of Significant Accounting Policies

Basis of Presentation

We prepared the accompanying unaudited consolidated financial statements in accordance with U.S. generally accepted accounting principles, or U.S. GAAP, pursuant to the rules and regulations of the Securities and Exchange Commission, or SEC, regarding interim financial reporting. Accordingly, the unaudited consolidated financial statements do not include all of the information and footnotes required by U.S. GAAP for complete financial statements and should be read in conjunction with our audited consolidated financial statements and related notes thereto for the year ended December 31, 2021. The accompanying unaudited consolidated financial statements reflect all adjustments that are, in our opinion, necessary for a fair presentation of results for the interim periods presented. We believe that these adjustments are of a normal recurring nature.

Preparing financial statements in conformity with U.S. GAAP requires us to make certain estimates and assumptions that affect the amounts that are reported in our consolidated financial statements and accompanying disclosures. Actual amounts could differ from those estimates. The results of operations for the nine and three months ended September 30, 2022 are not necessarily indicative of the results to be expected for any future period or the full fiscal year.

These statements include the accounts of our wholly-owned and controlled subsidiaries. All intercompany balances and transactions between us and our wholly-owned and controlled subsidiaries have been eliminated in consolidation. For consolidated subsidiaries in which our ownership is less than 100% and for which we have control over the assets and liabilities and the management of the entity, the outside stockholders' interests are shown as non-controlling interests.

We have considered the impacts of the ongoing conflict between Russia, Belarus and Ukraine on our financial statements. As of September 30, 2022, our businesses and operations, including our exchanges, clearing houses, listings venues, data services businesses and mortgage platforms, have not suffered a material negative impact as a result of these events. There continues to be uncertainty surrounding the extent and duration of this ongoing conflict and the impact that it may have on the global economy and on our business.

Consolidated Statements of Cash Flows Presentation

As of December 31, 2021, we revised our consolidated statements of cash flows to include changes in cash and cash equivalent margin within cash flows from financing activities and changes in invested margin deposits within cash flows from investing activities. This immaterial revision did not have an effect on our previously reported consolidated balance sheets, statements of income, statements of comprehensive income, or statements of changes in equity and redeemable non-controlling interest or the related disclosures. Cash and cash equivalent margin amounts cannot be used to satisfy the Company's operating or other liabilities, as further discussed in Note 12. The following table summarizes the

immaterial revisions to our historical consolidated statements of cash flows for the nine months ended September 30, 2021 (in millions):

	Nine Months Ended September 30, 2021		
	As Previously Presented	Adjustment	As Adjusted
Purchases of invested margin deposits (within investing activities)	\$ —	\$ (2,828)	\$ (2,828)
Proceeds from sales of invested margin deposits (within investing activities)	—	3,281	3,281
Net cash provided by investing activities	872	453	1,325
Change in cash and cash equivalent margin deposits and guaranty funds (within financing activities)	—	24,094	24,094
Net cash provided by/(used in) financing activities	(2,928)	24,094	21,166
Net increase in cash and cash equivalents, restricted cash and cash equivalents, and cash and cash equivalent margin deposits and guaranty funds	70	24,547	24,617
Cash and cash equivalents, restricted cash and cash equivalents, and cash and cash equivalent margin deposits and guaranty funds at beginning of period	1,991	81,628	83,619
Cash and cash equivalents, restricted cash and cash equivalents, and cash and cash equivalent margin deposits and guaranty funds at end of period	\$ 2,061	\$ 106,175	\$ 108,236

Recently Adopted Accounting Pronouncements

During the nine months ended September 30, 2022, there were no significant changes to the new and recently adopted accounting pronouncements applicable to us from those disclosed in Note 2 to the consolidated financial statements in Part II, Item 8 of our Annual Report on Form 10-K for the year ended December 31, 2021, or the 2021 Form 10-K.

3. Acquisitions and Divestitures

Pending Acquisition of Black Knight, Inc.

On May 4, 2022, we announced that we had entered into a definitive agreement to acquire Black Knight, Inc., or Black Knight, a software, data and analytics company that serves the housing finance continuum, including real estate data, mortgage lending and servicing, as well as the secondary markets. Pursuant to that certain Agreement and Plan of Merger, dated as of May 4, 2022, among ICE, Sand Merger Sub Corporation, a wholly owned subsidiary of ICE, or Sub, and Black Knight, which we refer to as the “merger agreement,” Sub will merge with and into Black Knight, which we refer to as the “merger,” with Black Knight surviving as a wholly owned subsidiary of ICE. As of May 4, 2022, the transaction was valued at approximately \$13.1 billion, or \$85 per share of Black Knight common stock, with cash comprising 80% of the value of the aggregate transaction consideration and shares of our common stock comprising 20% of the value of the aggregate transaction consideration at that time. The aggregate cash component of the transaction consideration is fixed at \$10.5 billion, and the value of the aggregate stock component of the transaction consideration will fluctuate with the market price of our common stock and will be determined based on the average of the volume weighted averages of the trading prices of our common stock on each of the ten consecutive trading days ending three trading days prior to the closing of the merger. This transaction builds on our position as a provider of end-to-end electronic workflow solutions for the rapidly evolving U.S. residential mortgage industry.

Black Knight provides a comprehensive and integrated ecosystem of software, data and analytics solutions serving the real estate and housing finance markets. We believe the Black Knight ecosystem adds value for clients of all sizes across the mortgage and real estate lifecycles by helping organizations lower costs, increase efficiencies, grow their businesses, and reduce risk.

On August 19, 2022, our preliminary proxy statement/prospectus on Form S-4 was declared effective by the SEC, and on September 21, 2022, Black Knight stockholders approved the transaction. The transaction is expected to close in the first half of 2023 following the receipt of regulatory approvals and the satisfaction of customary closing conditions.

Bakkt Transaction

As discussed in Note 3 to the consolidated financial statements included in Part II, Item 8 of our 2021 Form 10-K, on October 15, 2021, Bakkt Holdings, LLC, or Bakkt, completed its merger with VPC Impact Acquisition Holdings, or VIH, a

special purpose acquisition company sponsored by Victory Park Capital, or VPC. The newly combined company was renamed Bakkt Holdings, Inc. and is listed on the New York Stock Exchange, or NYSE.

Following the transaction, we held an approximate 68% economic interest in the combined company. As a result of limitations on ICE from the Bakkt voting agreement entered into in connection with the transaction, we hold a minority voting interest in the combined company. Prior to the closing, Bakkt revenues and operating expenses were reported within our consolidated revenues and operating expenses. Following the closing, as a consequence of holding a minority voting interest in the combined company, during the fourth quarter of 2021 we deconsolidated Bakkt and treat it as an equity method investment within our financial statements.

4. Investments

Equity Investments

Our equity investments are subject to valuation under ASU 2016-01, *Financial Instruments- Overall (Subtopic 825-10): Recognition and Measurement of Financial Assets and Financial Liabilities*, or ASU 2016-01. See Note 14 for a discussion of our determination of fair value of our financial instruments.

Investment in Euroclear

We previously owned a 9.8% stake in Euroclear, plc, or Euroclear, that we originally purchased for \$631 million. We participated on the Euroclear Board of Directors, and classified our investment in Euroclear as an equity investment.

On May 20, 2022, we completed the sale of our 9.8% stake in Euroclear. The carrying value of our investment was \$700 million at the time of the sale and was classified within other current assets on our balance sheet. We recorded a net gain on the sale of \$41 million, which is included in other income during the nine months ended September 30, 2022.

We did not receive a dividend from Euroclear during the nine months ended September 30, 2022. We recognized dividend income of \$60 million and \$30 million during the nine and three months ended September 30, 2021 from Euroclear, which is included in other income.

Investment in Coinbase

On December 1, 2014, we acquired preferred stock of Coinbase Global, Inc., or Coinbase, which operates a cryptocurrency exchange platform, for \$10 million, representing a 1.4% ownership share on a fully-diluted, as-converted basis. On April 14, 2021, Coinbase completed an initial public offering, or IPO. On April 15, 2021, we completed the sale of our investment in Coinbase for \$1.24 billion and recorded a gain of \$1.23 billion, or \$892 million net of tax, as other income in our consolidated statement of income during the nine months ended September 30, 2021.

Equity Method Investments

Our equity method investments include the Options Clearing Corporation, or OCC, and Bakkt, among others. Our equity method investments are included in other non-current assets in the accompanying consolidated balance sheet. We carry our equity method investments at cost and assess the carrying value periodically if impairment indicators are present. At the end of each reporting period, we record our share of profits or losses of our equity method investments as equity earnings included in other income. We recognized (\$1.1 billion) and \$42 million as our share of estimated (losses)/profits, net, from our equity method investments during the nine months ended September 30, 2022 and 2021, respectively, and (\$1.1 billion) and \$8 million as our share of (losses)/profits, net, from our equity method investments during the three months ended September 30, 2022 and 2021, respectively. The estimated losses during the nine and three months ended September 30, 2022 are primarily related to our investment in Bakkt, and the estimated profits during the nine and three months ended September 30, 2021 are related to our investment in OCC. Both periods include adjustments to reflect the difference between reported prior period actual results from our original estimates.

When performing our assessment of the carrying value of our investments, we consider, among other things, the length of time and the extent to which the market value has been less than our cost basis, if applicable, the investee's financial condition and near-term prospects, the economic or technological environment in which our investees operate, weakening of the general market condition of the related industry, whether an investee can continue as a going concern, any impairment charges recorded by an investee on goodwill, intangible or long-lived assets, and our intent and ability to retain the investment for a period of time sufficient to allow for any anticipated recovery in market value.

Investment in OCC

We own a 40% interest in OCC through a direct investment by the NYSE. OCC is regulated by the SEC as a registered clearing agency and by the Commodity Futures Trading Commission, or CFTC, as a derivatives clearing organization. OCC serves as a clearing house for securities options, security futures, commodity futures and options on futures traded on various independent exchanges. OCC clears securities options traded on NYSE Arca and NYSE Amex Options, along with other non-affiliated exchanges.

Investment in Bakkt

Following Bakkt's October 15, 2021 merger with VIH, we held an approximate 68% economic interest in Bakkt and treat it as an equity method investment (see Note 3). During the three months ended September 30, 2022, Bakkt reported an impairment of goodwill and intangible assets of approximately \$1.5 billion, of which \$1.0 billion is included in our share of estimated losses. As of September 30, 2022, after recording our share of Bakkt's equity method losses, which included Bakkt's impairment charge, the carrying value was determined to be \$439 million. We also recorded an impairment of \$40 million in our investment in Bakkt to its fair value as of September 30, 2022 as other expense. This was based on what we consider to be an other than temporary decline in fair value as a result of the factors noted above, including consideration for the impairment charge recorded by Bakkt.

5. Revenue Recognition

Substantially all of our revenues are considered to be revenues from contracts with customers. The related accounts receivable balances are recorded in our balance sheets as customer accounts receivable. We do not have obligations for warranties, returns or refunds to customers, other than rebates, which are settled each period and therefore do not result in variable consideration. We do not have significant revenue recognized from performance obligations that were satisfied in prior periods, and we do not have any transaction price allocated to unsatisfied performance obligations other than in our deferred revenue.

Deferred revenue represents our contract liabilities related to our annual, original and other listings revenues, certain data services, clearing services, mortgage technology services and other revenues. Deferred revenue is our only significant contract liability. See Note 7 for our discussion of deferred revenue balances, activity, and expected timing of recognition.

For all of our contracts with customers, except for listings and certain data, clearing and mortgage services, our performance obligations are short term in nature and there is no significant variable consideration. In addition, we have elected the practical expedient of excluding sales taxes from transaction prices. We have assessed the costs incurred to obtain or fulfill a contract with a customer, which are primarily our sales commissions.

Certain judgments and estimates are used in the identification and timing of satisfaction of performance obligations and the related allocation of transaction price. We believe that these represent a faithful depiction of the transfer of services to our customers. Refer to Note 5 to the consolidated financial statements included in Part II, Item 8 of our 2021 Form 10-K where our primary revenue contract classifications are described in detail.

The following table depicts the disaggregation of our revenue according to business line and segment (in millions). Amounts here have been aggregated as they follow consistent revenue recognition patterns, and are consistent with the segment information in Note 15:

	Exchanges Segment	Fixed Income and Data Services Segment	Mortgage Technology Segment	Total Consolidated
Nine Months Ended September 30, 2022:				
Total revenues	\$ 4,824	\$ 1,555	\$ 880	\$ 7,259
Transaction-based expenses	1,735	—	—	1,735
Total revenues, less transaction-based expenses	\$ 3,089	\$ 1,555	\$ 880	\$ 5,524
Timing of Revenue Recognition				
Services transferred at a point in time	\$ 1,760	\$ 263	\$ 374	\$ 2,397
Services transferred over time	1,329	1,292	506	3,127
Total revenues, less transaction-based expenses	\$ 3,089	\$ 1,555	\$ 880	\$ 5,524

	Exchanges Segment	Fixed Income and Data Services Segment	Mortgage Technology Segment	Total Consolidated
Nine Months Ended September 30, 2021:				
Total revenues	\$ 4,376	\$ 1,403	\$ 1,061	\$ 6,840
Transaction-based expenses	1,534	—	—	1,534
Total revenues, less transaction-based expenses	\$ 2,842	\$ 1,403	\$ 1,061	\$ 5,306
Timing of Revenue Recognition				
Services transferred at a point in time	\$ 1,590	\$ 162	\$ 631	\$ 2,383
Services transferred over time	1,252	1,241	430	2,923
Total revenues, less transaction-based expenses	\$ 2,842	\$ 1,403	\$ 1,061	\$ 5,306
	Exchanges Segment	Fixed Income and Data Services Segment	Mortgage Technology Segment	Total Consolidated
Three Months Ended September 30, 2022:				
Total revenues	\$ 1,577	\$ 534	\$ 276	\$ 2,387
Transaction-based expenses	576	—	—	576
Total revenues, less transaction-based expenses	\$ 1,001	\$ 534	\$ 276	\$ 1,811
Timing of Revenue Recognition				
Services transferred at a point in time	\$ 563	\$ 104	\$ 104	\$ 771
Services transferred over time	438	430	172	1,040
Total revenues, less transaction-based expenses	\$ 1,001	\$ 534	\$ 276	\$ 1,811
	Exchanges Segment	Fixed Income and Data Services Segment	Mortgage Technology Segment	Total Consolidated
Three Months Ended September 30, 2021:				
Total revenues	\$ 1,434	\$ 477	\$ 366	\$ 2,277
Transaction-based expenses	475	—	—	475
Total revenues, less transaction-based expenses	\$ 959	\$ 477	\$ 366	\$ 1,802
Timing of Revenue Recognition				
Services transferred at a point in time	\$ 534	\$ 55	\$ 214	\$ 803
Services transferred over time	425	422	152	999
Total revenues, less transaction-based expenses	\$ 959	\$ 477	\$ 366	\$ 1,802

The Exchanges segment and the Fixed Income and Data Services segment revenues above include data services revenues. Our data services revenues are transferred over time, and a majority of those revenues are performed over a short period of time of one month or less and relate to subscription-based data services billed monthly, quarterly or annually in advance. These revenues are recognized ratably over time as our data delivery performance obligations are met consistently throughout the period.

The Exchanges segment revenues transferred over time in the table above also include services related to listings, services related to risk management of open interest performance obligations and services related to regulatory fees, trading permits, and software licenses.

The Fixed Income and Data Services segment revenues transferred over time in the table above also include services related to risk management of open interest performance obligations, primarily in our CDS business.

The Mortgage Technology segment revenues transferred over time in the table above primarily relate to our origination technology revenue where performance obligations consist of a series of distinct services and are recognized over the

contract terms as subscription performance obligations are satisfied, and to a lesser extent, professional services revenues and revenues from certain of our data and analytics offerings.

The components of services transferred over time for each of our segments are as follows:

	Nine Months Ended September 30,		Three Months Ended September 30,	
	2022	2021	2022	2021
Exchanges Segment:				
Data services revenues	\$ 651	\$ 623	\$ 219	\$ 208
Services transferred over time related to risk management of open interest performance obligations	\$ 201	\$ 190	\$ 62	\$ 65
Services transferred over time related to listings	\$ 388	\$ 356	\$ 128	\$ 123
Services transferred over time related to regulatory fees, trading permits, and software licenses	\$ 89	\$ 83	\$ 29	\$ 29
Total	\$ 1,329	\$ 1,252	\$ 438	\$ 425
Fixed Income Data Services Segment:				
Data services revenues	\$ 1,263	\$ 1,220	\$ 420	\$ 414
Services transferred over time related to risk management of open interest performance obligations in our CDS business	\$ 29	\$ 21	\$ 10	\$ 8
Total	\$ 1,292	\$ 1,241	\$ 430	\$ 422
Mortgage Technology Segment:				
Subscription revenues	\$ 479	\$ 404	\$ 163	\$ 143
Professional service revenues and other	\$ 27	\$ 26	\$ 9	\$ 9
Total	\$ 506	\$ 430	\$ 172	\$ 152
Total consolidated revenues transferred over time	\$ 3,127	\$ 2,923	\$ 1,040	\$ 999

6. Goodwill and Other Intangible Assets

The following is a summary of the activity in the goodwill balance for the nine months ended September 30, 2022 (in millions):

Goodwill balance at December 31, 2021	\$ 21,123
Acquisitions	46
Foreign currency translation	(90)
Other activity, net	(4)
Goodwill balance at September 30, 2022	\$ 21,075

The following is a summary of the activity in the other intangible assets balance for the nine months ended September 30, 2022 (in millions):

Other intangible assets balance at December 31, 2021	\$ 13,736
Acquisitions	14
Foreign currency translation	(84)
Amortization of other intangible assets	(459)
Other activity, net	3
Other intangible assets balance at September 30, 2022	\$ 13,210

Foreign currency translation adjustments result from a portion of our goodwill and other intangible assets being held at our U.K., EU and Canadian subsidiaries, whose functional currencies are not the U.S. dollar. The changes in other activity, net, in the table above primarily relate to adjustments to the fair value of the net tangible and intangible assets made within one year of acquisitions, with a corresponding adjustment to goodwill.

During the three months ended September 30, 2022, we considered potential indicators of impairment to goodwill and other intangible assets for each of our reporting units, which included declines in stock price, recent inflation spikes and rising interest rates, among others. As such, we performed a qualitative assessment to determine whether it was more-

likely-than-not that goodwill and indefinite lived intangibles within each of our reportable business segments were impaired. Additionally, we evaluated whether the carrying value of the finite lived intangible assets within our reportable business segments may not be recoverable. After evaluating events, circumstances and factors which could affect the significant inputs used in our evaluation of cash flows and related fair value, we determined it was not more-likely-than-not that an impairment existed in our goodwill and indefinite lived intangible assets or that the carrying amount of our finite lived intangible assets was not recoverable. We plan to perform our annual impairment testing in the fourth quarter.

7. Deferred Revenue

Our contract liabilities, or deferred revenue, represent consideration received that is yet to be recognized as revenue. Total deferred revenue was \$410 million as of September 30, 2022, including \$315 million in current deferred revenue and \$95 million in other non-current liabilities. The changes in our deferred revenue during the nine months ended September 30, 2022 are as follows (in millions):

	Annual Listings Revenues	Original Listings Revenues	Other Listings Revenues	Data Services and Other Revenues	Mortgage Technology	Total
Deferred revenue balance at December 31, 2021	\$ —	\$ 19	\$ 93	\$ 93	\$ 79	\$ 284
Additions	434	32	39	330	62	897
Amortization	(327)	(28)	(33)	(301)	(82)	(771)
Deferred revenue balance at September 30, 2022	\$ 107	\$ 23	\$ 99	\$ 122	\$ 59	\$ 410

The changes in our deferred revenue during the nine months ended September 30, 2021 are as follows (in millions):

	Annual Listings Revenues	Original Listings Revenues	Other Listings Revenues	Data Services and Other Revenues	Mortgage Technology	Total
Deferred revenue balance at December 31, 2020	\$ —	\$ 13	\$ 92	\$ 95	\$ 59	\$ 259
Additions	397	26	42	352	60	877
Amortization	(301)	(24)	(31)	(319)	(39)	(714)
Deferred revenue balance at September 30, 2021	\$ 96	\$ 15	\$ 103	\$ 128	\$ 80	\$ 422

Included in the amortization recognized during the nine months ended September 30, 2022 is \$144 million related to the deferred revenue balance as of December 31, 2021. Included in the amortization recognized for the nine months ended September 30, 2021 is \$128 million related to the deferred revenue balance as of December 31, 2020. As of September 30, 2022, the remaining deferred revenue balance will be recognized over the period of time we satisfy our performance obligations as described in Note 5.

8. Debt

Our total debt, including short-term and long-term debt, consisted of the following (in millions):

	As of September 30, 2022	As of December 31, 2021
Debt:		
Short-term debt:		
Commercial Paper	\$ —	\$ 1,012
2022 Senior Notes (2.35% senior unsecured notes due September 15, 2022)	—	499
Other short-term debt	7	10
Total short-term debt	7	1,521
Long-term debt:		
2023 Senior Notes (0.70% senior unsecured notes due June 15, 2023)	—	997
2023 Senior Notes (3.45% senior unsecured notes due September 21, 2023)	—	399
2023 Senior Notes (4.00% senior unsecured notes due October 15, 2023)	—	797
2025 Senior Notes (3.65% senior unsecured notes due May 23, 2025)	1,243	—
2025 Senior Notes (3.75% senior unsecured notes due December 1, 2025)	1,247	1,246
2027 Senior Notes (4.00% senior unsecured notes due September 15, 2027)	1,485	—
2027 Senior Notes (3.10% senior unsecured notes due September 15, 2027)	497	497
2028 Senior Notes (3.75% senior unsecured notes due September 21, 2028)	594	594
2029 Senior Notes (4.35% senior unsecured notes due June 15, 2029)	1,239	—
2030 Senior Notes (2.10% senior unsecured notes due June 15, 2030)	1,235	1,234
2032 Senior Notes (1.85% senior unsecured notes due September 15, 2032)	1,484	1,483
2033 Senior Notes (4.60% senior unsecured notes due March 15, 2033)	1,488	—
2040 Senior Notes (2.65% senior unsecured notes due September 15, 2040)	1,231	1,230
2048 Senior Notes (4.25% senior unsecured notes due September 21, 2048)	1,231	1,230
2050 Senior Notes (3.00% senior unsecured notes due June 15, 2050)	1,221	1,220
2052 Senior Notes (4.95% senior unsecured notes due June 15, 2052)	1,464	—
2060 Senior Notes (3.00% senior unsecured notes due September 15, 2060)	1,471	1,470
2062 Senior Notes (5.20% senior unsecured notes due June 15, 2062)	983	—
Total long-term debt	18,113	12,397
Total debt	\$ 18,120	\$ 13,918

Our senior notes of \$18.1 billion have a weighted average maturity of 17 years and a weighted average cost of 3.6% per annum.

Credit Facilities

We have a \$3.9 billion senior unsecured revolving credit facility, or the Credit Facility, with future capacity to increase our borrowings under the Credit Facility by an additional \$1.0 billion, subject to the consent of the lenders funding the increase and certain other conditions. On May 25, 2022, we agreed with the lenders to extend the maturity date of the Credit Facility from October 15, 2026, to May 25, 2027, among other items. We also exercised our option to increase the amount of the Credit Facility from \$3.8 billion to \$3.9 billion. We incurred new debt issuance costs of \$4 million relating to the Credit Facility and these costs are represented in the accompanying consolidated balance sheet as other non-current assets and will be amortized over the remaining life of the Credit Facility. No amounts were outstanding under the Credit Facility as of September 30, 2022.

As of September 30, 2022, of the \$3.9 billion that was available for borrowing under the Credit Facility, \$170 million was required to support certain broker-dealer and other subsidiary commitments. As there was no commercial paper outstanding as of September 30, 2022, there was no required amount to backstop our U.S. dollar commercial paper program, or the Commercial Paper Program. The amount required to backstop the amounts outstanding under the Commercial Paper Program will fluctuate as we increase or decrease our commercial paper borrowings. The remaining \$3.7 billion was available for working capital and general corporate purposes including, but not limited to, acting as a backstop to future increases in the amounts outstanding under the Commercial Paper Program.

On May 4, 2022, we entered into a 364-day senior unsecured bridge facility in an aggregate principal amount not to exceed \$14.0 billion, or the Bridge Facility. The commitments that the Company obtained for the Bridge Facility have been permanently reduced from \$14.0 billion and there were no amounts outstanding as of September 30, 2022 as a result of (i) the amendment and extension of the Credit Facility, (ii) the issuance by the Company of certain senior unsecured notes on May 23, 2022, (iii) Euroclear divestment proceeds, (iv) the generation of cash internally by the Company, and (v) the effectiveness of our term loan facility.

On May 25, 2022, we entered into a \$2.4 billion two-year senior unsecured delayed draw term loan facility, or Term Loan. Draws under the Term Loan bear interest on the principal amount outstanding at either (a) Term Secured Overnight Financing Rate, or Term SOFR, plus an applicable margin plus a credit spread adjustment of 10 basis points or (b) a "base rate" plus an applicable margin. The applicable margin ranges from 0.625% to 1.125% for Term SOFR loans and from 0.000% to 0.125% for base rate loans, in each case, based on a ratings-based pricing grid. The proceeds from borrowings under the Term Loan will be used to fund a portion of the purchase price for the Black Knight acquisition. We incurred new debt issuance costs of \$4 million relating to the Term Loan and these costs are represented in the accompanying consolidated balance sheet as other non-current assets and will be amortized over the remaining life of the Term Loan. We have the option to prepay outstanding amounts under the Term Loan in whole or in part at any time. No amounts were outstanding under the Term Loan as of September 30, 2022.

Our India subsidiaries maintain \$14 million of credit lines for their general corporate purposes. As of September 30, 2022, they had borrowed \$7 million, which is reflected as "other short-term debt" in the table above.

Commercial Paper Program

Our Commercial Paper Program is currently backed by the borrowing capacity available under the Credit Facility, as described above. The effective interest rate of commercial paper issuances does not materially differ from short-term interest rates, which fluctuate due to market conditions and as a result may impact our interest expense. During the nine months ended September 30, 2022, we had net paydowns of \$1.0 billion under the Commercial Paper Program and did not have any notes outstanding under our Commercial Paper Program as of September 30, 2022.

New Senior Notes

On May 23, 2022, we issued \$8.0 billion in aggregate principal amount of new senior notes, comprised of the following:

- \$1.25 billion in aggregate principal amount of 3.65% senior notes due in 2025, or the 2025 Notes;
- \$1.5 billion in aggregate principal amount of 4.00% senior notes due in 2027, or the 2027 Notes;
- \$1.25 billion in aggregate principal amount of 4.35% senior notes due in 2029, or the 2029 Notes;
- \$1.5 billion in aggregate principal amount of 4.60% senior notes due in 2033, or the 2033 Notes;
- \$1.5 billion in aggregate principal amount of 4.95% senior notes due in 2052, or the 2052 Notes; and
- \$1.0 billion in aggregate principal amount of 5.20% senior notes due in 2062, or the 2062 Notes, collectively, the Notes.

We intend to use the net proceeds of \$4.9 billion from the offering of the 2025 Notes, the 2027 Notes, the 2029 Notes and the 2062 Notes, or collectively, the SMR Notes, together with the issuance of commercial paper and/or borrowings under the Credit Facility, cash on hand or other immediately available funds and borrowings under the Term Loan, to finance the cash portion of the purchase price for Black Knight. The SMR Notes are subject to a special mandatory redemption feature pursuant to which we will be required to redeem all of the outstanding SMR Notes at a redemption price equal to 101% of the aggregate principal amount of the SMR Notes, plus accrued and unpaid interest, in the event that the Black Knight acquisition is not consummated on or prior to May 4, 2023, subject to two automatic extensions of three months each, to August 4, 2023 and to November 4, 2023, respectively, if U.S. antitrust clearance or a related law, injunction, order or other judgment, in each case whether temporary, preliminary or permanent, that restrains, enjoins or otherwise prohibits the consummation of the Black Knight merger remains outstanding and all other conditions to closing are satisfied (or in the case of conditions that by their terms are to be satisfied at the closing, are capable of being satisfied if the closing were to occur on such date) at each extension date, or if the Black Knight merger agreement is terminated at any time prior to such date. The \$4.9 billion net proceeds from the SMR Notes are separately invested and recorded as short-term restricted cash and cash equivalents in our consolidated balance sheet as of September 30, 2022.

We used the \$3.0 billion of net proceeds from the offering of the 2033 Notes and the 2052 Notes to redeem \$2.7 billion aggregate principal amount of four series of senior notes that would have matured in 2022 and 2023. The balance of the net proceeds was used for general corporate purposes, which included paying down a portion of the amounts outstanding under our Commercial Paper Program. We recorded \$30 million in costs associated with the extinguishment and re-financing of our existing debt in connection with our May 2022 debt refinancing. These costs are included in interest

expense in our consolidated statements of income for the nine months ended September 30, 2022.

We incurred debt issuance costs of \$67 million relating to the issuance of the Notes and these costs are presented in the accompanying consolidated balance sheet as a deduction from the carrying amount of the related debt liability and will be amortized over the remaining term of each series of the Notes. The Notes contain affirmative and negative covenants, including, but not limited to, certain redemption rights, limitations on liens and indebtedness and limitations on certain mergers, sales, dispositions and lease-back transactions.

9. Share-Based Compensation

We currently sponsor employee and director stock option, restricted stock and employee stock purchase plans. Stock options and restricted stock are granted at the discretion of the Compensation Committee of our Board of Directors, or Board, based on the estimated fair value on the date of grant. The fair value of the stock options and restricted stock on the date of grant is recognized as expense over the vesting period, net of forfeitures. The non-cash compensation expenses recognized in our consolidated statements of income for stock options, restricted stock and under our employee stock purchase plan, net of amounts classified as capitalized software, were \$116 million and \$112 million for the nine months ended September 30, 2022 and 2021, respectively, and \$40 million and \$39 million for the three months ended September 30, 2022 and 2021, respectively.

Stock Option Plans

We use the Black-Scholes option pricing model to value our stock option awards. During the nine months ended September 30, 2022 and 2021, we used the assumptions in the table below to compute the value:

Assumptions:	Nine Months Ended September 30,	
	2022	2021
Risk-free interest rate	1.72%	0.64%
Expected life in years	6.0	5.7
Expected volatility	23%	24%
Expected dividend yield	1.17%	1.16%
Estimated weighted-average fair value of options granted per share	\$28.18	\$22.70

The risk-free interest rate is based on the zero-coupon U.S. Treasury yield curve in effect at the date of grant. The expected life is derived from historical and anticipated future exercise patterns. Expected volatility is based on historical volatility data of our stock.

Restricted Stock Plans

Restricted shares are used as an incentive to attract and retain qualified employees and to align our and our stockholders' interests by linking actual performance to both short and long-term stockholder return. We issue awards that may contain a combination of time, performance and/or market conditions. The grant date fair value of each award is based on the closing stock price of our stock at the date of grant. For time-based restricted stock, we recognize expense ratably over the vesting period, which is typically three or four years, net of forfeitures.

In February 2022, we reserved a maximum of 0.7 million restricted shares for potential issuance as performance-based restricted shares to certain of our employees. The number of shares ultimately granted under this award will be based on our actual financial performance as compared to financial performance targets set by our Board and the Compensation Committee for the year ending December 31, 2022, and will also be subject to a market condition reduction based on how our 2022 total stockholder return, or TSR, compares to that of the S&P 500 Index. The maximum compensation expense to be recognized under these performance-based restricted shares is \$84 million if the maximum financial performance target is met and all 0.7 million shares vest. The compensation expense to be recognized under these performance-based restricted shares will be \$42 million if the target financial performance is met, which would result in 0.3 million shares vesting. For these awards with performance conditions, we recognize expense on an accelerated basis over the three-year vesting period based on our quarterly assessment of the probable 2022 actual financial performance as compared to the 2022 financial performance targets. As of September 30, 2022, our best estimate is that the financial performance level will be below target for 2022. Based on this assessment, we recorded non-cash compensation expense of \$14 million and \$5 million for the nine and three months ended September 30, 2022, respectively, related to these awards and the remaining \$26 million in non-cash compensation expense will be recorded on an accelerated basis over the remaining vesting period, including \$8 million which will be recorded over the remainder of 2022.

We also issue awards with a market condition but no performance condition. The fair value of these awards is estimated based on a simulation of various outcomes and includes inputs such as our stock price on the grant date, the valuation of historical awards with market conditions, the relatively low likelihood that the market condition will affect the number of shares granted (as the market condition only affects shares granted in excess of certain financial performance targets), and our expectation of achieving the financial performance targets.

10. Equity

Stock Repurchase Program

In December 2021, our Board approved an aggregate of \$3.15 billion for future repurchases of our common stock with no fixed expiration date that became effective on January 1, 2022. The \$3.15 billion replaced the previous amount approved by the Board. The approval of our Board for the share repurchases does not obligate us to acquire any particular amount of our common stock. In addition, our Board may increase or decrease the amount available for repurchases from time to time. We fund repurchases from our operating cash flow or borrowings under our debt facilities or our Commercial Paper Program. Repurchases may be made from time to time on the open market, through established trading plans, in privately-negotiated transactions or otherwise, in accordance with all applicable securities laws, rules and regulations. We may begin or discontinue stock repurchases at any time and may amend or terminate a Rule 10b5-1 trading plan at any time or enter into additional plans.

During the nine months ended September 30, 2022, we repurchased a total of 5.0 million shares of our outstanding common stock at a cost of \$632 million, consisting of 4.6 million shares at a cost of \$582 million under our Rule 10b5-1 trading plan and 0.4 million shares at a cost of \$50 million on the open market during an open trading period. We did not have any stock repurchases during the three months ended September 30, 2022 or during the nine months ended September 30, 2021. As of September 30, 2022, the remaining balance of Board approved funds for future repurchases was \$2.5 billion. In connection with our pending acquisition of Black Knight, on May 4, 2022 we terminated our Rule 10b5-1 trading plan and suspended share repurchases.

Dividends

During the nine months ended September 30, 2022 and 2021, we declared and paid cash dividends per share of \$1.14 and \$0.99, respectively, for an aggregate payout of \$640 million and \$561 million, respectively. During the three months ended September 30, 2022 and 2021, we declared and paid cash dividends per share of \$0.38 and \$0.33, respectively, for an aggregate payout of \$213 million and \$187 million, respectively. The declaration of dividends is subject to the discretion of our Board. Our Board has adopted a quarterly dividend declaration policy providing that the declaration of any dividends will be determined quarterly by the Board or the Audit Committee, taking into account such factors as our evolving business model, prevailing business conditions, our financial results and capital requirements and other considerations which our Board deems relevant, without a predetermined annual net income payout ratio.

Accumulated Other Comprehensive Income/(Loss)

The following tables present changes in the accumulated balances for each component of other comprehensive income/ (loss) (in millions):

	Changes in Accumulated Other Comprehensive Income/(Loss) by Component			
	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of December 31, 2021	\$ (150)	\$ 2	\$ (48)	\$ (196)
Other comprehensive income/(loss)	(208)	—	—	(208)
Income tax benefit/(expense)	1	—	—	1
Net current period other comprehensive income/(loss)	(207)	—	—	(207)
Balance, as of September 30, 2022	\$ (357)	\$ 2	\$ (48)	\$ (403)

Changes in Accumulated Other Comprehensive Income/(Loss) by Component

	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of June 30, 2022	\$ (259)	\$ 2	\$ (48)	\$ (305)
Other comprehensive income/(loss)	(98)	—	—	(98)
Income tax benefit/(expense)	—	—	—	—
Net current period other comprehensive income/(loss)	(98)	—	—	(98)
Balance, as of September 30, 2022	<u>\$ (357)</u>	<u>\$ 2</u>	<u>\$ (48)</u>	<u>\$ (403)</u>

Changes in Accumulated Other Comprehensive Income/(Loss) by Component

	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of December 31, 2020	\$ (134)	\$ 1	\$ (59)	\$ (192)
Other comprehensive income/(loss)	(16)	2	—	(14)
Income tax benefit/(expense)	1	(1)	—	—
Net current period other comprehensive income/(loss)	(15)	1	—	(14)
Balance, as of September 30, 2021	<u>\$ (149)</u>	<u>\$ 2</u>	<u>\$ (59)</u>	<u>\$ (206)</u>

Changes in Accumulated Other Comprehensive Income/(Loss) by Component

	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of June 30, 2021	\$ (117)	\$ 2	\$ (59)	\$ (174)
Other comprehensive income/(loss)	(33)	—	—	(33)
Income tax benefit/(expense)	1	—	—	1
Net current period other comprehensive income/(loss)	(32)	—	—	(32)
Balance, as of September 30, 2021	<u>\$ (149)</u>	<u>\$ 2</u>	<u>\$ (59)</u>	<u>\$ (206)</u>

11. Income Taxes

Our effective tax rate was 15% and 29% during the nine months ended September 30, 2022 and 2021, respectively, and 47% and 23% during the three months ended September 30, 2022 and 2021, respectively. The effective tax rate for the nine months ended September 30, 2022 was lower than the effective tax rate for the comparable period in 2021 primarily due to the deferred income tax benefit from the impairment to our equity method investment in Bakkt in the current year period, and the deferred income tax expense from U.K. tax law changes in the prior year period. During the nine months ended September 30, 2021, the U.K. Finance Act 2021 was enacted, which increased the U.K. corporate income tax rate from 19% to 25%, effective April 1, 2023. The effective tax rate for the three months ended September 30, 2022, was higher than the effective tax rate for the comparable period in 2021 primarily due to the deferred income tax benefit from the impairment to our equity method investment in Bakkt.

12. Clearing Operations

We operate six clearing houses, each of which acts as a central counterparty that becomes the buyer to every seller and the seller to every buyer for its clearing members or participants, or Members. Through this central counterparty function, the clearing houses provide financial security for each transaction for the duration of the position by limiting counterparty credit risk.

Our clearing houses are responsible for providing clearing services to each of our futures exchanges, and in some cases to third-party execution venues, and are as follows, referred to herein collectively as "the ICE Clearing Houses":

Clearing House	Products Cleared	Exchange where Executed	Location
ICE Clear Europe	Energy, agricultural, interest rates and equity index futures and options contracts and OTC European CDS instruments	ICE Futures Europe, ICE Futures U.S., ICE Endex, ICE Futures Abu Dhabi and third-party venues	U.K.
ICE Clear U.S.	Agricultural, metals, foreign exchange, or FX, interest rate, equity index and digital asset futures and/or options contracts	ICE Futures U.S.	U.S.
ICE Clear Credit	OTC North American, European, Asian-Pacific and Emerging Market CDS instruments	Creditex and third-party venues	U.S.
ICE Clear Netherlands	Derivatives on equities and equity indices traded on regulated markets	ICE Endex	The Netherlands
ICE Clear Singapore	Energy, metals and financial futures products and digital assets futures contracts	ICE Futures Singapore	Singapore
ICE NGX	Physical North American natural gas and electricity	ICE NGX	Canada

In June 2022, we announced our decision to cease our CDS clearing service at ICE Clear Europe, our clearing house in the U.K., and our CDS clearing offering will therefore be consolidated at our ICE Clear Credit clearing house in the U.S.

Original and Variation Margin

Each of the ICE Clearing Houses generally requires all Members to deposit collateral in cash or certain pledged assets. The collateral deposits are known as “original margin.” In addition, the ICE Clearing Houses may make intraday original margin calls in circumstances where market conditions require additional protection. The daily profits and losses to and from the ICE Clearing Houses due to the marking-to-market of open contracts is known as “variation margin.” With the exception of ICE NGX’s physical natural gas and physical power products discussed separately below, the ICE Clearing Houses mark all outstanding contracts to market, and therefore pay and collect variation margin, at least once daily.

The amounts that Members are required to maintain are determined by proprietary risk models established by each ICE Clearing House and reviewed by the relevant regulators, independent model validators, risk committees and the boards of directors of the respective ICE Clearing House. The amounts required may fluctuate over time. Each of the ICE Clearing Houses is a separate legal entity and is not subject to the liabilities of the others, or the obligations of Members of the other ICE Clearing Houses.

Should a particular Member fail to deposit its original margin or fail to make a variation margin payment, when and as required, the relevant ICE Clearing House may liquidate or hedge the defaulting Member’s open positions and use their original margin and guaranty fund deposits to pay any amount owed. In the event that the defaulting Member’s deposits are not sufficient to pay the amount owed in full, the ICE Clearing Houses will first use their respective contributions to the guaranty fund, often referred to as Skin In The Game, or SITG, to pay any remaining amount owed. In the event that the SITG is not sufficient, the ICE Clearing Houses may utilize the respective guaranty fund deposits and default insurance, or collect limited additional funds from their respective non-defaulting Members on a pro-rata basis, to pay any remaining amount owed.

As of September 30, 2022 and December 31, 2021, the ICE Clearing Houses had received or had been pledged \$290.2 billion and \$239.9 billion, respectively, in cash and non-cash collateral in original margin and guaranty fund deposits to cover price movements of underlying contracts for both periods.

Guaranty Funds and ICE Contribution

As described above, mechanisms have been created, called guaranty funds, to provide partial protection in the event of a Member default. With the exception of ICE NGX, each of the ICE Clearing Houses requires that each Member make deposits into a guaranty fund.

In addition, we have contributed our own capital that could be used if a defaulting Member’s original margin and guaranty fund deposits are insufficient. Such amounts are recorded as long-term restricted cash and cash equivalents in our balance sheets and are as follows (in millions):

Clearing House	ICE Portion of Guaranty Fund Contribution		Default insurance	
	As of September 30, 2022	As of December 31, 2021	As of September 30, 2022	As of December 31, 2021
ICE Clear Europe	\$247	\$247	\$100	\$75
ICE Clear U.S.	90	83	25	25
ICE Clear Credit	50	50	75	50
ICE Clear Netherlands	2	2	N/A	N/A
ICE Clear Singapore	1	1	N/A	N/A
ICE NGX	15	15	200	100
Total	\$405	\$398	\$400	\$250

Of our total contribution to ICE Clear U.S. above, as of September 30, 2022, \$15 million was solely applicable to any losses associated with a default in Bitcoin contracts and other digital assets that ICE Clear U.S. may clear in the future.

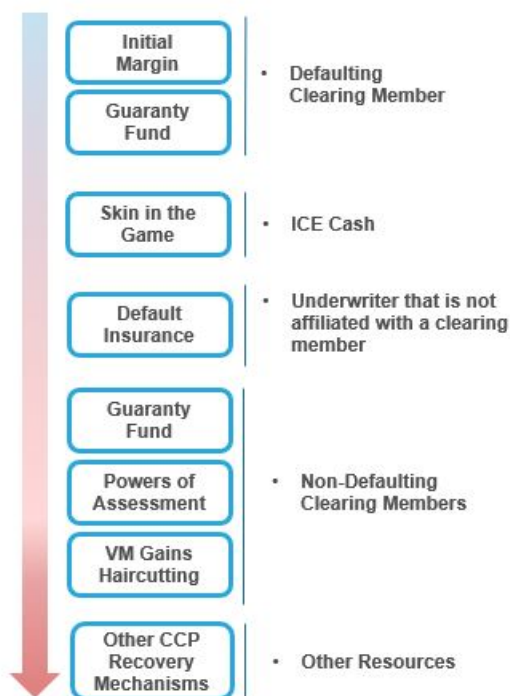
We also maintain default insurance as an additional layer of clearing member default protection. The default insurance was renewed in September 2022 and has a three-year term for the following clearing houses in the following amounts: ICE Clear Europe - \$100 million; ICE Clear U.S. - \$25 million and ICE Clear Credit - \$75 million. The default insurance layer resides after and in addition to the ICE Clear Europe, ICE Clear U.S. and ICE Clear Credit SITG contributions and before the guaranty fund contributions of the non-defaulting Members.

Similar to SITG, the default insurance layer is not intended to replace or reduce the position risk-based amount of the guaranty fund. As a result, the default insurance layer is not a factor that is included in the calculation of the Members' guaranty fund contribution requirement. Instead, it serves as an additional, distinct, and separate default resource that should serve to further protect the non-defaulting Members' guaranty fund contributions from being mutualized in the event of a default.

As of September 30, 2022, ICE NGX maintained a guaranty fund of \$215 million, comprising \$15 million in cash and a \$200 million letter of credit backed by a default insurance policy of the same amount, discussed below.

Below is a depiction of our Default Waterfall which summarizes the lines of defense and layers of protection we maintain at ICE Clear Europe, ICE Clear U.S. and ICE Clear Credit.

ICE Clearing House Default Waterfall



Cash and Invested Deposits

We have recorded cash and invested margin and guaranty fund deposits and amounts due in our balance sheets as current assets with corresponding current liabilities to the Members. As of September 30, 2022, our cash and invested margin and guaranty fund deposits were as follows (in millions):

	ICE Clear Europe ⁽¹⁾	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Other ICE Clearing Houses	Total
Original margin	\$ 113,708	\$ 37,563	\$ 3,331	\$ —	\$ 4	\$ 154,606
Unsettled variation margin, net	—	—	—	747	—	747
Guaranty fund	3,928	3,476	588	—	4	7,996
Delivery contracts receivable/payable, net	—	—	—	1,342	—	1,342
Total	\$ 117,636	\$ 41,039	\$ 3,919	\$ 2,089	\$ 8	\$ 164,691

As of December 31, 2021, our cash and invested margin and guaranty fund deposits were as follows (in millions):

	ICE Clear Europe ⁽²⁾	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Other ICE Clearing Houses	Total
Original margin	\$ 94,010	\$ 39,372	\$ 6,963	\$ —	\$ 27	\$ 140,372
Unsettled variation margin, net	—	—	—	226	—	226
Guaranty fund	4,175	3,952	597	—	4	8,728
Delivery contracts receivable/payable, net	—	—	—	1,103	—	1,103
Total	\$ 98,185	\$ 43,324	\$ 7,560	\$ 1,329	\$ 31	\$ 150,429

⁽¹⁾ \$110.8 billion and \$6.8 billion is related to futures/options and CDS, respectively.

⁽²⁾ \$92.0 billion and \$6.2 billion is related to futures/options and CDS, respectively.

Our cash and invested margin and guaranty fund deposits are maintained in accounts with national banks and highly-rated financial institutions or secured through direct investments, primarily in U.S. Treasury and other highly-rated foreign government securities, or reverse repurchase agreements with primarily overnight maturities. We primarily use Level 1 inputs when evaluating the fair value of the non-cash equivalent direct investments, as highly-rated government securities are quoted in active markets. The carrying value of these deposits is deemed to approximate fair value.

To provide a tool to address the liquidity needs of our clearing houses and manage the liquidation of margin and guaranty fund deposits held in the form of cash and high quality sovereign debt, ICE Clear Europe, ICE Clear Credit and ICE Clear U.S. have entered into Committed Repurchase Agreement Facilities, or Committed Repo. Additionally, ICE Clear Credit and ICE Clear Netherlands have entered into Committed FX Facilities to support these liquidity needs. As of September 30, 2022, the following facilities were in place:

- **ICE Clear Europe:** \$1.0 billion in Committed Repo to finance U.S. dollar, euro and pound sterling deposits.
- **ICE Clear Credit:** \$300 million in Committed Repo (U.S. dollar based) to finance U.S. dollar denominated sovereign debt and euro deposits, €250 million in Committed Repo (euro based) to finance euro and U.S. dollar denominated sovereign debt deposits, and €1.9 billion in Committed FX Facilities to finance euro payment obligations.
- **ICE Clear U.S.:** \$250 million in Committed Repo to finance U.S. dollar denominated sovereign debt deposits.
- **ICE Clear Netherlands:** €10 million in Committed FX Facilities to finance euro payment obligations.

Details of our deposits are as follows (in millions):

Cash and Cash Equivalent Margin Deposits and Guaranty Funds				
Clearing House	Investment Type	As of September 30, 2022		As of December 31, 2021
ICE Clear Europe	National bank account ⁽¹⁾	\$	13,541	\$ 59,948
ICE Clear Europe	Reverse repo		71,360	25,518
ICE Clear Europe	Sovereign debt		26,830	9,324
ICE Clear Europe	Demand deposits		92	231
ICE Clear Credit	National bank account		33,745	37,282
ICE Clear Credit	Reverse repo		4,387	3,639
ICE Clear Credit	Demand deposits		2,907	2,403
ICE Clear U.S.	Reverse repo		3,445	6,485
ICE Clear U.S.	Sovereign Debt		473	1,075
Other ICE Clearing Houses	Demand deposits		9	31
Total cash and cash equivalent margin deposits and guaranty funds		\$	156,789	\$ 145,936

Clearing House	Investment Type	As of September 30, 2022		As of December 31, 2021
ICE NGX	Unsettled variation margin and delivery contracts receivable/payable		2,088	1,329
ICE Clear Europe	Invested deposits - sovereign debt		5,814	3,164
Total invested deposits, delivery contracts receivable and unsettled variation margin		\$	7,902	\$ 4,493

⁽¹⁾ As of September 30, 2022, ICE Clear Europe held €9.3 billion (\$9.2 billion based on the euro/U.S. dollar exchange rate of 0.9802 as of September 30, 2022) at the European Central Bank, or ECB, £3.9 billion (\$4.4 billion based on the pound sterling/U.S. dollar exchange rate of 1.1165 as of September 30, 2022) at the Bank of England, or BOE, and €10 million (\$10 million based on the above exchange rate) at the BOE. As of December 31, 2021, ICE Clear Europe held €47.2 billion (\$53.7 billion based on the euro/U.S. dollar exchange rate of 1.1372 as of December 31, 2021) at ECB, £1.7 billion (\$2.3 billion based on the pound sterling/U.S. dollar exchange rate of 1.3524 as of December 31, 2021), as well as \$4.0 billion at the BOE, and €10 million (\$11 million based on the above exchange rate) at the BOE.

Other Deposits

Non-cash original margin and guaranty fund deposits are not reflected in the accompanying consolidated balance sheets as the risks and rewards of these assets remain with the clearing members unless the clearing houses have sold or re-pledged the assets or in the event of a clearing member default, where the clearing member is no longer entitled to redeem the assets. Any income, gain or loss accrues to the clearing members.

In addition to the cash and invested deposits above, the ICE Clearing Houses have also received other assets from Members, which include government obligations, and may include other non-cash collateral such as letters of credit at ICE NGX, or gold on rare occasions at ICE Clear Europe, to mitigate credit risk. For certain deposits, we may impose discount or "haircut" rates to ensure adequate collateral if market values fluctuate. The value-related risks and rewards of these assets remain with the Members. Any gain or loss accrues to the Member. The ICE Clearing Houses do not, in the ordinary course, rehypothecate or re-pledge these assets. These pledged assets are not reflected in our balance sheets, and are as follows (in millions):

As of September 30, 2022					
	ICE Clear Europe	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Total
Original margin:					
Government securities at face value	\$ 81,061	\$ 23,480	\$ 13,842	\$ —	\$ 118,383
Letters of credit	—	—	—	3,869	3,869
ICE NGX cash deposits	—	—	—	1,621	1,621
Total	\$ 81,061	\$ 23,480	\$ 13,842	\$ 5,490	\$ 123,873
Guaranty fund:					
Government securities at face value	\$ 742	\$ 618	\$ 291	\$ —	\$ 1,651

	As of December 31, 2021				
	ICE Clear Europe	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Total
Original margin:					
Government securities at face value	\$ 58,156	\$ 8,425	\$ 17,211	\$ —	\$ 83,792
Letters of credit	—	—	—	3,566	3,566
ICE NGX cash deposits	—	—	—	987	987
Total	<u>\$ 58,156</u>	<u>\$ 8,425</u>	<u>\$ 17,211</u>	<u>\$ 4,553</u>	<u>\$ 88,345</u>
Guaranty fund:					
Government securities at face value	<u>\$ 740</u>	<u>\$ 152</u>	<u>\$ 273</u>	<u>\$ —</u>	<u>\$ 1,165</u>

ICE NGX

ICE NGX owns a clearing house which primarily administers the physical delivery of energy trading contracts. ICE NGX is the central counterparty to Members on opposite sides of its physically-settled contracts, and the balance related to delivered but unpaid contracts is recorded as a delivery contract net receivable, with an offsetting delivery contract net payable in our balance sheets. Unsettled variation margin equal to the fair value of open contracts is recorded as of each balance sheet date. There is no impact on our consolidated statements of income as an equal amount is recognized as both an asset and a liability. ICE NGX marks all its outstanding physical natural gas and physical power contracts to market daily, but only collects variation margin when a Member's open position falls outside a specified percentage of its pledged collateral. Due to the highly liquid nature and the short period of time to maturity, the fair values of our delivery contract net payable and net receivable are determined to approximate carrying value.

ICE NGX requires Members to maintain cash or letters of credit to serve as collateral in the event of default. The cash is maintained in a segregated bank account for the benefit of the Member, and remains the property of the Member, therefore, it is not included in our balance sheets. ICE NGX maintains a committed daylight-overnight liquidity facility in the amount of \$100 million with an additional \$150 million uncommitted with a third-party Canadian chartered bank which provides liquidity in the event of a settlement shortfall, subject to certain conditions.

During the nine months ended September 30, 2022, ICE NGX increased its default insurance by \$100 million, and as of September 30, 2022, ICE NGX maintains a guaranty fund of \$215 million funded by a \$200 million letter of credit issued by a major Canadian chartered bank, and backed by default insurance underwritten by Export Development Canada, or EDC, a Crown corporation operated at arm's length from the Canadian government, plus \$15 million held as restricted cash to fund the first loss amount that ICE NGX is responsible for under the default insurance policy. In the event of a participant default where the Member's collateral is depleted, the shortfall would be covered by a draw down on the letter of credit following which ICE NGX would file a claim under the default insurance to recover additional losses up to \$200 million beyond the \$15 million first-loss amount that ICE NGX is responsible for under the default insurance policy.

Clearing House Exposure

The net notional value of unsettled contracts was \$2.8 trillion as of September 30, 2022. Each ICE Clearing House bears financial counterparty credit risk and provides a central counterparty guarantee, or performance guarantee, to its Members. To reduce their exposure, the ICE Clearing Houses have a risk management program with both initial and ongoing membership standards. Excluding the effects of original and variation margin, guaranty fund and collateral requirements and default insurance, the ICE Clearing Houses' maximum estimated exposure for this guarantee is \$244.6 billion as of September 30, 2022, which represents the maximum estimated value by the ICE Clearing Houses of a hypothetical one-day movement in pricing of the underlying unsettled contracts. This value was determined using proprietary risk management software that simulates gains and losses based on historical market prices, volatility and other factors present at that point in time for those particular unsettled contracts. Future actual market price volatility could result in the exposure being significantly different than this amount.

13. Legal Proceedings

In the ordinary course of our business, from time to time we are subject to legal proceedings, lawsuits, government investigations and other claims with respect to a variety of matters. In addition, we are subject to periodic reviews, inspections, examinations and investigations by regulators in the U.S. and other jurisdictions, any of which may result in claims, legal proceedings, assessments, fines, penalties, restrictions on our business or other sanctions. We record estimated expenses and reserves for legal or regulatory matters or other claims when these matters present loss contingencies that are probable and the related amount is reasonably estimable. Any such accruals may be adjusted as circumstances change. Assessments of losses are inherently subjective and involve unpredictable factors. While the

outcome of legal and regulatory matters is inherently difficult to predict and/or the range of loss often cannot be reasonably estimable, we do not believe that the liabilities, if any, which may ultimately result from the resolution of the various legal and regulatory matters that arise in the ordinary course of our business, including the matters described below and those described in Note 15 to the consolidated financial statements in Part II, Item 8 of our 2021 Form 10-K, are likely to have a material adverse effect on our consolidated financial condition, results of operations, or liquidity. It is possible, however, that future results of operations for any particular quarterly or annual period could be materially and adversely affected by any developments relating to these legal and regulatory matters. A range of possible losses related to certain cases cannot be reasonably estimated at this time, except as otherwise disclosed below and in Note 15 to the consolidated financial statements in Part II, Item 8 of our 2021 Form 10-K. Individual matter disclosures in this Form 10-Q are limited to new significant matters or significant updates on existing matters since our most recent Form 10-K.

City of Providence Litigation

On March 28, 2022, the district court entered an order granting the defendant exchanges' (including New York Stock Exchange LLC and NYSE Arca, Inc., two of our subsidiaries) motion for summary judgment on the ground that the plaintiffs lack standing under Article III of the U.S. Constitution, and dismissing without prejudice the plaintiffs' claims on this basis. Among other things, the district court found that the plaintiffs failed to show that they have been injured and that, even putting aside this defect, the plaintiffs failed to produce evidence from which a jury could reasonably conclude that they suffered an injury traceable to any conduct of the exchanges. The district court also held that the opinions of the plaintiffs' principal expert witness in this matter were fundamentally flawed and unreliable, and therefore inadmissible. In light of these holdings, the district court denied as moot the plaintiffs' motion for class certification and the exchanges' motion for summary judgment on the basis of preclusion. On April 25, 2022, the plaintiffs filed a notice of appeal of the district court's dismissal order. Effective as of June 6, 2022, the parties to the litigation executed settlement agreements pursuant to which, among other things, the plaintiffs withdrew their appeal with prejudice and provided the defendants a covenant not to sue (and additionally provided the NYSE and Cboe Global Markets, Inc. defendants a release of claims). No monetary payment was made by the defendants. The settlement agreements and dismissal of the appeal constitute the final resolution of this matter.

LIBOR Litigation

On February 14, 2022, the U.S. Court of Appeals for the Second Circuit, or the Second Circuit, issued a decision in the appeal of the March 2020 dismissal of the underlying complaint against the defendants, which include ICE and several of our subsidiaries. In its decision, the Second Circuit dismissed the appeal for lack of jurisdiction, holding that DYJ Holdings, LLC, the sole entity attempting to pursue the appeal, lacked standing to do so. The dismissal of the appeal constitutes the final resolution of this matter.

ICE Data Pricing & Reference Data Matter

As of April 28, 2022, our subsidiary ICE Data Pricing & Reference Data, LLC, or PRD, resolved the last known remaining claim of a Live Well Financial, Inc., or Live Well, financial institution creditor relating to PRD's legacy business practices with respect to broker quotes received from Live Well. With the resolution of this putative claim, there are no known unresolved assertions of liability against PRD relating to broker quotes PRD received from Live Well.

For further information on our legal and regulatory matters, please see Note 15 to the consolidated financial statements in Part II, Item 8 of our 2021 Form 10-K.

14. Fair Value Measurements

Fair value is the price that would be received from selling an asset or paid to transfer a liability (an exit price) in the principal or most advantageous market for the asset or liability in an orderly transaction between market participants at the measurement date. Our financial instruments consist primarily of certain short-term and long-term assets and liabilities, customer accounts receivable, margin deposits and guaranty funds, equity and equity method investments, and short-term and long-term debt.

The fair value of our financial instruments is measured based on a three-level hierarchy:

- **Level 1 inputs** — quoted prices for identical assets or liabilities in active markets.
- **Level 2 inputs** — observable inputs other than Level 1 inputs such as quoted prices for similar assets and liabilities in active markets or inputs other than quoted prices that are directly observable.
- **Level 3 inputs** — unobservable inputs supported by little or no market activity and that are significant to the fair value of the assets or liabilities.

Financial assets and liabilities recorded or disclosed at fair value in the accompanying consolidated balance sheets as of September 30, 2022 and December 31, 2021 were classified in their entirety based on the lowest level of input that is significant to the asset or liability's fair value measurement.

Our mutual funds are equity and fixed income mutual funds held for the purpose of providing future payments for our supplemental executive savings plan and the supplemental executive retirement plan. These mutual funds are classified as equity investments and measured at fair value using Level 1 inputs with adjustments recorded in net income.

The fair values of all other financial instruments are determined to materially approximate carrying value due to the short period of time to their maturities.

We did not use Level 3 inputs to determine the fair value of assets or liabilities measured at fair value on a recurring basis as of September 30, 2022 or December 31, 2021.

We measure certain assets, such as intangible assets and equity method investments, at fair value on a non-recurring basis. These assets are recognized at fair value if they are deemed to be impaired. As of September 30, 2022, none of our intangible assets were required to be recorded at fair value since no impairments were recorded.

We measure certain equity investments at fair value on a non-recurring basis using our policy election under ASU 2016-01. During the nine months ended September 30, 2022, we evaluated these investments and determined that the value of our equity method investment in Bakkt was impaired using a Level 1 input, which was the publicly-traded closing stock price of Bakkt on September 30, 2022 (Note 4). No other fair value adjustments were required under our accounting policy election related to these investments.

See Note 12 for the fair value considerations related to our margin deposits, guaranty funds and delivery contracts receivable.

The table below displays the fair value of our debt as of September 30, 2022. The fair values of our fixed rate notes were estimated using quoted market prices for these instruments. The fair value of other short-term debt approximates par value since the interest rates on this short-term debt approximate market rates as of September 30, 2022.

As of September 30, 2022 (in millions)			
Debt:	Carrying Amount		Fair value
	\$	7	\$ 7
Other short-term debt			
2025 Senior Notes (3.65% senior unsecured notes due May 23, 2025)		1,243	1,210
2025 Senior Notes (3.75% senior unsecured notes due December 1, 2025)		1,247	1,207
2027 Senior Notes (4.00% senior unsecured notes due September 15, 2027)		1,485	1,423
2027 Senior Notes (3.10% senior unsecured notes due September 15, 2027)		497	455
2028 Senior Notes (3.75% senior unsecured notes due September 21, 2028)		594	553
2029 Senior Notes (4.35% senior unsecured notes due June 15, 2029)		1,239	1,183
2030 Senior Notes (2.10% senior unsecured notes due June 15, 2030)		1,235	999
2032 Senior Notes (1.85% senior unsecured notes due September 15, 2032)		1,484	1,096
2033 Senior Notes (4.60% senior unsecured notes due March 15, 2033)		1,488	1,400
2040 Senior Notes (2.65% senior unsecured notes due September 15, 2040)		1,231	840
2048 Senior Notes (4.25% senior unsecured notes due September 21, 2048)		1,231	1,019
2050 Senior Notes (3.00% senior unsecured notes due June 15, 2050)		1,221	817
2052 Senior Notes (4.95% senior unsecured notes due June 15, 2052)		1,464	1,332
2060 Senior Notes (3.00% senior unsecured notes due September 15, 2060)		1,471	893
2062 Senior Notes (5.20% senior unsecured notes due June 15, 2062)		983	900
Total debt	\$	18,120	\$ 15,334

15. Segment Reporting

Our business is conducted through three reportable business segments, comprised of the following:

- In our Exchanges segment, we operate regulated marketplaces for the listing, trading and clearing of a broad array of derivatives contracts and financial securities;
- In our Fixed Income and Data Services segment, we provide fixed income pricing, reference data, indices, analytics and execution services as well as global CDS clearing and multi-asset class data delivery solutions; and
- In our Mortgage Technology segment, we provide an end-to-end technology platform that offers customers comprehensive, digital workflow tools that aim to address the inefficiencies that exist in the U.S. residential mortgage market, from application through closing and the secondary market.

While revenues are recorded specifically in the segment in which they are earned or to which they relate, a significant portion of our operating expenses are not solely related to a specific segment because the expenses serve functions that are necessary for the operation of more than one segment. We directly allocate expenses when reasonably possible to do so. Otherwise, we use a pro-rata revenue approach as the allocation method for the expenses that do not relate solely to one segment and serve functions that are necessary for the operation of all segments.

Our chief operating decision maker does not review total assets or statements of income below operating income by segments; therefore, such information is not presented below. Our three segments do not engage in intersegment transactions.

Financial data for our business segments is as follows for the nine and three months ended September 30, 2022 and 2021 (in millions):

	Nine Months Ended September 30, 2022			
	Exchanges	Fixed Income and Data Services	Mortgage Technology	Consolidated
Revenues:				
Energy futures and options	\$ 884	\$ —	\$ —	\$ 884
Agricultural and metals futures and options	179	—	—	179
Financial futures and options	375	—	—	375
Cash equities and equity options	2,021	—	—	2,021
OTC and other	326	—	—	326
Data and connectivity services	651	—	—	651
Listings	388	—	—	388
Fixed income execution	—	66	—	66
CDS clearing	—	226	—	226
Fixed income data and analytics	—	824	—	824
Other data and network services	—	439	—	439
Origination technology	—	—	586	586
Closing solutions	—	—	187	187
Data and analytics	—	—	66	66
Other	—	—	41	41
Revenues	4,824	1,555	880	7,259
Transaction-based expenses	1,735	—	—	1,735
Revenues, less transaction-based expenses	3,089	1,555	880	5,524
Operating expenses	904	1,029	817	2,750
Operating income	\$ 2,185	\$ 526	\$ 63	\$ 2,774

Nine Months Ended September 30, 2021				
	Exchanges	Fixed Income and Data Services	Mortgage Technology	Consolidated
Revenues:				
Energy futures and options	\$ 900	\$ —	\$ —	\$ 900
Agricultural and metals futures and options	177	—	—	177
Financial futures and options	281	—	—	281
Cash equities and equity options	1,800	—	—	1,800
OTC and other	239	—	—	239
Data and connectivity services	623	—	—	623
Listings	356	—	—	356
Fixed income execution	—	39	—	39
CDS clearing	—	144	—	144
Fixed income data and analytics	—	804	—	804
Other data and network services	—	416	—	416
Origination technology	—	—	740	740
Closing solutions	—	—	227	227
Data and analytics	—	—	55	55
Other	—	—	39	39
Revenues	4,376	1,403	1,061	6,840
Transaction-based expenses	1,534	—	—	1,534
Revenues, less transaction-based expenses	2,842	1,403	1,061	5,306
Operating expenses	977	1,010	750	2,737
Operating income	\$ 1,865	\$ 393	\$ 311	\$ 2,569

Three Months Ended September 30, 2022				
	Exchanges	Fixed Income and Data Services	Mortgage Technology	Consolidated
Revenues:				
Energy futures and options	\$ 266	\$ —	\$ —	\$ 266
Agricultural and metals futures and options	57	—	—	57
Financial futures and options	122	—	—	122
Cash equities and equity options	664	—	—	664
OTC and other	121	—	—	121
Data and connectivity services	219	—	—	219
Listings	128	—	—	128
Fixed income execution	—	26	—	26
CDS clearing	—	88	—	88
Fixed income data and analytics	—	273	—	273
Other data and network services	—	147	—	147
Origination technology	—	—	187	187
Closing solutions	—	—	53	53
Data and analytics	—	—	22	22
Other	—	—	14	14
Revenues	1,577	534	276	2,387
Transaction-based expenses	576	—	—	576
Revenues, less transaction-based expenses	1,001	534	276	1,811
Operating expenses	301	337	260	898
Operating income	\$ 700	\$ 197	\$ 16	\$ 913

Three Months Ended September 30, 2021				
	Exchanges	Fixed Income and Data Services	Mortgage Technology	Consolidated
Revenues:				
Energy futures and options	\$ 316	\$ —	\$ —	\$ 316
Agricultural and metals futures and options	56	—	—	56
Financial futures and options	93	—	—	93
Cash equities and equity options	554	—	—	554
OTC and other	84	—	—	84
Data and connectivity services	208	—	—	208
Listings	123	—	—	123
Fixed income execution	—	12	—	12
CDS clearing	—	51	—	51
Fixed income data and analytics	—	272	—	272
Other data and network services	—	142	—	142
Origination technology	—	—	245	245
Closing solutions	—	—	88	88
Data and analytics	—	—	19	19
Other	—	—	14	14
Revenues	1,434	477	366	2,277
Transaction-based expenses	475	—	—	475
Revenues, less transaction-based expenses	959	477	366	1,802
Operating expenses	330	338	256	924
Operating income	\$ 629	\$ 139	\$ 110	\$ 878

No customers or clearing members accounted for more than 10% of our Exchange revenues, less transaction-based expenses during the nine and three months ended September 30, 2022. Revenue from one member of the Exchanges segment comprised \$326 million, or 11%, and \$118 million, or 12%, of our Exchange revenues, less transaction-based expenses for the nine and three months ended September 30, 2021, respectively. Clearing members are primarily intermediaries and represent a broad range of principal trading firms. If a clearing member ceased its operations, we believe that the trading firms would continue to conduct transactions and would clear those transactions through another clearing member firm. No additional customers or clearing members accounted for more than 10% of our segment revenues or consolidated revenues during the nine and three months ended September 30, 2022 or 2021.

16. Earnings/(Loss) Per Common Share

The following is a reconciliation of the numerators and denominators of the basic and diluted earnings/(loss) per common share computations for the nine and three months ended September 30, 2022 and 2021 (in millions, except per share amounts):

	Nine Months Ended September 30,		Three Months Ended September 30,	
	2022	2021	2022	2021
Basic:				
Net income/(loss) attributable to Intercontinental Exchange, Inc.	\$ 1,021	\$ 2,531	\$ (191)	\$ 633
Weighted average common shares outstanding	559	563	558	563
Basic earnings/(loss) per common share	\$ 1.83	\$ 4.50	\$ (0.34)	\$ 1.12
Diluted:				
Weighted average common shares outstanding	559	563	558	563
Effect of dilutive securities - stock options and restricted stock	2	2	2	3
Diluted weighted average common shares outstanding	561	565	560	566
Diluted earnings/(loss) per common share	\$ 1.82	\$ 4.48	\$ (0.34)	\$ 1.12

Basic earnings/(loss) per common share is calculated using the weighted average common shares outstanding during the period.

Common equivalent shares from stock options and restricted stock awards, calculated using the treasury stock method, are included in the diluted per share calculations unless the effect of their inclusion would be antidilutive. During the nine months ended September 30, 2022 and 2021, 1 million and 0.3 million outstanding stock options and restricted stock awards, respectively, were not included in the computation of diluted earnings/(loss) per common share, because to do so would have had an antidilutive effect. In addition, for the nine months ended September 30, 2021, we excluded warrants and preferred and common incentive units under the Bakkt Equity Incentive Plan because they were also antidilutive. Certain figures in the table above may not recalculate due to rounding.

17. Subsequent Events

We have evaluated subsequent events, and determined that no events or transactions met the definition of a subsequent event for purposes of recognition or disclosure in the accompanying consolidated financial statements.

ITEM 2. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

In this Quarterly Report on Form 10-Q, or Quarterly Report, and unless otherwise indicated, the terms "Intercontinental Exchange," "ICE," "we," "us," "our," "our company" and "our business" refer to Intercontinental Exchange, Inc., together with its consolidated subsidiaries. All references to "options" or "options contracts" in the context of our futures products refer to options on futures contracts. Solely for convenience, references in this Quarterly Report to any trademarks, service marks and trade names owned by ICE are listed without the ®, ™ and © symbols, but we will assert, to the fullest extent under applicable law, our rights to these trademarks, service marks and trade names.

We also include references to third-party trademarks, trade names and service marks in this Quarterly Report. Except as otherwise expressly noted, our use or display of any such trademarks, trade names or service marks is not an endorsement or sponsorship and does not indicate any relationship between us and the parties that own such marks and names.

The following discussion should be read in conjunction with our consolidated financial statements and related notes included elsewhere in this Quarterly Report. Due to rounding, figures in tables may not sum exactly.

Forward-Looking Statements

This Quarterly Report, including the sections entitled "Notes to Consolidated Financial Statements," "Legal Proceedings" and "Management's Discussion and Analysis of Financial Condition and Results of Operations," contains "forward-looking statements" as defined in the Private Securities Litigation Reform Act of 1995. Any statements contained herein that are not statements of historical fact may be forward-looking statements.

These forward-looking statements relate to future events or our future financial performance and are based on our present beliefs and assumptions as well as the information currently available to us. They involve known and unknown risks, uncertainties and other factors that may cause our results, levels of activity, performance, cash flows, financial position or achievements to differ materially from those expressed or implied by these statements.

Forward-looking statements may be introduced by or contain terminology such as "may," "will," "should," "could," "would," "targets," "goal," "expect," "intend," "plan," "anticipate," "believe," "estimate," "predict," "potential," "continue," or the antonyms of these terms or other comparable terminology. Although we believe that the expectations reflected in the forward-looking statements are reasonable, we cannot guarantee future results, levels of activity, performance, cash flows, financial position or achievements. Accordingly, we caution you not to place undue reliance on any forward-looking statements we may make.

Factors that may affect our performance and the accuracy of any forward-looking statements include, but are not limited to, those listed below:

- conditions in global financial markets, domestic and international economic and social conditions, inflation, political uncertainty and discord, geopolitical events or conflicts, international trade policies and sanctions laws;
- the impact of the introduction of or any changes in laws, regulations, rules or government policies with respect to financial markets, climate change, increased regulatory scrutiny or enforcement actions and our ability to comply with these requirements;
- volatility in commodity prices and equity prices, and price volatility of financial benchmarks and instruments such as interest rates, credit spreads, equity indices, foreign exchange rates, and mortgage origination trends;
- the impact of climate change and the transition to renewable energy and a net zero economy;
- the business environment in which we operate and trends in our industry, including trading volumes, prevalence of clearing, demand for data services, mortgage lending activity, fees, changing regulations, competition and consolidation;
- our ability to minimize the risks associated with operating clearing houses in multiple jurisdictions;
- our exchanges' and clearing houses' compliance with their respective regulatory and oversight responsibilities;
- the resilience of our electronic platforms and soundness of our business continuity and disaster recovery plans;
- our ability to realize the expected benefits of our acquisitions and our investments, including our ability to close the Black Knight acquisition on the terms and timing expected;
- our ability to execute our growth strategy, identify and effectively pursue, implement and integrate acquisitions and strategic alliances and realize the synergies and benefits of such transactions within the expected time frame;
- the performance and reliability of our trading, clearing and mortgage technologies and those of third-party service providers;
- our ability to keep pace with technological developments and client preferences;

- our ability to ensure that the technology we utilize is not vulnerable to cyberattacks, hacking and other cybersecurity risks or other disruptive events or to minimize the impact of any such events;
- our ability to keep information and data relating to the customers of the users of the software and services provided by our ICE Mortgage Technology business confidential;
- the impacts of the COVID-19 pandemic on our business, results of operations and financial condition as well as the broader business environment;
- our ability to identify trends and adjust our business to benefit from such trends, including trends in the U.S. mortgage industry such as inflation rates, interest rates, new home purchases, refinancing activity, and home builder and buyer sentiment, among others;
- our ability to evolve our benchmarks and indices in a manner that maintains or enhances their reliability and relevance;
- the accuracy of our cost and other financial estimates and our belief that cash flows from operations will be sufficient to service our debt and to fund our operational and capital expenditure needs;
- our ability to incur additional debt and pay off our existing debt in a timely manner;
- our ability to maintain existing market participants and data and mortgage technology customers, and to attract new ones;
- our ability to offer additional products and services, leverage our risk management capabilities and enhance our technology in a timely and cost-effective fashion;
- our ability to attract, develop and retain key talent;
- our ability to protect our intellectual property rights and to operate our business without violating the intellectual property rights of others; and
- potential adverse results of threatened or pending litigation and regulatory actions and proceedings.

These risks and other factors include, among others, those set forth in Part 1, Item 1(A) under the caption “Risk Factors” in our 2021 Form 10-K, as filed with the SEC on February 3, 2022. Due to the uncertain nature of these factors, management cannot assess the impact of each factor on the business or the extent to which any factor, or combination of factors, may cause actual results to differ materially from those contained in any forward-looking statements.

Any forward-looking statement speaks only as of the date on which such statement is made, and we undertake no obligation to update any of these statements to reflect events or circumstances occurring after the date of this Quarterly Report. New factors may emerge and it is not possible to predict all factors that may affect our business and prospects.

Overview

We are a provider of market infrastructure, data services and technology solutions to a broad range of customers including financial institutions, corporations and government entities. Our products, which span major asset classes including futures, equities, fixed income and residential mortgages in the U.S., provide our customers with access to mission critical tools that are designed to increase asset class transparency and workflow efficiency. While we report our results in three reportable business segments, we operate as one business, leveraging the collective expertise, particularly in data services and technology, that exists across our platforms to inform and enhance our operations.

- In our Exchanges segment, we operate regulated marketplaces for the listing, trading and clearing of a broad array of derivatives contracts and financial securities.
- In our Fixed Income and Data Services segment, we provide fixed income pricing, reference data, indices, analytics and execution services as well as global CDS clearing and multi-asset class data delivery solutions.
- In our Mortgage Technology segment, we provide an end-to-end technology platform that offers customers comprehensive, digital workflow tools that aim to address the inefficiencies that exist in the U.S. residential mortgage market, from application through closing and the secondary market.

Recent Developments

Pending Acquisition of Black Knight, Inc.

On May 4, 2022, we announced that we had entered into a definitive agreement to acquire Black Knight, Inc., or Black Knight, a software, data and analytics company that serves the housing finance continuum, including real estate data, mortgage lending and servicing, as well as the secondary markets. Pursuant to the merger agreement, Sub will merge with and into Black Knight, with Black Knight surviving as a wholly owned subsidiary of ICE. As of May 4, 2022, the transaction was valued at approximately \$13.1 billion, or \$85 per share of Black Knight common stock, with cash comprising 80% of the value of the aggregate transaction consideration and shares of our common stock comprising 20% of the value of the aggregate transaction consideration at that time. The aggregate cash component of the transaction

consideration is fixed at \$10.5 billion, and the value of the aggregate stock component of the transaction consideration will fluctuate with the market price of our common stock and will be determined based on the average of the volume weighted averages of the trading prices of our common stock on each of the ten consecutive trading days ending three trading days prior to the closing of the merger. This transaction builds on our position as a provider of end-to-end electronic workflow solutions for the rapidly evolving U.S. residential mortgage industry.

Black Knight provides a comprehensive and integrated ecosystem of software, data and analytics solutions serving the real estate and housing finance markets. We believe the Black Knight ecosystem adds value for clients of all sizes across the mortgage and real estate lifecycles by helping organizations lower costs, increase efficiencies, grow their businesses, and reduce risk.

On August 19, 2022, our preliminary proxy statement/prospectus on Form S-4 was declared effective by the SEC, and on September 21, 2022, Black Knight stockholders approved the transaction. The transaction is expected to close in the first half of 2023 following the receipt of regulatory approvals and the satisfaction of customary closing conditions.

Global Market Conditions

Our results of operations are affected by global economic conditions, including macroeconomic conditions and geopolitical events or conflicts. During 2022, macroeconomic conditions, including rising interest rates, recent spikes in inflation rates and market volatility, along with geopolitical concerns, including the war in Ukraine and the sanctions and other measures that have been and continue to be imposed in response to the war, created uncertainty and volatility in the global economy and resulted in a dynamic operating environment.

Our business has been impacted positively and negatively by these global economic conditions. For instance, due to market volatility and rising interest rates, we have seen increased trading across a number of our products, such as interest rate & equity futures, credit default swaps and bonds. Conversely, increases in mortgage interest rates in 2022 have resulted in reduced consumer and investor demand for mortgages and adversely impacted the transaction-based revenues in our Mortgage Technology segment.

We have suspended all services in Russia except for limited offerings to non-sanctioned entities. From an operational perspective, our businesses, including our exchanges, clearing houses, listings venues, data services businesses and mortgage platforms, have not suffered a material negative impact as a result of these events in Ukraine and the surrounding region.

We expect the macro environment to remain dynamic in the near-term, and we continue to monitor macroeconomic conditions, including interest rates and inflation rates, as well as the uncertainty surrounding the extent and duration of the ongoing conflict between Russia and Ukraine, and the impact that any of the foregoing may have on the global economy and on our business.

Tax Policy Changes

In July and August 2022, the CHIPS and Science Act, or CHIPS, and the Inflation Reduction Act of 2022, or IRA, were signed into law. The IRA introduced a 15% corporate alternative minimum tax, or CAMT, on adjusted financial statement income for corporations with profits in excess of \$1 billion, effective for tax years after December 31, 2022. While further guidance on the implementation of the CAMT is expected, we do not expect it will have a material impact to our 2023 effective tax rate. We also do not expect that CHIPS will have a material impact. The IRA also includes a stock buyback excise tax of 1%, which will apply to net stock buybacks after December 31, 2022. We do not expect this to have a material impact once share repurchases are resumed.

Regulation

Our activities and the markets in which we operate are subject to regulations that impact us as well as our customers, and, in turn, meaningfully influence our activities, the manner in which we operate and our strategy. We are primarily subject to the jurisdiction of regulatory agencies in the U.S., U.K., EU, Canada, Singapore and Abu Dhabi. Failure to satisfy regulatory requirements can or may give rise to sanctions by the applicable regulator.

Global policy makers have undertaken reviews of their existing legal framework governing financial markets in connection with regulatory reform, and have either passed new laws and regulations, or are in the process of debating and/or enacting new laws and regulations that apply to our business and to our customers' businesses. Legislative and regulatory actions may impact the way in which we or our customers conduct business and may create uncertainty, which could affect trading volumes or demand for market data. See Part 1, Item 1 "Business - Regulation" and Part 1, Item 1(A)

"Risk Factors" included in our 2021 Form 10-K for a discussion of the primary regulations applicable to our business and certain risks associated with those regulations.

Domestic and foreign policy makers continue to review their legal frameworks governing financial markets, and periodically change the laws and regulations that apply to our business and to our customers' businesses. Our key areas of focus on these evolving efforts are:

- **Regulatory Structure Applicable to Non-EU Clearing Houses.** On January 1, 2020, the European Markets Infrastructure Regulation, or EMIR 2.2, became effective, which revises the EU's current regulatory and supervisory structure for EU and non-EU clearing houses. The European Securities and Markets Authority, or ESMA, has recognized ICE Clear Europe as a third-country central counterparty, or CCP, under EMIR and determined that it is a Tier 2 CCP on the basis that it is systemically important to the financial stability of the EU or one or more of its Member States. ESMA has recognized all other ICE clearing houses as third-country CCPs and determined that they are Tier 1 CCPs on the basis that they are not systemically-important to the financial stability of the EU or one or more of its Member States. ESMA's continuing implementation of these delegated regulations could still impact one or more of our other non-EU clearing houses. In February 2022, the European Commission extended the temporary equivalence for U.K. CCPs until June 2025. In March 2022, ESMA extended the ICE Clear Europe recognition decision and tiering determination until June 2025 and confirmed the recognition and tiering determination of all other ICE clearing houses.
- **Benchmarks Regulation.** The Financial Conduct Authority, or FCA, used its legal powers under the U.K. Benchmarks Regulation, or U.K. BMR, to require ICE Benchmark Administration Limited, or IBA, as the administrator of the London Interbank Offered Rate, or LIBOR, to publish certain Sterling and Japanese Yen LIBOR settings under a changed "synthetic" methodology until the end of 2022. As a result of the FCA's June 2022 consultation, the FCA will require IBA to continue publishing 1 and 6-month "synthetic" Sterling LIBOR and is considering whether to require IBA to publish 3-month "synthetic" Sterling LIBOR until the end of March 2023. "Synthetic" Japanese Yen LIBOR settings will cease at the end of 2022. Any settings published under the "synthetic" methodology are not representative of the underlying market or economic reality the setting is intended to measure as those terms are used in the U.K. BMR. The FCA has confirmed that it expects certain U.S. Dollar LIBOR settings to continue being published on a representative basis until the end of June 2023. The FCA stated that it will consider requiring IBA to publish certain U.S. Dollar LIBOR settings beyond June 30, 2023 under a changed "synthetic" methodology. Usage of the "synthetic" LIBOR and continuing U.S. Dollar LIBOR settings may be restricted or prohibited in certain circumstances under applicable law.

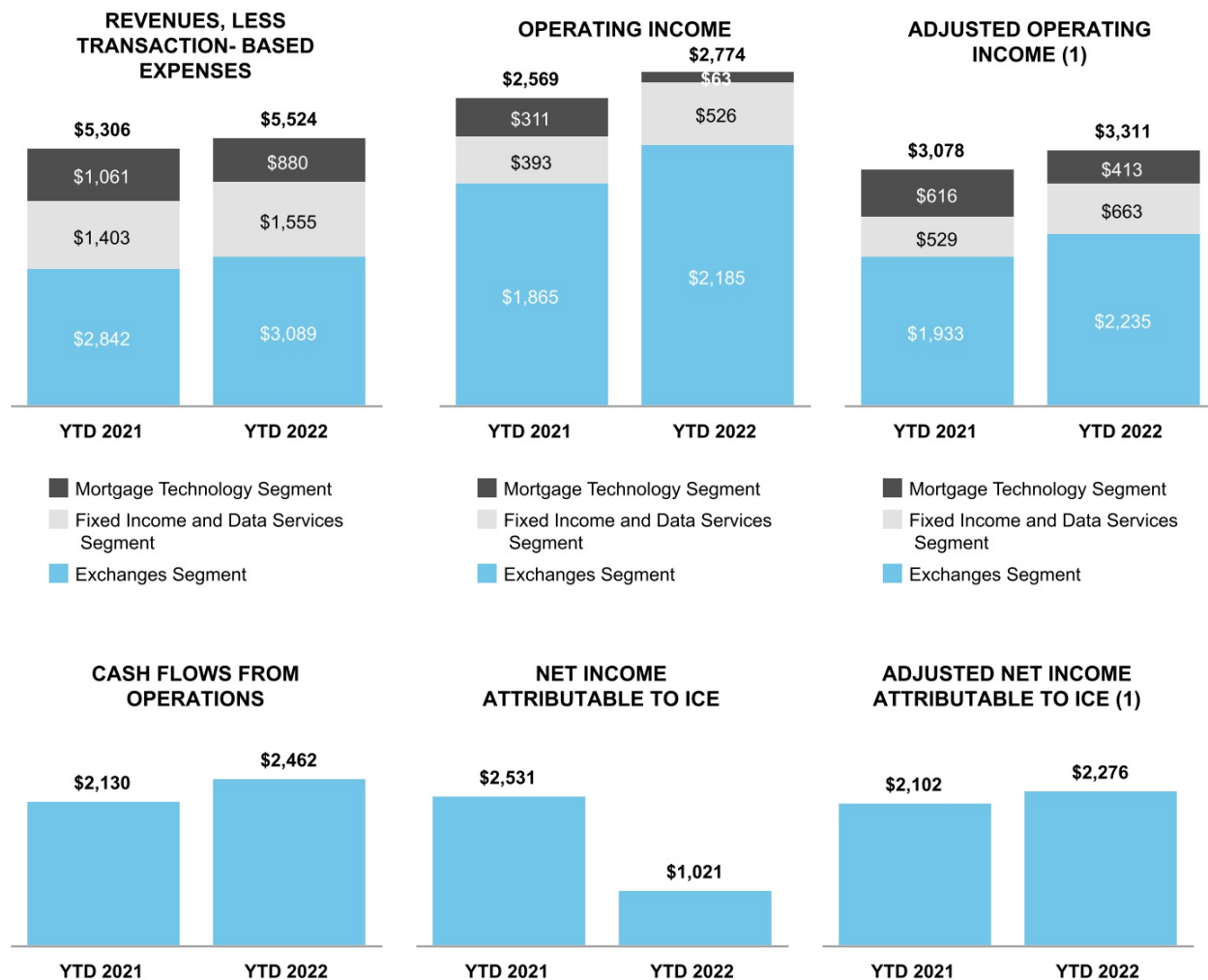
The European Commission used its powers under the EU Benchmarks Regulation, or EU BMR, to designate replacement benchmarks for certain Swiss franc LIBOR settings and the Euro Overnight Index Average, or EONIA, which cover all references to the relevant benchmark. The transition period for the use of benchmarks provided by third-country administrators has been extended until at least December 31, 2023. In May 2022, the European Commission published a consultation on the third-country regime of the EU BMR to prepare for the development of a legislative proposal.

In March 2022, President Biden signed into law federal LIBOR legislation, referred to as the LIBOR Act, designed to reduce uncertainty and economic impacts of the permanent cessation of LIBOR for specified contracts, securities and other agreements that are economically linked to LIBOR. The LIBOR Act provides a statutory framework to replace U.S. Dollar LIBOR with a benchmark rate based on the SOFR for contracts governed by U.S. law that have no fallbacks or fallbacks that would require the use of a poll or LIBOR-based rate.
- **Policy intervention to address high energy prices.** In March 2022, EU leaders agreed to reduce the EU's dependency on Russian gas, oil and coal imports and invited the European Commission to put forward legislative proposals to ensure security of supply and affordable energy prices. Various options for regulatory intervention have been adopted by the European Commission to allow EU countries to jointly buy strategic reserves of gas. In June 2022, the EU imposed a partial embargo on Russian crude oil and petroleum products. In July 2022, EU Member States reached a political agreement on a voluntary reduction of natural gas demand in the EU. In September 2022, the European Commission approved legislative proposals to address the energy crisis including reducing Member States' energy consumption, imposing a cap on revenues for electricity producers and requiring a contribution on excess profits generated from oil, gas, coal and refinery activities. The European Commission is also exploring additional measures including a price cap on imported gas and an LNG import benchmark. The potential impact of these measures on the functioning of European energy wholesale markets remains uncertain at this time.
- **CCP Resolution.** In March 2022, the U.K. Treasury published a feedback statement and status update on its plans to enhance the U.K.'s regime for resolution of CCPs in the event that they fail. This is intended to expand the prior regime which was not in line with U.K. Financial Stability Board guidance issued subsequently. Many of the parameters of the new regime have yet to be finalized and will be subject to a consultation process by the Bank of

England which will be the resolution authority for CCPs in the U.K. However, they will include increased CCP contributions (known as "second skin in the game") to the default fund.

Consolidated Financial Highlights

The following summarizes our results and significant changes in our consolidated financial performance for the periods presented (dollars in millions, except per share amounts and YTD represents the nine-month periods ended September 30th).



⁽¹⁾ The adjusted figures exclude items that are not reflective of our ongoing core operations and business performance. Adjusted net income attributable to ICE is presented net of taxes. These adjusted numbers are not calculated in accordance with U.S. GAAP. See "- Non-GAAP Financial Measures" below.

	Nine Months Ended September 30,			Three Months Ended September 30,		
	2022	2021	Change	2022	2021	Change
Revenues, less transaction-based expenses	\$ 5,524	\$ 5,306	4 %	\$ 1,811	\$ 1,802	1 %
Recurring revenues ⁽¹⁾	\$ 2,781	\$ 2,603	7 %	\$ 930	\$ 888	5 %
Transaction revenues, net ⁽¹⁾	\$ 2,743	\$ 2,703	1 %	\$ 881	\$ 914	(4) %
Operating expenses	\$ 2,750	\$ 2,737	— %	\$ 898	\$ 924	(3) %
Adjusted operating expenses ⁽²⁾	\$ 2,213	\$ 2,228	(1) %	\$ 727	\$ 755	(4) %
Operating income	\$ 2,774	\$ 2,569	8 %	\$ 913	\$ 878	4 %
Adjusted operating income ⁽²⁾	\$ 3,311	\$ 3,078	8%	\$ 1,084	\$ 1,047	4%
Operating margin	50 %	48 %	2 pts	50 %	49 %	1 pt
Adjusted operating margin ⁽²⁾	60 %	58 %	2 pts	60 %	58 %	2 pts
Other income/(expense), net	\$ (1,530)	\$ 1,020	n/a	\$ (1,240)	\$ (54)	n/a
Income tax expense/(benefit)	\$ 186	\$ 1,049	(82) %	\$ (152)	\$ 187	(182) %
Effective tax rate	15 %	29 %	(14 pts)	47 %	23 %	24 pts
Net income/(loss) attributable to ICE	\$ 1,021	\$ 2,531	(60) %	\$ (191)	\$ 633	(130) %
Adjusted net income attributable to ICE ⁽²⁾	\$ 2,276	\$ 2,102	8 %	\$ 733	\$ 711	3 %
Diluted earnings/(loss) per share attributable to ICE common stockholders	\$ 1.82	\$ 4.48	(59) %	\$ (0.34)	\$ 1.12	(130) %
Adjusted diluted earnings per share attributable to ICE common stockholders ⁽²⁾	\$ 4.06	\$ 3.72	9 %	\$ 1.31	\$ 1.26	4 %
Cash flows from operating activities	\$ 2,462	\$ 2,130	16 %			

*Percentage changes in the table above deemed "n/a" are not meaningful.

⁽¹⁾ We define recurring revenues as the portion of our revenues that are generally predictable, stable, and can be expected to occur at regular intervals in the future with a relatively high degree of certainty and visibility. We define transaction revenues as those associated with a more specific point-in-time service, such as a trade execution.

⁽²⁾ The adjusted figures exclude items that are not reflective of our ongoing core operations and business performance. Adjusted net income attributable to ICE and adjusted diluted earnings per share attributable to ICE common stockholders are presented net of taxes. These adjusted numbers are not calculated in accordance with U.S. GAAP. See "- Non-GAAP Financial Measures" below.

- Revenues, less transaction-based expenses, increased \$218 million and \$9 million for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021. See "-Exchanges Segment", "Fixed Income and Data Services Segment" and "Mortgage Technology Segment" below for a discussion of the significant changes in our revenues. The increase in revenues during the nine and three months ended September 30, 2022 includes \$86 million and \$42 million, respectively, in unfavorable foreign exchange effects arising from fluctuations in the U.S. dollar from the comparable periods in 2021. See Item 3 "Quantitative and Qualitative Disclosures About Market Risk-Foreign Currency Exchange Rate Risk" below for additional information on the impact of currency fluctuations.
- Operating expenses increased \$13 million and decreased \$26 million for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021. See "-Consolidated Operating Expenses" below for a discussion of the significant changes in our operating expenses. The increase in operating expenses during the nine months ended September 30, 2022 and the decrease during the three months ended September 30, 2022 includes \$27 million and \$13 million, respectively, in favorable foreign exchange effects arising from fluctuations in the U.S. dollar from the comparable periods in 2021. See Item 3 "Quantitative and Qualitative Disclosures About Market Risk-Foreign Currency Exchange Rate Risk" below for additional information on the impact of currency fluctuations.

Variability in Quarterly Comparisons

Our business environment has been characterized by:

- globalization of marketplaces, customers and competitors;
- growing customer demand for workflow efficiency and automation;
- commodity, interest rate and financial markets uncertainty;
- growing demand for data to inform customers' risk management and investment decisions;

- evolving, increasing and disparate regulation across multiple jurisdictions;
- price volatility increasing customers' demand for risk management services;
- increasing focus on capital and cost efficiencies;
- customers' preference to manage risk in markets demonstrating the greatest depth of liquidity and product diversity;
- the evolution of existing products and new product innovation to serve emerging customer needs and changing industry agreements;
- rising demand for speed, data, data capacity and connectivity by market participants, necessitating increased investment in technology; and
- consolidation and increasing competition among global markets for trading, clearing and listings.

For additional information regarding the factors that affect our results of operations, see Item 1(A) "Risk Factors" included in our 2021 Form 10-K, and Part II, Item 1(A) "Risk Factors" below.

Segment Results

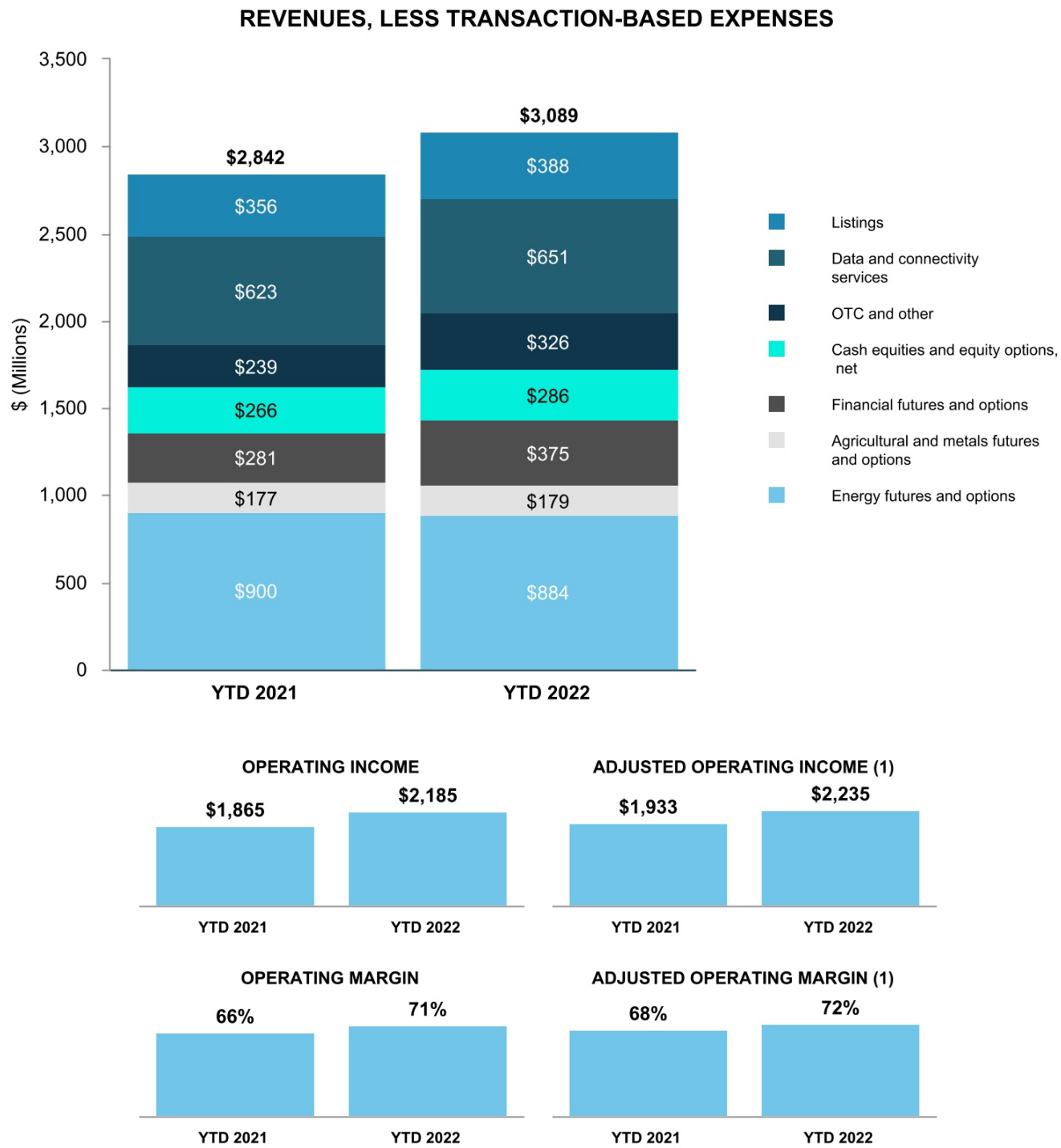
Our business is conducted through three reportable business segments, comprised of the following:

- In our Exchanges segment, we operate regulated marketplaces for the listing, trading and clearing of a broad array of derivatives contracts and financial securities;
- In our Fixed Income and Data Services segment, we provide fixed income pricing, reference data, indices, analytics and execution services as well as global CDS clearing and multi-asset class data delivery solutions; and
- In our Mortgage Technology segment, we provide an end-to-end technology platform that offers customers comprehensive, digital workflow tools that aim to address the inefficiencies that exist in the U.S. residential mortgage market, from application through closing and the secondary market.

While revenues are recorded specifically in the segment in which they are earned or to which they relate, a significant portion of our operating expenses are not solely related to a specific segment because the expenses serve functions that are necessary for the operation of more than one segment. We directly allocate expenses when reasonably possible to do so. Otherwise, we use a pro-rata revenue approach as the allocation method for the expenses that do not relate solely to one segment and serve functions that are necessary for the operation of all segments. Our segments do not engage in intersegment transactions.

Exchanges Segment

The following presents selected statements of income data for our Exchanges segment (dollars in millions and YTD represents the nine-month periods ended September 30th):



(1) The adjusted numbers in the charts above are calculated by excluding items that are not reflective of our cash operations and core business performance. As a result, these adjusted numbers are not calculated in accordance with U.S. GAAP. See “- Non-GAAP Financial Measures” below.

	Nine Months Ended September 30,			Three Months Ended September 30,		
	2022	2021	Change	2022	2021	Change
Revenues:						
Energy futures and options	\$ 884	\$ 900	(2)%	\$ 266	\$ 316	(16)%
Agricultural and metals futures and options	179	177	1	57	56	1
Financial futures and options	375	281	33	122	93	30
Futures and options	1,438	1,358	6	445	465	(5)
Cash equities and equity options	2,021	1,800	12	664	554	20
OTC and other	326	239	37	121	84	45
Transaction and clearing, net	3,785	3,397	11	1,230	1,103	11
Data and connectivity services	651	623	4	219	208	6
Listings	388	356	9	128	123	3
Revenues	4,824	4,376	10	1,577	1,434	10
Transaction-based expenses ⁽¹⁾	1,735	1,534	13	576	475	21
Revenues, less transaction-based expenses	3,089	2,842	9	1,001	959	4
Other operating expenses	725	778	(7)	241	265	(9)
Depreciation and amortization	178	186	(4)	60	62	(2)
Acquisition-related transaction and integration costs	1	13	(92)	—	3	(103)
Operating expenses	904	977	(7)	301	330	(9)
Operating income	\$ 2,185	\$ 1,865	17 %	\$ 700	\$ 629	11 %
Recurring revenues	\$ 1,039	\$ 979	6 %	\$ 347	\$ 331	5 %
Transaction revenues, net	\$ 2,050	\$ 1,863	10 %	\$ 654	\$ 628	4 %

⁽¹⁾Transaction-based expenses are largely attributable to our cash equities and options business.

Exchanges Revenues

Our Exchanges segment includes transaction and clearing revenues from our futures and NYSE exchanges, related data and connectivity services, and our listings business. Transaction and clearing revenues consist of fees collected from derivatives, cash equities and equity options trading and derivatives clearing, and are reported on a net basis, except for the NYSE transaction-based expenses discussed below. Rates per-contract, or RPC, are driven by the number of contracts or securities traded and the fees charged per contract, net of certain rebates. Our per-contract transaction and clearing revenues will depend upon many factors, including, but not limited to, market conditions, transaction and clearing volume, product mix, pricing, applicable revenue sharing and market making agreements, and new product introductions.

Transaction and clearing revenues are generally assessed on a per-contract basis and revenues and profitability fluctuate with changes in contract volume and product mix. We consider data and connectivity services revenues and listings revenues to be recurring revenues. Our data and connectivity services revenues are recurring subscription fees related to the various data and connectivity services that we provide which are directly attributable to our exchange venues. Our listings revenues are also recurring subscription fees that we earn for the provision of NYSE listings services for public companies and ETFs, and related corporate actions for listed companies.

For the nine months ended September 30, 2022 and 2021, 19% and 16%, respectively, of our Exchanges segment revenues, less transaction-based expenses, were billed in pounds sterling or euros. For the three months ended September 30, 2022 and 2021, 19% and 18%, respectively, of our Exchanges segment revenues, less transaction-based expenses, were billed in pounds sterling or euros. Due to the fluctuations of the pound sterling and euro compared to the U.S. dollar, our Exchanges segment revenues, less transaction-based expenses, were lower by \$66 million and \$33 million for the nine and three months ended September 30, 2022, from the comparable periods in 2021.

Our exchange transaction and clearing revenues are presented net of rebates. We recorded rebates of \$665 million and \$790 million for the nine months ended September 30, 2022 and 2021, respectively, and \$201 million and \$264 million for the three months ended September 30, 2022 and 2021, respectively. We offer rebates in certain of our markets primarily to support market liquidity and trading volume by providing qualified participants in those markets a discount to the applicable commission rate. Such rebates are calculated based on volumes traded. The decrease in rebates for the

nine and three months ended September 30, 2022 is primarily due to lower volumes as compared to the prior year and the migration of Sterling futures rebates into the Sterling Overnight Index Average, or SONIA, and a change in the pricing and structure of SONIA products.

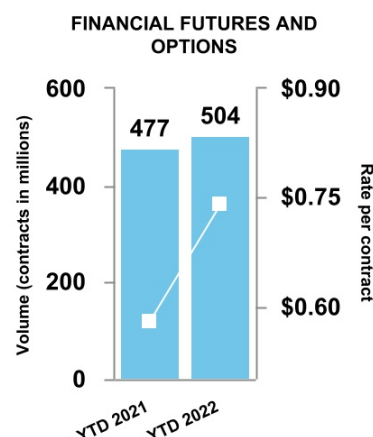
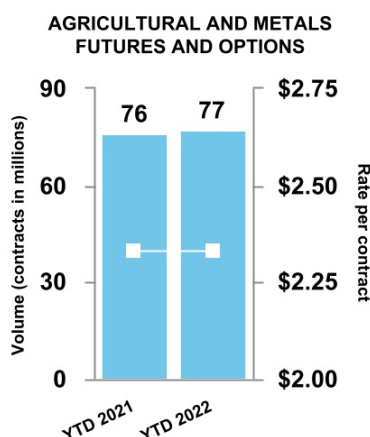
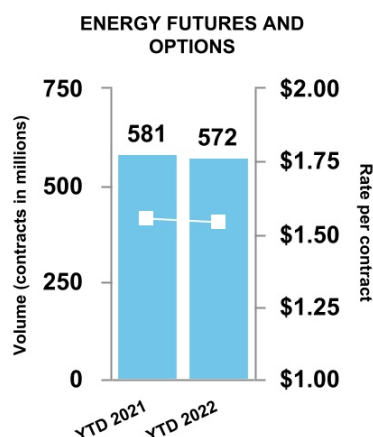
- **Energy Futures and Options:** Total energy volume decreased 1% and revenues decreased 2% for the nine months ended September 30, 2022 from the comparable period in 2021 and volume decreased 13% and revenues decreased 16% for the three months ended September 30, 2022 from the comparable period in 2021.
 - Total oil futures and options volume decreased 10% and 18% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021, driven, in part, by lower Gasoil volumes which are impacted by the uncertainty around Russian sanctions and the conflict in Ukraine.
 - Our global natural gas futures and options volume increased 18% for the nine months ended September 30, 2022 and decreased 4% for the three months ended September 30, 2022, from the comparable periods in 2021, as the first half of 2022 benefited from elevated price volatility related to geopolitical events, including the conflict in Ukraine. In the third quarter of 2022, increased volumes in our North American gas complex were offset by muted activity in our Dutch TTF natural gas complex.
 - Our environmentals and other futures and options volume decreased 6% and 20% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021, due in part to lower power and environmental options volumes.
- **Agricultural and Metals Futures and Options:** Total volumes in our agricultural and metals futures and options markets increased 1% and 4% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021 and revenues increased 1% for both the nine and three months ended September 30, 2022, from the comparable periods in 2021. The third quarter of 2022 benefited from elevated price volatility and price inflation driving an increased need to manage risk across our commodity markets.
 - Sugar futures and options volumes were flat for the nine months ended September 30, 2022 and increased 4% for the three months ended September 30, 2022, from the comparable periods in 2021.
 - Other agricultural and metal futures and options volume increased 2% and 3% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021.
- **Financial Futures and Options:** Total volumes in our financial futures and options markets increased 5% and 15% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021 and revenues increased 33% and 30% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021. The nine months ended September 30, 2022 benefited from elevated volatility across global markets driven by geopolitical events, central bank activity and inflationary concerns.
 - Interest rate futures and options volume increased 3% and 15% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021, and revenue increased 44% and 37% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021. Adjusting for the transition of the LIBOR-based Sterling contract to the alternative rate-based SONIA contract, which is half the notional size of the Sterling contract, interest rate volumes increased 27% and 40% for the nine and three months ended September 30, 2022, respectively from the comparable periods in 2021 driven by interest rate volatility and increased speculation of central bank activity due to inflation concerns. Interest rate futures and options revenues were \$235 million and \$163 million for the nine months ended September 30, 2022 and 2021, respectively, and \$77 million and \$55 million for the three months ended September 30, 2022 and 2021, respectively.
 - Other financial futures and options volume, which includes our MSCI®, FTSE® and NYSE FANG+ equity index products, increased 16% for both the nine and three months ended September 30, 2022 from the comparable periods in 2021. Financial futures and options revenue increased 19% and 20% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021. The nine months ended September 30, 2022 benefited from elevated volatility across global markets driven by geopolitical events, central bank activity and inflationary concerns. Other financial futures and options revenues were \$140 million and \$118 million for the nine months ended September 30, 2022 and 2021, respectively and \$45 million and \$38 million for the three months ended September 30, 2022 and 2021, respectively.

- **Cash Equities and Equity Options:** Cash equities volume increased 3% and 7% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021 due to higher total market volumes driven by elevated volatility related to inflationary, recessionary and geopolitical concerns. Cash equities revenues, net of transaction-based expenses, were \$211 million and \$184 million for the nine months ended September 30, 2022 and 2021, respectively, and \$64 million and \$54 million for the three months ended September 30, 2022 and 2021, respectively. Equity options volume increased 10% for the nine months ended September 30, 2022, and decreased 1% for the three months ended September 30, 2022, from the comparable periods in 2021. The overall increase in equity options volume for the nine months ended September 30, 2022 was driven by increased market share. The overall decrease in equity options volume for the three months ended September 30, 2022 was primarily due to the Arca Options Pillar migration. Equity options revenues, net of transaction-based expenses, were \$75 million and \$82 million for the nine months ended September 30, 2022 and 2021, respectively, and \$24 million and \$25 million for the three months ended September 30, 2022 and 2021, respectively.
- **OTC and Other:** OTC and other transactions include revenues from our OTC energy business and other trade confirmation services, as well as interest income on certain clearing margin deposits, regulatory penalties and fines, fees for use of our facilities, regulatory fees charged to member organizations of our U.S. securities exchanges, designated market maker service fees, exchange membership fees and agricultural grading and certification fees. Our OTC and other revenues increased 37% and 45% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021 primarily due to an increase in interest income on clearing margin deposits. Following the October 2021 Bakkt transaction, Bakkt revenues are no longer included within our OTC and other revenues.
- **Data and Connectivity Services:** Our data and connectivity services revenues increased 4% and 6% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021. The increase in revenue was driven by the strong retention rate of existing customers, the addition of new customers and increased purchases by existing customers.
- **Listings Revenues:** Through NYSE, NYSE American and NYSE Arca, we generate listings revenue related to the provision of listings services for public companies and ETFs, and related corporate actions for listed companies. Listings revenues increased 9% and 3% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021, driven by the full impact of strong equity capital markets activity in 2021. All listings fees are billed upfront and revenues are recognized over time as the identified performance obligations are satisfied.

Selected Operating Data

The following charts and tables present trading activity in our futures and options markets by commodity type based on the total number of contracts traded, as well as futures and options rate per contract (in millions, except for percentages and rate per contract amounts and YTD represents the nine-month periods ended September 30th):

Volume and Rate per Contract



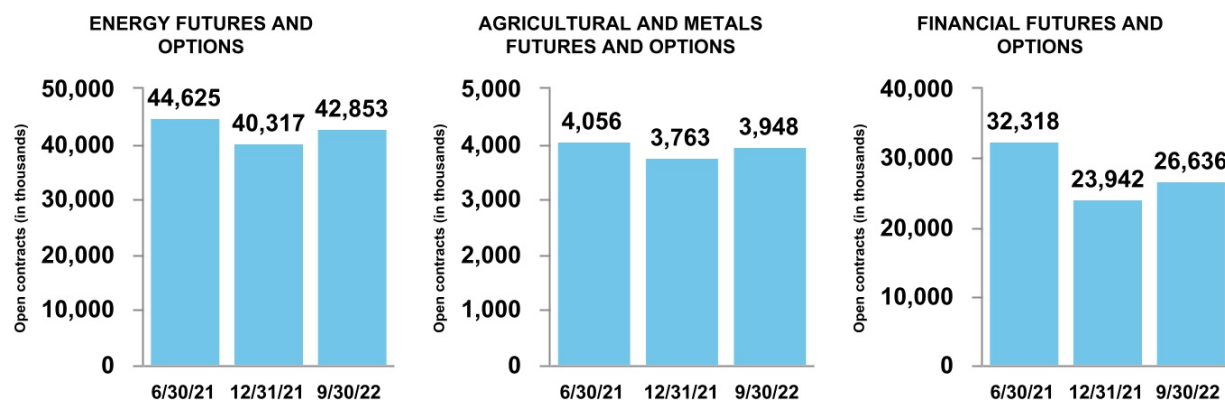
	Nine Months Ended September 30,			Three Months Ended September 30,		
	2022	2021	Change	2022	2021	Change
Number of contracts traded (in millions):						
Energy futures and options	572	581	(1)%	173	200	(13)%
Agricultural and metals futures and options	77	76	1	25	24	4
Financial futures and options	504	477	5	176	152	15
Total	1,153	1,134	2 %	374	376	(1)%

	Nine Months Ended September 30,			Three Months Ended September 30,		
	2022	2021	Change	2022	2021	Change
Average daily volume of contracts traded (in thousands):						
Energy futures and options	3,046	3,089	(1)%	2,706	3,126	(13)%
Agricultural and metals futures and options	408	404	1	393	379	4
Financial futures and options	2,622	2,495	5	2,668	2,320	15
Total	6,076	5,988	1 %	5,767	5,825	(1)%

	Nine Months Ended September 30,			Three Months Ended September 30,		
	2022	2021	Change	2022	2021	Change
Rate per contract:						
Energy futures and options	\$ 1.54	\$ 1.55	(1)%	\$ 1.54	\$ 1.58	(3)%
Agricultural and metals futures and options	\$ 2.33	\$ 2.33	— %	\$ 2.26	\$ 2.33	(3)%
Financial futures and options	\$ 0.74	\$ 0.58	27 %	\$ 0.69	\$ 0.61	13 %

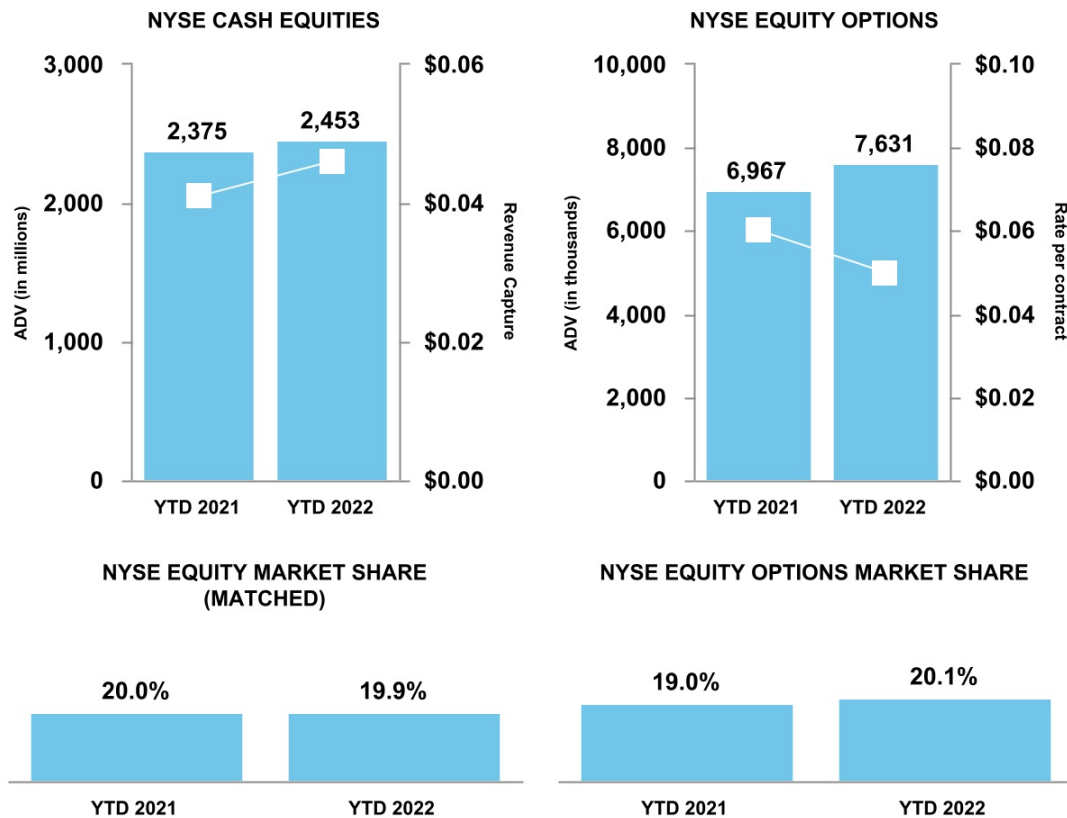
Open interest is the aggregate number of contracts (long or short) that clearing members hold either for their own account or on behalf of their clients. Open interest refers to the total number of contracts that are currently “open,” – in other words, contracts that have been entered into but not yet liquidated by either an offsetting trade, exercise, expiration or assignment. Open interest is also a measure of the future activity remaining to be closed out in terms of the number of contracts that members and their clients continue to hold in the particular contract and by the number of contracts held for each contract month listed by the exchange. The following charts and table present our quarter-end open interest for our futures and options contracts (in thousands, except for percentages):

Open Interest



	As of September 30,		Change
	2022	2021	
Open interest — in thousands of contracts:			
Energy futures and options	42,853	44,625	(4)%
Agricultural and metals futures and options	3,948	4,056	(3)
Financial futures and options	26,636	32,318	(18)
Total	73,437	80,999	(9)%

The following charts and tables present selected cash and equity options trading data. All trading volume below is presented as average net daily trading volume, or ADV, and is single counted and YTD represents the nine-month periods ended September 30th:



	Nine Months Ended September 30,			Three Months Ended September 30,		
	2022	2021	Change	2022	2021	Change
NYSE cash equities (shares in millions):						
Total cash handled volume	2,453	2,375	3 %	2,158	2,022	7 %
Total cash market share matched	19.9 %	20.0 %	(0.1 pts)	19.4 %	20.3 %	(0.9 pts)
NYSE equity options (contracts in thousands):						
NYSE equity options volume	7,631	6,967	10 %	7,037	7,078	(1)%
Total equity options volume	37,888	36,684	3 %	36,994	35,546	4 %
NYSE share of total equity options	20.1 %	19.0 %	1.1 pts	19.0 %	19.9 %	(0.9 pts)
Revenue capture or rate per contract:						
Cash equities rate per contract (per 100 shares)	\$0.046	\$0.041	11 %	\$0.046	\$0.042	11 %
Equity options rate per contract	\$0.05	\$0.06	(16)%	\$0.05	\$0.05	1 %

Handled volume represents the total number of shares of equity securities, ETFs and crossing session activity internally matched on our exchanges or routed to and executed on an external market center. Matched volume represents the total number of shares of equity securities, ETFs and crossing session activity executed on our exchanges.

Transaction-Based Expenses

Our equities and equity options markets pay fees to the SEC pursuant to Section 31 of the Exchange Act. Section 31 fees are recorded on a gross basis as a component of transaction and clearing fee revenue. These Section 31 fees are assessed to recover the government's costs of supervising and regulating the securities markets and professionals and are subject to change. We, in turn, collect corresponding activity assessment fees from member organizations clearing or settling trades on the equities and options exchanges, and recognize these amounts in our transaction and clearing revenues when invoiced. The activity assessment fees are designed to equal the Section 31 fees. As a result, activity assessment fees and the corresponding Section 31 fees do not have an impact on our net income, although the timing of payment by us will vary from collections. Section 31 fees were \$332 million and \$204 million for the nine months

ended September 30, 2022 and 2021, respectively, and \$158 million and \$38 million for the three months ended September 30, 2022 and 2021, respectively. The increase in Section 31 fees was primarily due to an increase in rates. The fees we collect are included in cash at the time of receipt and we remit the amounts to the SEC semi-annually as required. The total amount is included in current liabilities and was \$58 million as of September 30, 2022.

We make liquidity payments to cash and options trading customers, as well as routing charges made to other exchanges which are included in transaction-based expenses. We incur routing charges when we do not have the best bid or offer in the market for a security that a customer is trying to buy or sell on one of our securities exchanges. In that case, we route the customer's order to the external market center that displays the best bid or offer. The external market center charges us a fee per share (denominated in tenths of a cent per share) for routing to its system. We record routing charges on a gross basis as a component of transaction and clearing fee revenue. Cash liquidity payments, routing and clearing fees were \$1.4 billion and \$1.3 billion for the nine months ended September 30, 2022 and 2021, respectively, and \$418 million and \$437 million for the three months ended September 30, 2022 and 2021, respectively.

Operating Expenses, Operating Income and Operating Margin

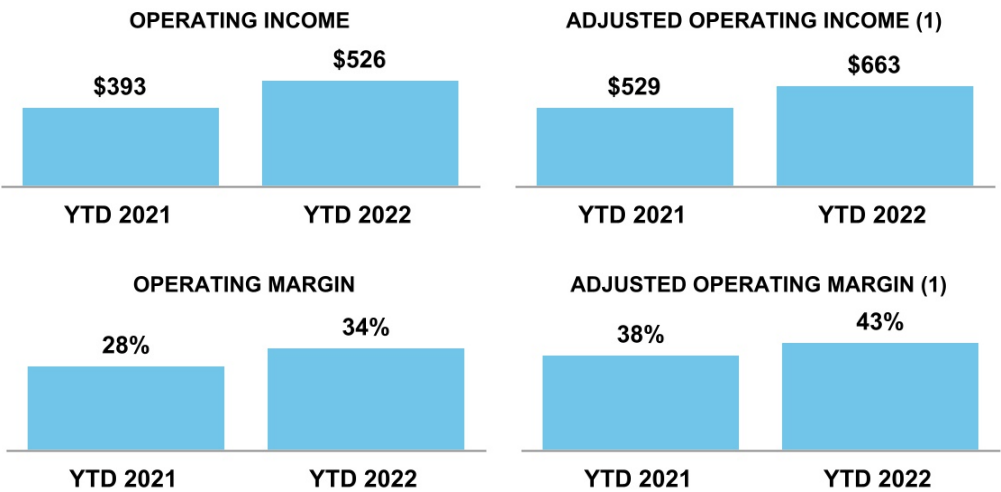
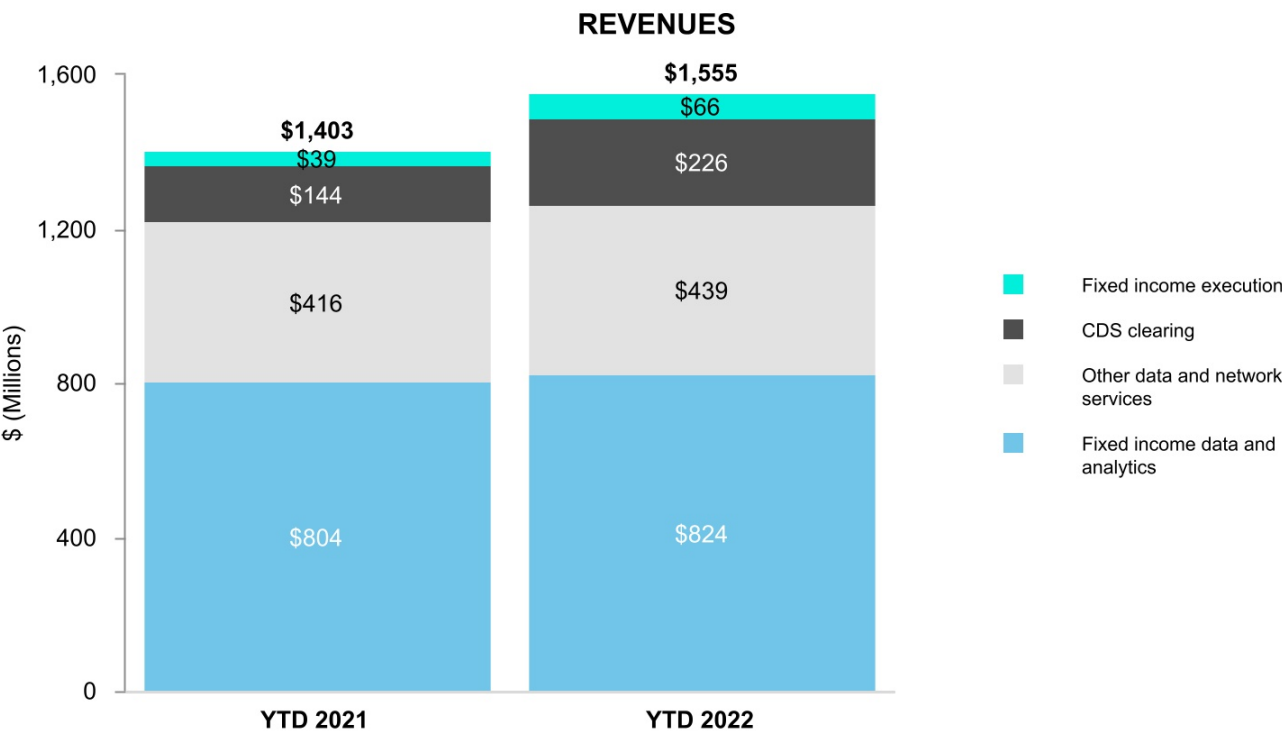
The following chart summarizes our Exchanges segment's operating expenses, operating income and operating margin (dollars in millions). See "- Consolidated Operating Expenses" below for a discussion of the significant changes in our operating expenses.

Exchanges Segment:	Nine Months Ended September 30,			Three Months Ended September 30,		
	2022	2021	Change	2022	2021	Change
Operating expenses	\$ 904	\$ 977	(7)%	\$ 301	\$ 330	(9)%
Adjusted operating expenses ⁽¹⁾	\$ 854	\$ 909	(6)%	\$ 284	\$ 309	(8)%
Operating income	\$ 2,185	\$ 1,865	17 %	\$ 700	\$ 629	11 %
Adjusted operating income ⁽¹⁾	\$ 2,235	\$ 1,933	16 %	\$ 717	\$ 650	10 %
Operating margin	71 %	66 %	5 pts	70 %	66 %	4 pts
Adjusted operating margin ⁽¹⁾	72 %	68 %	4 pts	72 %	68 %	4 pts

⁽¹⁾ The adjusted figures exclude items that are not reflective of our ongoing core operations and business performance. These adjusted numbers are not calculated in accordance with U.S. GAAP. See "- Non-GAAP Financial Measures" below.

Fixed Income and Data Services Segment

The following charts and table present our selected statements of income data for our Fixed Income and Data Services segment (dollars in millions and YTD represents the nine-month periods ended September 30th):



(1) The adjusted figures in the charts above are calculated by excluding items that are not reflective of our cash operations and core business performance. As a result, these adjusted numbers are not calculated in accordance with U.S. GAAP. See “- Non-GAAP Financial Measures” below.

	Nine Months Ended September 30,			Three Months Ended September 30,		
	2022	2021	Change	2022	2021	Change
Revenues:						
Fixed income execution	\$ 66	\$ 39	69 %	\$ 26	\$ 12	121 %
CDS clearing	226	144	57	88	51	72
Fixed income data and analytics	824	804	2	273	272	—
Fixed income and credit	1,116	987	13	387	335	15
Other data and network services	439	416	6	147	142	4
Revenues	1,555	1,403	11	534	477	12
Other operating expenses	766	752	2	250	252	(1)
Depreciation and amortization	262	257	2	86	85	1
Acquisition-related transaction and integration costs	1	1	45	1	1	(43)
Operating expenses	1,029	1,010	2	337	338	—
Operating income	\$ 526	\$ 393	34 %	\$ 197	\$ 139	42 %
Recurring revenues	\$ 1,263	\$ 1,220	3 %	\$ 420	\$ 414	1 %
Transaction revenues	\$ 292	\$ 183	59 %	\$ 114	\$ 63	81 %

In the table above, we consider fixed income data and analytics revenues and other data and network services revenues to be recurring revenues.

For the nine months ended September 30, 2022 and 2021, 12% and 14%, respectively, of our Fixed Income and Data Services segment revenues were billed in pounds sterling or euros and for the three months ended September 30, 2022 and 2021, 10% and 13%, respectively, of our Fixed Income and Data Services segment revenues were billed in pounds sterling or euros. As the pound sterling or euro exchange rate changes, the U.S. equivalent of revenues denominated in foreign currencies changes accordingly. Due to the fluctuations of the pound sterling and euro compared to the U.S. dollar, our Fixed Income and Data Services revenues were lower by \$20 million and \$9 million for the nine and three months ended September 30, 2022, respectively, than the comparable periods in 2021.

Fixed Income and Data Services Revenues

Our Fixed Income and Data Services revenues increased 11% and 12% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021. The increase in revenue was primarily due to strength in our fixed income execution and CDS clearing businesses due to elevated volatility across global markets driven by geopolitical events, central bank activity and inflationary concerns as well as increased market share.

- **Fixed Income Execution:** Fixed income execution includes revenues from ICE Bonds. Execution fees are reported net of rebates, which were nominal for both the nine and three months ended September 30, 2022 and 2021. Our fixed income execution revenues increased 69% and 121% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021, due to elevated volatility across global markets driven by geopolitical events, central bank activity and inflationary concerns.
- **CDS Clearing:** CDS clearing revenues increased 57% and 72% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021. The notional value of CDS cleared was \$19.7 trillion and \$12.6 trillion for the nine months ended September 30, 2022 and 2021, respectively, and \$6.1 trillion and \$4.5 trillion for the three months ended September 30, 2022 and 2021, respectively. The increases in the notional value of CDS cleared were primarily driven by heightened volatility related to geopolitical events and inflationary concerns.
- **Fixed Income Data and Analytics:** Our fixed income data and analytics revenues increased 2% for the nine months ended September 30, 2022 and were flat for the three months ended September 30, 2022 from the comparable periods in 2021. The increase in revenue for the nine months ended September 30, 2022 was due to strength in our index business in the first half of 2022 and continued growth in our pricing and reference data business driven by the strong retention rate of existing customers, the addition of new customers and increased purchases by existing customers. This was partially offset by unfavorable foreign exchange effects arising from fluctuations in the U.S. dollar from the comparable periods in 2021.

- **Other Data and Network Services:** Our other data and network services revenues increased 6% and 4% for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021. The increase in revenues was driven primarily by growth in our ICE Global Network offering, coupled with strength in our consolidated feeds and stronger desktop revenues.

Annual Subscription Value, or ASV, represents, at a point in time, the data services revenues, which includes Fixed Income Data and Analytics as well as other data and network services, subscribed for the succeeding 12 months. ASV does not include new sales, contract terminations or price changes that may occur during that 12-month period. However, while it is an indicative forward-looking metric, it does not provide a precise growth forecast of the next 12 months of data services revenues.

As of September 30, 2022, ASV was \$1.643 billion, which increased 1.1% compared to the ASV as of September 30, 2021. ASV represents nearly 100% of total data services revenues for this segment. This does not adjust for year-over-year foreign exchange fluctuations.

Operating Expenses, Operating Income and Operating Margin

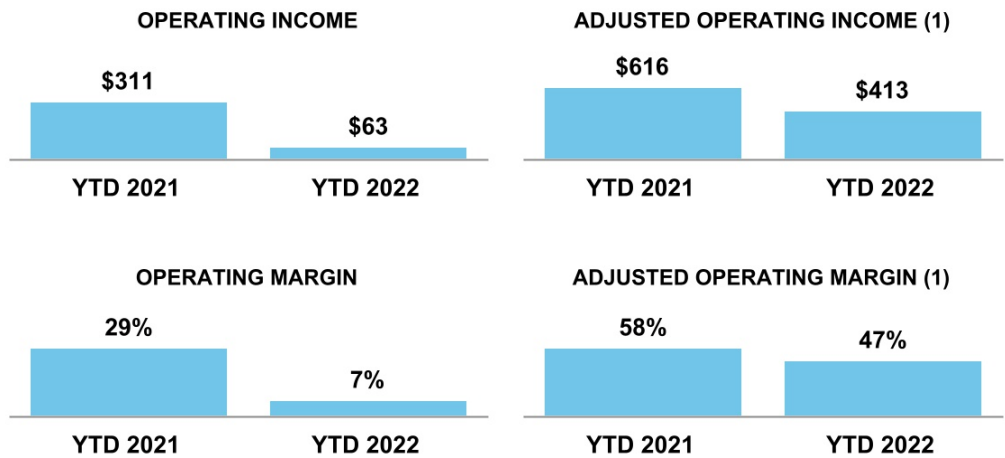
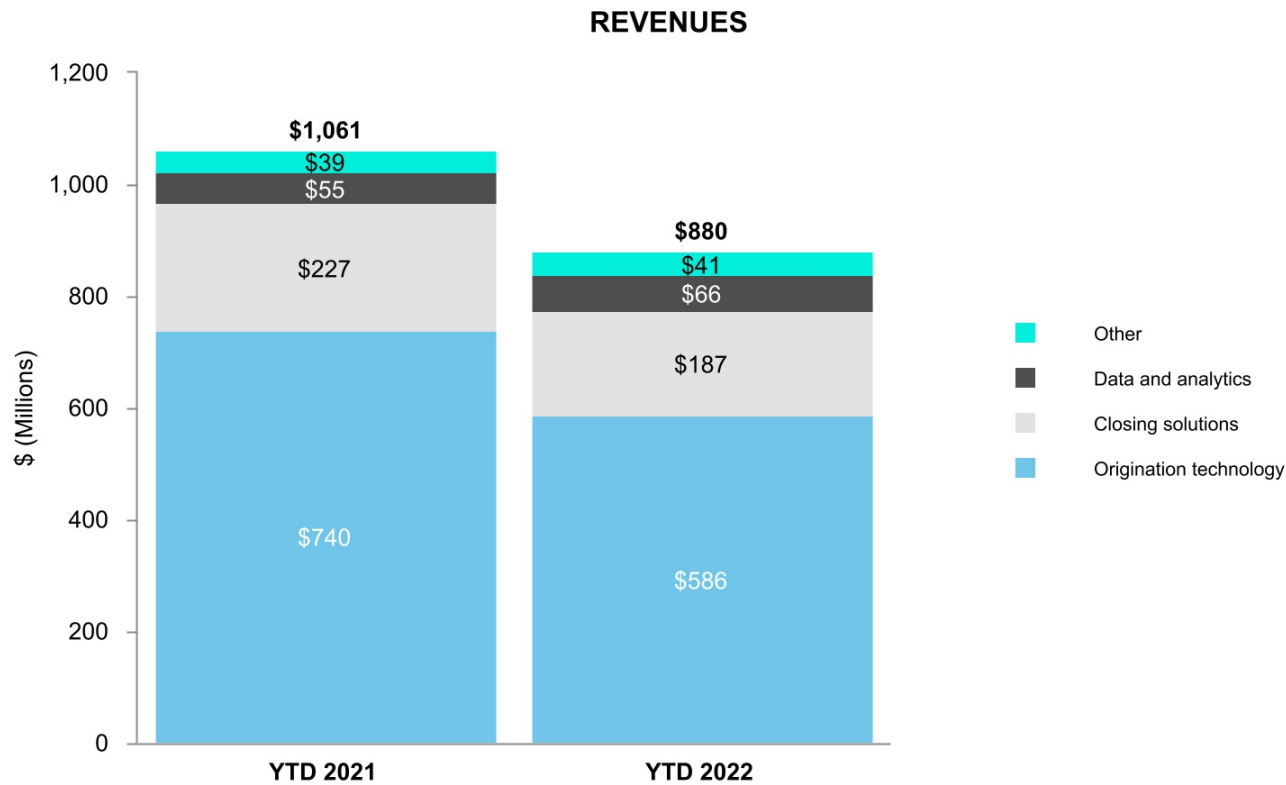
The following chart summarizes our Fixed Income and Data Services segment's operating expenses, operating income and operating margin (dollars in millions). See "- Consolidated Operating Expenses" below for a discussion of the significant changes in our operating expenses.

Fixed Income and Data Services Segment:	Nine Months Ended September 30,			Three Months Ended September 30,		
	2022	2021	Change	2022	2021	Change
Operating expenses	\$ 1,029	\$ 1,010	2 %	\$ 337	\$ 338	— %
Adjusted operating expenses ⁽¹⁾	\$ 892	\$ 874	2 %	\$ 293	\$ 293	— %
Operating income	\$ 526	\$ 393	34 %	\$ 197	\$ 139	42 %
Adjusted operating income ⁽¹⁾	\$ 663	\$ 529	25 %	\$ 241	\$ 184	31 %
Operating margin	34 %	28 %	6 pts	37 %	29 %	8 pts
Adjusted operating margin ⁽¹⁾	43 %	38 %	5 pts	45 %	39 %	6 pts

⁽¹⁾ The adjusted figures exclude items that are not reflective of our ongoing core operations and business performance. These adjusted numbers are not calculated in accordance with U.S. GAAP. See "- Non-GAAP Financial Measures" below.

Mortgage Technology Segment

The following charts and table present our selected statements of income data for our Mortgage Technology segment (dollars in millions and YTD represents the nine-month periods ended September 30th):



(1) The adjusted figures in the charts above are calculated by excluding items that are not reflective of our cash operations and core business performance. As a result, these adjusted numbers are not calculated in accordance with U.S. GAAP. See “- Non-GAAP Financial Measures” below.

	Nine Months Ended September 30,			Three Months Ended September 30,		
	2022	2021	Change	2022	2021	Change
Revenues:						
Origination technology	586	740	(21)%	187	245	(24)%
Closing solutions	187	227	(17)	53	88	(39)
Data and analytics	66	55	22	22	19	22
Other	41	39	2	14	14	(8)
Revenues	880	1,061	(17)	276	366	(25)
Other operating expenses	410	406	1	130	140	(6)
Depreciation and amortization	328	316	4	112	106	5
Acquisition-related transaction and integration costs	79	28	182	18	10	67
Operating expenses	817	750	9	260	256	2
Operating income	\$ 63	\$ 311	(80)%	\$ 16	\$ 110	(86)%
Recurring revenues	\$ 479	\$ 404	19%	\$ 163	\$ 143	14%
Transaction revenues	\$ 401	\$ 657	(39)%	\$ 113	\$ 223	(49)%

In the table above, we consider subscription fee and certain other revenues to be recurring revenues. Each revenue classification, above, contains a mix of recurring and transaction revenues, based on the various service offerings described in more detail, below.

Mortgage Technology Revenues

Our mortgage technology revenues are derived from our comprehensive, end-to-end U.S. residential mortgage platform. Our mortgage technology business is intended to enable greater workflow efficiency for customers focused on originating U.S. residential mortgage loans. Mortgage technology revenues decreased \$181 million and \$90 million for the nine and three months ended September 30, 2022 from the comparable periods in 2021 due to lower mortgage origination volumes driven by rising interest rates. See Note 6 of our consolidated financial statements in this Quarterly Report where discussed further.

- **Origination technology:** Our origination technology acts as a system of record for the mortgage transaction, automating the gathering, reviewing, and verifying of mortgage-related information and enabling automated enforcement of rules and business practices designed to help ensure that each completed loan transaction is of high quality and adheres to secondary market standards. These revenues are based on recurring Software as a Service, or SaaS, subscription fees, with an additive transaction-based or success-based pricing fee as lenders exceed the number of loans closed that are included with their monthly base subscription.

In addition, the ICE Mortgage Technology network provides originators connectivity to the mortgage supply chain and facilitates the secure exchange of information between our customers and a broad ecosystem of third-party service providers, as well as lenders and investors that are critical to consummating the millions of loan transactions that occur on our origination network each year. Revenue from the ICE Mortgage Technology network is largely transaction-based.

- **Closing solutions:** Our closing solutions connect key participants, such as lenders, title and settlement agents and individual county recorders, to digitize the closing and recording process. Closing solutions also include revenues from our Mortgage Electronic Registrations Systems, Inc., or MERS database, which provides a system of record for recording and tracking changes and servicing rights and beneficial ownership interests in loans secured by U.S. residential real estate. Revenues from closing solutions are largely transaction-based and are based on volume of loan closings.
- **Data and Analytics:** Revenues include those related to ICE Mortgage Technology's Automation, Intelligence, Quality, or AIQ, offering which applies machine learning to the entire loan origination process, offering customers greater efficiency by streamlining data collection and validation through our automated document recognition and data extraction capabilities. AIQ revenues can be both recurring and transaction-based in nature. In addition, our data offerings include real-time industry and peer benchmarking tools, which provide originators a granular view into the real-time trends of nearly half the U.S. residential mortgage market. We also provide a Data as a Service, or DaaS, offering through private data clouds for lenders to access their own data and origination information. Revenues related to our data products are largely subscription-based and recurring in nature.

- **Other:** Other revenues include professional services fees, as well as revenues from ancillary products. Other revenues can be both recurring and transaction-based in nature.

Operating Expenses, Operating Income and Operating Margin

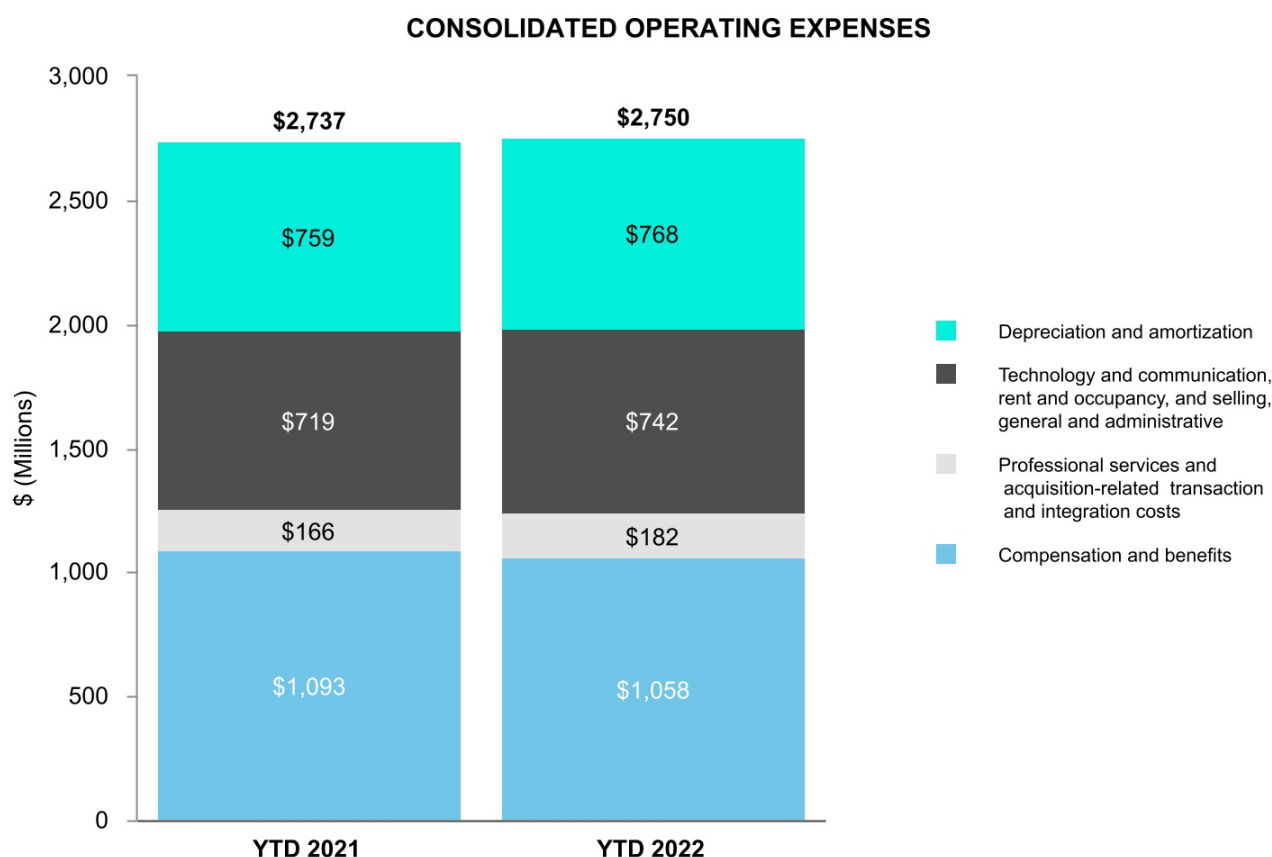
The following chart summarizes our Mortgage Technology segment's operating expenses, operating income and operating margin (dollars in millions). See “- Consolidated Operating Expenses” below for a discussion of the significant changes in our operating expenses.

Mortgage Technology Segment:	Nine Months Ended September 30,			Three Months Ended September 30,		
	2022	2021	Change*	2022	2021	Change
Operating expenses	\$ 817	\$ 750	9%	\$ 260	\$ 256	2%
Adjusted operating expenses ⁽¹⁾	\$ 467	\$ 445	5%	\$ 150	\$ 153	(2)%
Operating income	\$ 63	\$ 311	(80)%	\$ 16	\$ 110	(86)%
Adjusted operating income ⁽¹⁾	\$ 413	\$ 616	(33)%	\$ 126	\$ 213	(41)%
Operating margin	7 %	29 %	(22 pts)	6 %	30 %	(24 pts)
Adjusted operating margin ⁽¹⁾	47 %	58 %	(11 pts)	46 %	58 %	(12 pts)

⁽¹⁾ The adjusted figures exclude items that are not reflective of our ongoing core operations and business performance. These adjusted numbers are not calculated in accordance with GAAP. See “- Non-GAAP Financial Measures”

Consolidated Operating Expenses

The following presents our consolidated operating expenses (dollars in millions and YTD represents the nine-month periods ended September 30th):



	Nine Months Ended September 30,			Three Months Ended September 30,		
	2022	2021	Change	2022	2021	Change
Compensation and benefits	\$ 1,058	\$ 1,093	(3)%	\$ 344	\$ 374	(8)%
Professional services	101	124	(19)	32	43	(27)
Acquisition-related transaction and integration costs	81	42	93	19	14	38
Technology and communication	513	495	4	169	168	1
Rent and occupancy	63	61	2	22	20	5
Selling, general and administrative	166	163	2	54	52	6
Depreciation and amortization	768	759	1	258	253	2
Total operating expenses	\$ 2,750	\$ 2,737	— %	\$ 898	\$ 924	(3)%

The majority of our operating expenses do not vary directly with changes in our volume and revenues, except for certain technology and communication expenses, including data acquisition costs, licensing and other fee-related arrangements and a portion of our compensation expense that is tied directly to our data sales or overall financial performance.

We expect our operating expenses to increase in absolute terms in future periods in connection with the growth of our business, and to vary from year-to-year based on the type and level of our acquisitions, integration of acquisitions and other investments.

For both the nine and three months ended September 30, 2022 and 2021, 10% of our operating expenses were billed in pounds sterling or euros, and for the three months ended September 30, 2022 and 2021, 9% and 10%, respectively, of

our operating expenses were billed in pounds sterling or euros. Due to fluctuations in the U.S. dollar compared to the pound sterling and euro, our consolidated operating expenses were \$27 million and \$13 million lower during the nine and three months ended September 30, 2022, respectively, than in the comparable periods in 2021. See Item 3 “— Quantitative and Qualitative Disclosures About Market Risk - Foreign Currency Exchange Rate Risk” below for additional information.

Compensation and Benefits Expenses

Compensation and benefits expense is our most significant operating expense and includes non-capitalized employee wages, bonuses, non-cash or stock compensation, certain severance costs, benefits and employer taxes. The bonus component of our compensation and benefits expense is based on both our financial performance and individual employee performance. The performance-based restricted stock compensation expense is also based on our financial performance. Therefore, our compensation and benefits expense will vary year-to-year based on our financial performance and fluctuations in our number of employees. The below chart summarizes the significant drivers of our compensation and benefits expense results for the periods presented (dollars in millions, except employee headcount).

	Nine Months Ended September 30,			Three Months Ended September 30,		
	2022	2021	Change	2022	2021	Change
Employee headcount	8,935	9,381	(5)%			
Stock-based compensation expenses	\$ 110	\$ 112	— %	\$ 37	\$ 39	(6)%

Headcount decreased primarily due to a reduction of 608 Bakkt employees following its deconsolidation, partially offset by 274 additional employees hired in India.

Compensation and benefits expense decreased \$35 million for the nine months ended September 30, 2022 from the comparable period in 2021, primarily due to \$51 million in expenses related to Bakkt during the nine months ended September 30, 2021, prior to deconsolidation. This was partially offset by a \$14 million increase for the nine months ended September 30, 2022, from the comparable period in 2021, related to additional headcount, increased commissions, merit pay increases, and higher employee insurance costs. The stock-based compensation expenses in the table above relate to employee stock option and restricted stock awards and exclude stock-based compensation related to acquisition-related transaction and integration costs.

Compensation and benefits expense decreased \$30 million for the three months ended September 30, 2022, primarily due to \$20 million in expenses related to Bakkt during the three months ended September 30, 2021, prior to deconsolidation.

Professional Services Expenses

Professional services expense includes fees for consulting services received on strategic and technology initiatives, temporary labor, as well as regulatory, legal and accounting fees, and may fluctuate as a result of changes in our use of these services in our business.

Professional services expenses decreased \$23 million and \$11 million for the nine and three months ended September 30, 2022, respectively, from the comparable periods in 2021, primarily due to \$11 million and \$6 million in decreased legal fees for the nine and three months ended September 30, 2022, respectively, as well as \$13 million and \$6 million of expenses incurred at Bakkt during the nine and three months ended September 30, 2021, respectively, prior to deconsolidation.

Acquisition-Related Transaction and Integration Costs

We incurred \$81 million and \$19 million in acquisition-related transaction and integration costs during the nine and three months ended September 30, 2022, primarily due to legal and consulting expenses related to our pending acquisition of Black Knight and our integration of Ellie Mae, Inc., or Ellie Mae. We incurred \$42 million and \$14 million in acquisition-related transaction costs for the nine and three months ended September 30, 2021, primarily related to our integration of Ellie Mae and the Bakkt transaction.

We expect to continue to explore and pursue various potential acquisitions and other strategic opportunities to strengthen our competitive position and support our growth. As a result, we may incur acquisition-related transaction costs in future periods.

Technology and Communication Expenses

Technology support services consist of costs for running our wholly-owned data centers, hosting costs paid to third-party data centers and maintenance of our computer hardware and software required to support our technology and cybersecurity. These costs are driven by system capacity, functionality and redundancy requirements. Communication expenses consist of costs or network connections for our electronic platforms and telecommunications costs.

Technology and communications expense also includes fees paid for access to external market data, licensing and other fee agreement expenses. Technology and communications expenses may be impacted by growth in electronic contract volume, our capacity requirements, changes in the number of telecommunications hubs and connections with customers to access our electronic platforms directly.

Technology and communications expenses increased \$18 million for the nine months ended September 30, 2022 from the comparable period in 2021, primarily due to \$17 million in increased hardware and software support costs, \$15 million in increased hosting costs and \$7 million in increased data services costs, partially offset by an \$11 million decrease in license expense and \$10 million in expenses during the nine months ended September 30, 2021 related to Bakkt prior to deconsolidation.

Technology and communications expenses increased \$1 million for the three months ended September 30, 2022 from the comparable period in 2021, primarily due to increased hardware and software support costs, hosting costs and data services costs, partially offset by a decrease in license expense and \$3 million in expenses during the three months ended September 30, 2021 related to Bakkt prior to deconsolidation.

Rent and Occupancy Expenses

Rent and occupancy expense relates to leased and owned property and includes rent, maintenance, real estate taxes, utilities and other related costs. We have significant operations located in the U.S., U.K., and India, with smaller offices located throughout the world.

Rent and occupancy expenses increased \$2 million for both the nine and three months ended September 30, 2022 from the comparable periods in 2021, primarily due to increased occupancy costs for the nine and three months ended September 30, 2022. During the nine and three months ended September 30, 2021, we incurred \$2 million and \$1 million in expenses related to Bakkt prior to deconsolidation.

Selling, General and Administrative Expenses

Selling, general and administrative expenses include marketing, advertising, public relations, insurance, bank service charges, dues and subscriptions, travel and entertainment, non-income taxes and other general and administrative costs.

Selling, general and administrative expenses increased \$3 million for the nine months ended September 30, 2022 from the comparable period in 2021 primarily due to \$11 million in increased marketing expenses and \$13 million in increased travel and entertainment expenses. This was partially offset by \$19 million in expenses related to Bakkt during the nine months ended September 30, 2021 prior to deconsolidation.

Selling, general and administrative expenses increased \$2 million for the three months ended September 30, 2022, from the comparable period in 2021, primarily due to increased travel and entertainment expense, dues and subscriptions and taxes and fees. This was partially offset by expenses related to Bakkt during the three months ended September 30, 2021, prior to deconsolidation.

Depreciation and Amortization Expenses

Depreciation and amortization expense results from depreciation of long-lived assets such as buildings, leasehold improvements, aircraft, hardware and networking equipment, software, furniture, fixtures and equipment over their estimated useful lives. This expense includes amortization of intangible assets obtained in our acquisitions of businesses, as well as on various licensing agreements, over their estimated useful lives. Intangible assets subject to amortization consist primarily of customer relationships, trading products with finite lives and technology. This expense also includes amortization of internally-developed and purchased software over its estimated useful life.

We recorded amortization expenses on intangible assets acquired as part of our acquisitions, as well as on other intangible assets, of \$459 million and \$470 million for the nine months ended September 30, 2022 and 2021, respectively, and \$153 million and \$156 million for the three months ended September 30, 2022 and 2021, respectively.

We recorded depreciation expenses on our fixed assets of \$309 million and \$289 million for the nine months ended September 30, 2022 and 2021, respectively, and \$105 million and \$97 million for the three months ended September 30, 2022 and 2021, respectively.

Consolidated Non-Operating Income/(Expense)

Income and expenses incurred through activities outside of our core operations are considered non-operating. The following tables present our non-operating income/(expenses) (dollars in millions):

	Nine Months Ended September 30,			Three Months Ended September 30,		
	2022	2021	Change*	2022	2021	Change*
Other income/(expense):						
Interest income	\$ 42	\$ —	n/a	\$ 33	\$ —	n/a
Interest expense	(440)	(321)	37	(176)	(108)	62
Other income/(expense), net	(1,132)	1,341	n/a	(1,097)	54	n/a
Total other income/(expense), net	\$ (1,530)	\$ 1,020	n/a	\$ (1,240)	\$ (54)	n/a
Net income attributable to non-controlling interest	\$ (37)	\$ (9)	300%	\$ (16)	\$ (4)	284%

*Percentage changes in the table above deemed "n/a" are not meaningful.

Interest Income

We recognized interest income of \$42 million and \$33 million during the nine and three months ended September 30, 2022, respectively. Interest income for the nine and three months ended September 30, 2022 primarily represents interest income on our short-term investments, including \$30 million and \$25 million in interest income recorded during the nine and three months ended September 30, 2022, respectively, in connection with the short-term investments related to the \$4.9 billion of the SMR Notes for the Black Knight acquisition.

Interest Expense

We recognized interest expense of \$440 million and \$321 million during the nine months ended September 30, 2022 and 2021, respectively, and \$176 million and \$108 million during the three months ended September 30, 2022 and 2021, respectively. Interest expense for the nine and three months ended September 30, 2022 primarily represents interest expense on our outstanding debt. We recognized \$79 million and \$56 million during the nine and three months ended September 30, 2022, respectively, in interest expense on the Black Knight acquisition-related-debt, and during the nine months ended September 30, 2022, we recognized \$30 million in costs associated with the extinguishment and re-financing of our existing debt in connection with our May 2022 debt refinancing. See "- Debt" below.

Other Income/(Expense), net

Our equity method investments include OCC and Bakkt, among others. We recognized (\$1.1 billion) and \$42 million during the nine months ended September 30, 2022 and 2021, respectively, and (\$1.1 billion) and \$8 million during the three months ended September 30, 2022 and 2021, respectively, of our share of estimated equity method investment (losses)/profits, net, which is included in other income. The estimated losses during the nine and three months ended September 30, 2022 are primarily related to our investment in Bakkt, and the estimated profits during the nine and three months ended September 30, 2021 are related to our investment in OCC. Both the nine month periods ended September 30, 2022 and 2021 include adjustments to reflect the difference between reported prior period actual results from our original estimates.

As of September 30, 2022, after recording our share of Bakkt's equity method losses, which included Bakkt's impairment charge, the carrying value was determined to be \$439 million. Based on our review, we determined that the decline in fair value was other than temporary in nature. Therefore, we recorded an impairment charge of \$40 million on our investment in Bakkt to its fair value as of September 30, 2022 as other expense. (see Notes 3 and 4 to our consolidated financial statements included elsewhere in this Quarterly Report).

During the nine months ended September 30, 2021, Coinbase completed an IPO and we sold our investment in Coinbase for \$1.24 billion, and recorded a gain of \$1.23 billion as other income.

During the nine months ended September 30, 2021, we recorded a gain of \$7 million related to the settlement of an acquisition-related indemnification claim from a prior acquisition as other income.

During the nine months ended September 30, 2022, we recorded a \$9 million accrual and during the nine and three months ended September 30, 2021, we recorded a \$16 million accrual for legal settlements, which are included in other expense.

We completed the sale of our Euroclear stake on May 20, 2022. The carrying value of our investment was \$700 million at the time of the sale. We recorded a net gain of \$41 million on the sale, which is included in other income during the nine

months ended September 30, 2022. We did not receive a Euroclear dividend during the nine months ended September 30, 2022 prior to the May 2022 sale of our investment. We recognized dividend income of \$60 million and \$30 million on our Euroclear investment during the nine and three months ended September 30, 2021, respectively, which is included in other income. In addition, during the three months ended September 30, 2021 we became aware of an observable price change in orderly transactions of similar Euroclear investments by a third party. The transactions resulted in a fair value adjustment of our Euroclear investment, and we recorded a gain of \$34 million in other income, including the impact of foreign currency exchange.

We incurred foreign currency transaction losses of \$9 million and \$11 million for the nine months ended September 30, 2022 and 2021, respectively, and \$2 million for the three months ended September 30, 2021, primarily attributable to the fluctuations of the pound sterling and euro relative to the U.S. dollar. We did not incur any significant foreign currency transaction gains/(losses) during the three months ended September 30, 2022. Foreign currency transaction gains/(losses) are recorded in other income/(expense), net, when the settlement of foreign currency assets, liabilities and payables occur in non-functional currencies and there is an increase or decrease in the period-end foreign currency exchange rates between periods. See Item 3 “- Quantitative and Qualitative Disclosures About Market Risk - Foreign Currency Exchange Rate Risk” included elsewhere in this Quarterly Report for more information on these items.

Non-controlling Interest

For consolidated subsidiaries in which our ownership is less than 100%, and for which we have control over the assets, liabilities and management of the entity, the outside stockholders' interests are shown as non-controlling interests. As of September 30, 2022, our non-controlling interests included those related to the non-ICE limited partners' 26.7% ownership interest in our CDS clearing subsidiaries, and non-controlling interests in ICE Futures Abu Dhabi. Prior to completion of the Bakkt transaction on October 15, 2021, our non-controlling interest also included the redeemable non-controlling interest of the non-ICE partners in Bakkt.

Consolidated Income Tax Provision

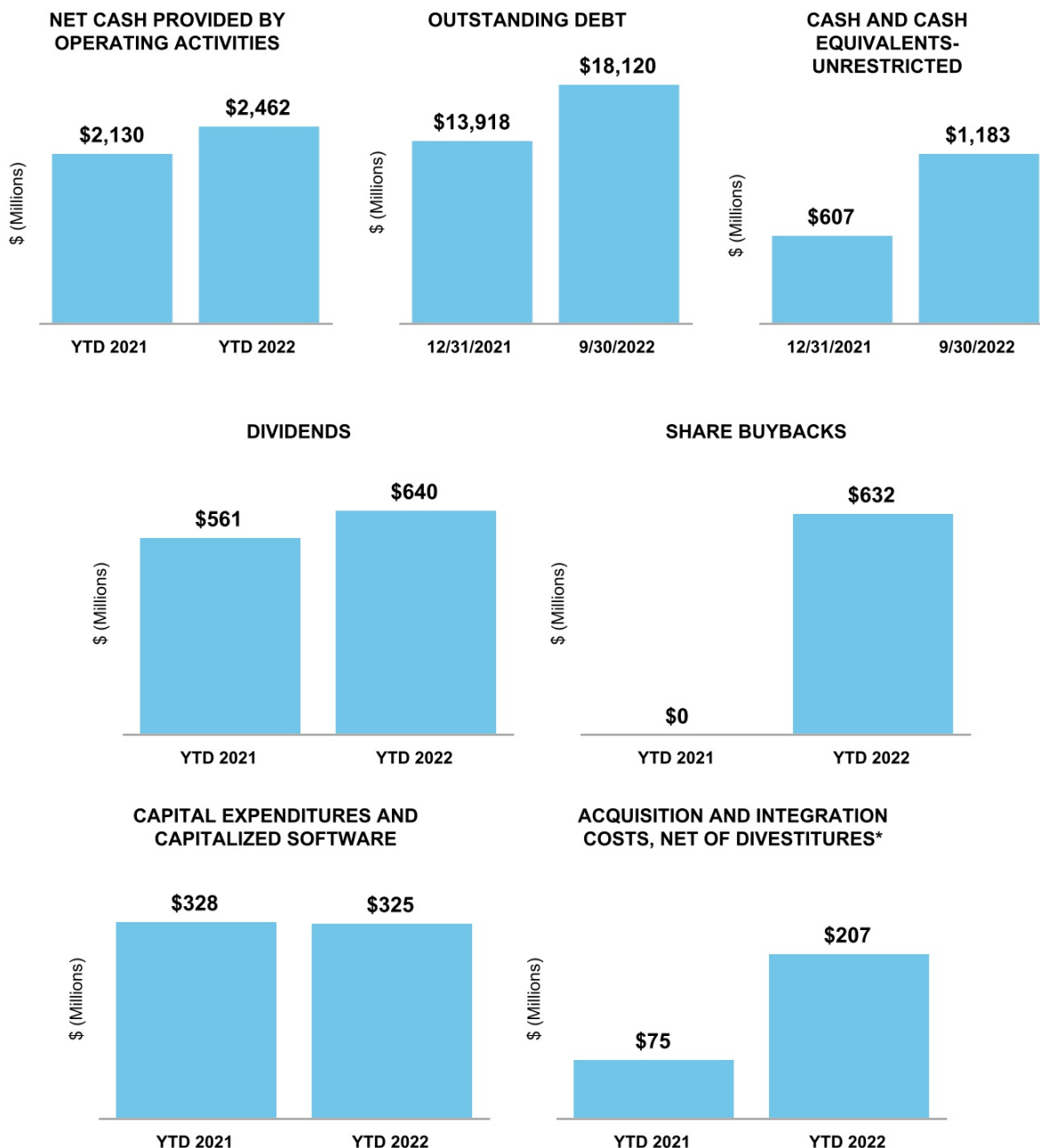
Consolidated income tax expense was \$186 million and \$1.0 billion for the nine months ended September 30, 2022 and 2021, respectively, and a \$152 million tax benefit and a \$187 million tax expense for the three months ended September 30, 2022 and 2021, respectively. The change in consolidated income tax expense between periods is primarily due to the tax impact of changes in our pre-tax income and the changes in our effective tax rate each period. The \$152 million tax benefit for the three months ended September 30, 2022 is primarily due to the deferred tax benefit associated with the impairment to our equity method investment in Bakkt more than offsetting the income tax expense on third quarter pre-tax income before the impairment. Our effective tax rate was 15% and 29% for the nine months ended September 30, 2022 and 2021, respectively, and 47% and 23% during the three months ended September 30, 2022 and 2021, respectively.

The effective tax rate for the nine months ended September 30, 2022 was lower than the effective tax rate for the comparable period in 2021 primarily due to the deferred income tax benefit associated with the impairment of our equity method investment in Bakkt in the current year period and the deferred income tax expense from U.K. tax law changes in the prior year period. During the nine months ended September 30, 2021, the U.K. Finance Act 2021 was enacted, which increased the U.K. corporate income tax rate from 19% to 25%, effective April 1, 2023.

The effective tax rate for the three months ended September 30, 2022 was higher than the effective tax rate for the comparable period in 2021 primarily due to the deferred tax benefit associated with the impairment of our equity method investment in Bakkt.

Liquidity and Capital Resources

Below are charts that reflect our outstanding debt and capital allocation. The acquisition and integration costs in the chart below include cash paid for acquisitions, net of cash received for divestitures, cash paid for equity and equity method investments, cash paid for non-controlling interest and redeemable non-controlling interest, and acquisition-related transaction and integration costs in each period (YTD represents the nine-month periods ended September 30th).



*Excludes the proceeds from the sales of our stakes in Euroclear (Q2 2022) and Coinbase (Q2 2021).

We have financed our operations, growth and cash needs primarily through income from operations and borrowings under our various debt facilities. Our principal capital requirements have been to fund capital expenditures, working capital, strategic acquisitions and investments, stock repurchases, dividends and the development of our technology platforms. We believe that our cash on hand and cash flows from operations will be sufficient to repay our outstanding debt, but we

may also need to incur additional debt or issue additional equity securities in the future. See “- Future Capital Requirements” below.

See “- Cash Flow” below for a discussion of our capital expenditures and capitalized software development costs.

Consolidated cash and cash equivalents were \$1.2 billion and \$607 million as of September 30, 2022 and December 31, 2021, respectively. We had \$6.4 billion and \$1.4 billion in short-term and long-term restricted cash and cash equivalents as of September 30, 2022 and December 31, 2021, respectively, the increase of which is related to the restricted \$4.9 billion net proceeds from the SMR Notes intended to be used to finance the Black Knight acquisition. We had \$156.8 billion and \$145.9 billion of cash and cash equivalent margin deposits and guaranty funds as of September 30, 2022 and December 31, 2021, respectively.

As of September 30, 2022, the amount of unrestricted cash held by our non-U.S. subsidiaries was \$430 million. Due to U.S. tax reform, the majority of our foreign earnings since January 1, 2018 have been subject to immediate U.S. income taxation, and the existing non-U.S. unrestricted cash balance can be distributed to the U.S. in the future with no material additional income tax consequences.

Our cash and cash equivalents and financial investments are managed as a global treasury portfolio of non-speculative financial instruments that are readily convertible into cash, such as overnight deposits, term deposits, money market funds, mutual funds for treasury investments, short duration fixed income investments and other money market instruments, thus ensuring high liquidity of financial assets. We may invest a portion of our cash in excess of short-term operating needs in investment-grade marketable debt securities, including government or government-sponsored agencies and corporate debt securities. As of September 30, 2022, we held \$4 million of unrestricted cash that was set aside for legal, regulatory, and surveillance operations at NYSE.

Cash Flow

The following table presents the major components of net changes in cash and cash equivalents, and restricted cash and cash equivalents (in millions):

	Nine Months Ended September 30,	
	2022	2021
Net cash provided by/(used in):		
Operating activities	\$ 2,462	\$ 2,130
Investing activities	(2,361)	1,325
Financing activities	16,373	21,166
Effect of exchange rate changes	(41)	(4)
Net increase in cash and cash equivalents, restricted cash and cash equivalents, and cash and cash equivalent margin deposits and guaranty funds	\$ 16,433	\$ 24,617

Operating Activities

Net cash provided by operating activities primarily consists of net income adjusted for certain items, including depreciation and amortization, deferred taxes, stock based compensation and the effects of changes in working capital.

The \$332 million increase in net cash provided by operating activities during the nine months ended September 30, 2022 from the comparable period in 2021 was driven by an increase in net income adjusted for the gain on our sale of our Coinbase investment during the nine months ended September 30, 2021 and a gain on the sale of our Euroclear investment during the nine months ended September 30, 2022, as well as \$1.1 billion in net losses from unconsolidated investees and an impairment loss of \$40 million on our investment in Bakkt. The remaining fluctuations are due to changes in our working capital and the timing of various payments such as an increase in Section 31 fees payable of \$194 million.

Investing Activities

Consolidated net cash used in investing activities for the nine months ended September 30, 2022 primarily relates to \$6.9 billion of purchases of invested margin deposits, \$125 million of capital expenditures and \$200 million of capitalized software development costs, partially offset by \$4.3 billion in proceeds from the sale of invested margin deposits and \$741 million in proceeds from the sale of our Euroclear investment.

Consolidated net cash provided by investing activities for the nine months ended September 30, 2021 primarily relates to \$1.2 billion of proceeds from the sale of our Coinbase investment and \$3.3 billion of proceeds from sales of invested

margin deposits, partially offset by \$2.8 billion purchases of invested margin deposits, \$117 million of capital expenditures and \$211 million of capitalized software development costs.

The capital expenditures primarily relate to hardware and software purchases to continue the development and expansion of our electronic platforms, data services and clearing houses, and leasehold improvements. The software development expenditures primarily relate to the development and expansion of our electronic trading platforms, data services, mortgage services and clearing houses.

Financing Activities

Consolidated net cash provided by financing activities for the nine months ended September 30, 2022 primarily relates to an increase in our cash and cash equivalent margin deposits and guaranty fund balances of \$13.5 billion due to increased volatility and \$7.9 billion in net proceeds from our debt facilities, partially offset by \$2.7 billion repayments of debt facilities, \$1.0 billion in net repayments under our Commercial Paper Program, \$632 million in repurchases of our common stock, \$640 million in dividend payments to stockholders and \$72 million in cash payments related to treasury shares received for restricted stock tax payments and stock option exercises.

Consolidated net cash provided by financing activities for the nine months ended September 30, 2021 primarily relates to an increase in our cash and cash equivalent margin deposits and guaranty fund balances of \$24.1 billion, partially offset by \$1.2 billion in repayments of debt facilities, \$1.1 billion in net repayments under our Commercial Paper Program, \$561 million in dividend payments to stockholders and \$69 million in cash payments related to treasury shares received for restricted stock tax payments and stock option exercises.

We have adjusted our historical presentation of opening and ending amounts of cash and cash equivalents, and restricted cash and cash equivalents in our consolidated statements of cash flows to include cash and cash equivalent margin deposits and guaranty funds. Changes in these balances are reflected as cash provided by financing activities.

Debt

As of September 30, 2022, we had \$18.1 billion in outstanding debt, all of which relates to our senior notes. We also have \$7 million under credit lines at our ICE India subsidiaries. As of September 30, 2022, our senior notes of \$18.1 billion had a weighted average maturity of 17 years and a weighted average cost of 3.6% per annum. There were no amounts outstanding under our Commercial Paper Program as of September 30, 2022. As of December 31, 2021, we had \$13.9 billion in outstanding debt, consisting of \$12.9 billion of senior notes, \$1.0 billion under our Commercial Paper Program and \$10 million under credit lines at our ICE India subsidiaries. As of December 31, 2021, our senior notes of \$12.9 billion had a weighted average maturity of 15 years and a weighted average cost of 2.9% per annum. The commercial paper notes had original maturities ranging from three to 73 days as of December 31, 2021, with a weighted average interest rate of 0.33% per annum, and a weighted average remaining maturity of 26 days.

We have a \$3.9 billion senior unsecured revolving credit facility, or the Credit Facility, with a maturity date of May 25, 2027 pursuant to a credit agreement with Wells Fargo Bank, N.A., as primary administrative agent, issuing lender and swing-line lender, Bank of America, N.A., as syndication agent, backup administrative agent and swing-line lender, and the lenders party thereto. As of September 30, 2022, of the \$3.9 billion that was available for borrowing under the Credit Facility, \$170 million was required to support certain broker-dealer and other subsidiary commitments. The remaining \$3.7 billion was available for working capital and general corporate purposes including, but not limited to, acting as a backstop to future increases in the amounts outstanding under the Commercial Paper Program.

On May 23, 2022, we issued \$8.0 billion in aggregate principal amount of new senior notes, comprised of the following:

- \$1.25 billion in aggregate principal amount of 3.65% senior notes due in 2025, or the 2025 Notes;
- \$1.5 billion in aggregate principal amount of 4.00% senior notes due in 2027, or the 2027 Notes;
- \$1.25 billion in aggregate principal amount of 4.35% senior notes due in 2029, or the 2029 Notes;
- \$1.5 billion in aggregate principal amount of 4.60% senior notes due in 2033, or the 2033 Notes;
- \$1.5 billion in aggregate principal amount of 4.95% senior notes due in 2052, or the 2052 Notes; and
- \$1.0 billion in aggregate principal amount of 5.20% senior notes due in 2062, or the 2062 Notes, collectively, the Notes.

We intend to use the net proceeds of \$4.9 billion from the offering of the 2025 Notes, the 2027 Notes, the 2029 Notes and the 2062 Notes, or collectively, the SMR Notes, together with the issuance of commercial paper and/or borrowings under the Credit Facility, cash on hand or other immediately available funds and borrowings under the Term Loan, to finance the cash portion of the purchase price for Black Knight. The SMR Notes are subject to a special mandatory redemption feature pursuant to which we will be required to redeem all of the outstanding SMR Notes at a redemption price equal to 101% of the aggregate principal amount of the SMR Notes, plus accrued and unpaid interest, in the event that the Black Knight acquisition is not consummated on or prior to May 4, 2023 (subject to two automatic extensions of three months each, to August 4, 2023 and to November 4, 2023, respectively, if U.S. antitrust clearance or a related law, injunction, order or other judgment, in each case whether temporary, preliminary or permanent, that restrains, enjoins or otherwise prohibits the consummation of the Black Knight merger remains outstanding and all other conditions to closing are satisfied (or in the case of conditions that by their terms are to be satisfied at the closing, are capable of being satisfied if the closing were to occur on such date) at each extension date), or if the Black Knight merger agreement is terminated at any time prior to such date. The \$4.9 billion net proceeds from the SMR Notes are separately invested and recorded as short-term restricted cash and cash equivalents in our consolidated balance sheet as of September 30, 2022.

We used the \$3.0 billion of net proceeds from the offering of the 2033 Notes and the 2052 Notes to redeem \$2.7 billion aggregate principal amount of four series of senior notes that would have matured in 2022 and 2023. The balance of the net proceeds was used for general corporate purposes, which included paying down a portion of the amounts outstanding under our Commercial Paper Program. We recorded \$30 million in costs associated with the extinguishment and re-financing of our existing debt in connection with our May 2022 debt refinancing. These costs are included in interest expense in our consolidated statements of income for the nine months ended September 30, 2022. For additional information regarding this transaction, refer to Note 3 to our consolidated unaudited financial statements, included in this Quarterly Report.

On May 4, 2022, we entered into a 364-day senior unsecured bridge facility in an aggregate principal amount not to exceed \$14.0 billion, or the Bridge Facility. The commitments that the Company obtained for the Bridge Facility were permanently reduced from \$14.0 billion and there were no amounts outstanding as of September 30, 2022 as a result of (i) the amendment and extension of the Credit Facility, (ii) the issuance by the Company of certain senior unsecured notes on May 23, 2022, (iii) Euroclear divestment proceeds, (iv) the generation of cash internally by the Company, and (v) the effectiveness of our term loan facility.

On May 25, 2022, we entered into a \$2.4 billion two-year senior unsecured delayed draw term loan facility, or the Term Loan. Draws under the Term Loan bear interest on the principal amount outstanding at either (a) Term SOFR plus an applicable margin plus a credit spread adjustment of 10 basis points or (b) a "base rate" plus an applicable margin. The applicable margin ranges from 0.625% to 1.125% for Term SOFR loans and from 0.000% to 0.125% for base rate loans, in each case, based on a ratings-based pricing grid. The proceeds from borrowings under the Term Loan will be used to fund a portion of the purchase price for the Black Knight acquisition. We have the option to prepay outstanding amounts under the Term Loan in whole or in part at any time. No amounts were outstanding under the Term Loan as of September 30, 2022.

Our Commercial Paper Program enables us to borrow efficiently at reasonable short-term interest rates and provides us with the flexibility to de-lever using our strong annual cash flows from operating activities whenever our leverage becomes elevated as a result of investment or acquisition activities. We had net paydowns of \$1.0 billion under our Commercial Paper Program during the nine months ended September 30, 2022 and did not have any notes outstanding under our Commercial Paper Program as of September 30, 2022.

Upon maturity of our commercial paper and to the extent old issuances are not repaid by cash on hand, we are exposed to the rollover risk of not being able to issue new commercial paper. To mitigate this risk, we maintain the Credit Facility for an aggregate amount which meets or exceeds the amount issued under our Commercial Paper Program at any time. If we were not able to issue new commercial paper, we have the option of drawing on the backstop revolving facility. However, electing to do so would result in higher interest expense.

For additional details of our debt instruments, refer to Note 8 to our consolidated unaudited financial statements, included in this Quarterly Report, and Note 10 to our consolidated financial statements included in our 2021 Form 10-K.

Capital Return

In December 2021, our Board approved an aggregate of \$3.15 billion for future repurchases of our common stock with no fixed expiration date that became effective January 1, 2022. The \$3.15 billion replaced the previous amount approved by the Board.

For the nine months ended September 30, 2022, we repurchased 5.0 million shares of our outstanding common stock at a cost of \$632 million, including 4.6 million shares at a cost of \$582 million under our Rule 10b5-1 trading plan and

0.4 million shares at a cost of \$50 million on the open market during an open trading period. For the nine months ended September 30, 2021, we did not repurchase any of our outstanding common stock. Shares repurchased are held in treasury stock.

The remaining balance of Board approved funds for future repurchases as of September 30, 2022 is \$2.5 billion. In connection with our pending acquisition of Black Knight, on May 4, 2022 we terminated our Rule 10b5-1 trading plan and suspended share repurchases. The approval of our Board for stock repurchases does not obligate us to acquire any particular amount of our common stock. In addition, our Board may increase or decrease the amount available for repurchases from time to time.

From time to time, we enter into Rule 10b5-1 trading plans, as authorized by our Board, to govern some or all of the repurchases of our shares of common stock. In December 2021 we entered into a new Rule 10b5-1 trading plan that became effective in February 2022. We may discontinue stock repurchases at any time and may amend or terminate a Rule 10b5-1 trading plan at any time. We expect funding for any stock repurchases to come from our operating cash flow or borrowings under our Commercial Paper Program or our debt facilities. Repurchases may be made from time to time on the open market, through established trading plans, in privately-negotiated transactions or otherwise, in accordance with all applicable securities laws, rules and regulations. The timing and extent of future repurchases that are not made pursuant to a Rule 10b5-1 trading plan will be at our discretion and will depend upon many conditions. In making a determination regarding any stock repurchases, management considers multiple factors, including overall stock market conditions, our common stock price performance, the remaining amount authorized for repurchases by our Board, the potential impact of a stock repurchase program on our corporate debt ratings, our expected free cash flow and working capital needs, our current and future planned strategic growth initiatives, and other potential uses of our cash and capital resources.

During the three months ended September 30, 2022, we paid a quarterly dividend of \$0.38 per share of our common stock for an aggregate payout of \$213 million, which includes the payment of dividend equivalents on unvested employee restricted stock units.

Future Capital Requirements

Our future capital requirements will depend on many factors, including the rate of growth across our segments, strategic plans and acquisitions, available sources for financing activities, required and discretionary technology and clearing initiatives, regulatory requirements, the timing and introduction of new products and enhancements to existing products, the geographic mix of our business and potential stock repurchases.

We currently expect to incur capital expenditures (including operational and real estate capital expenditures) and to incur software development costs that are eligible for capitalization ranging in the aggregate between \$490 million and \$520 million in 2022, which we believe will support the enhancement of our technology, business integration and the continued growth of our businesses.

As of September 30, 2022, we had \$2.5 billion authorized for future repurchases of our common stock. Refer to Note 10 to our consolidated financial statements included in this Quarterly Report for additional details on our stock repurchase program.

Our Board has adopted a quarterly dividend policy providing that dividends will be approved quarterly by the Board or the Audit Committee taking into account factors such as our evolving business model, prevailing business conditions, our current and future planned strategic growth initiatives and our financial results and capital requirements, without a predetermined net income payout ratio. On November 3, 2022, we announced a \$0.38 per share dividend for the fourth quarter of 2022 with the dividend payable on December 30, 2022 to stockholders of record as of December 15, 2022.

Other than the facilities for the ICE Clearing Houses, our Credit Facility and our Commercial Paper Program are currently the only significant agreements or arrangements that we have for liquidity and capital resources with third parties. See Notes 8 and 12 to our consolidated financial statements included in this Quarterly Report for further discussion. In the event of any strategic acquisitions, mergers or investments, or if we are required to raise capital for any reason or desire to return capital to our stockholders, we may incur additional debt, issue additional equity to raise necessary funds, repurchase additional shares of our common stock or pay a dividend. However, we cannot provide assurance that such financing or transactions will be available or successful, or that the terms of such financing or transactions will be favorable to us. See “—Debt” above.

Non-GAAP Measures

We use certain financial measures internally to evaluate our performance and make financial and operational decisions that are presented in a manner that adjusts from their equivalent GAAP measures or that supplement the information provided by our GAAP measures. We use these adjusted results because we believe they more clearly highlight trends in our business that may not otherwise be apparent when relying solely on GAAP financial measures, since these measures eliminate from our results specific financial items that have less bearing on our core operating performance.

We use these measures in communicating certain aspects of our results and performance, including in this Quarterly Report, and believe that these measures, when viewed in conjunction with our GAAP results and the accompanying reconciliation, can provide investors with greater transparency and a greater understanding of factors affecting our financial condition and results of operations than GAAP measures alone. In addition, we believe the presentation of these measures is useful to investors for making period-to-period comparisons of results because the adjustments to GAAP are not reflective of our core business performance.

These financial measures are not presented in accordance with, or as an alternative to, GAAP financial measures and may be different from non-GAAP measures used by other companies. We encourage investors to review the GAAP financial measures included in this Quarterly Report, including our consolidated financial statements, to aid in their analysis and understanding of our performance and in making comparisons.

The table below outlines our adjusted operating expenses, adjusted operating income, adjusted operating margin, adjusted net income attributable to ICE common stockholders and adjusted earnings per share, which are non-GAAP measures that are calculated by making adjustments for items we view as not reflective of our cash operations and core business performance. These measures, including the adjustments and their related income tax effect and other tax adjustments (in millions, except for percentages and per share amounts), are as follows:

	Exchanges Segment		Fixed Income and Data Services Segment		Mortgage Technology Segment		Consolidated	
	Nine Months Ended September 30,							
	2022	2021	2022	2021	2022	2021	2022	2021
Total revenues, less transaction-based expenses	\$ 3,089	\$ 2,842	\$ 1,555	\$ 1,403	\$ 880	\$ 1,061	\$ 5,524	\$ 5,306
Operating expenses	904	977	1,029	1,010	817	750	2,750	2,737
Less: Amortization of acquisition-related intangibles	50	56	137	136	271	277	458	469
Less: Transaction and integration costs	—	12	—	—	79	28	79	40
Adjusted operating expenses	\$ 854	\$ 909	\$ 892	\$ 874	\$ 467	\$ 445	\$ 2,213	\$ 2,228
Operating income	\$ 2,185	\$ 1,865	\$ 526	\$ 393	\$ 63	\$ 311	\$ 2,774	\$ 2,569
Adjusted operating income	\$ 2,235	\$ 1,933	\$ 663	\$ 529	\$ 413	\$ 616	\$ 3,311	\$ 3,078
Operating margin	71 %	66 %	34 %	28 %	7 %	29 %	50 %	48 %
Adjusted operating margin	72 %	68 %	43 %	38 %	47 %	58 %	60 %	58 %
Net income attributable to ICE common stockholders							\$ 1,021	\$ 2,531
Add: Amortization of acquisition-related intangibles							458	469
Add: Transaction and integration costs							79	40
Add: Accelerated unamortized costs related to the early payoff of the June 2023 floating rate senior notes							—	4
Add: Impairment on Bakkt equity method investment							40	—
Add: Accrual relating to legal settlement							9	16
Add: Net interest expense on pre-acquisition-related debt							49	—
Add: Extinguishment of 2022 and 2023 Senior Notes							30	—
Less: Gain on sale and fair value adjustment of Euroclear equity investment and dividends received							(41)	(94)
Less: Gain on sale of Coinbase equity investment							—	(1,227)
Less: Gain related to the settlement of an acquisition-related indemnification claim							—	(7)
Add/(Less): Net losses/(income) from unconsolidated investees							1,112	(42)
Add/(Less): Income tax effect for the above items							(478)	216
Add/(Less): Deferred tax adjustments on acquisition-related intangibles							(3)	196
Adjusted net income attributable to ICE common stockholders							\$ 2,276	\$ 2,102
Basic earnings per share attributable to ICE common stockholders							\$ 1.83	\$ 4.50
Diluted earnings per share attributable to ICE common stockholders							\$ 1.82	\$ 4.48
Adjusted basic earnings per share attributable to ICE common stockholders							\$ 4.07	\$ 3.74
Adjusted diluted earnings per share attributable to ICE common stockholders							\$ 4.06	\$ 3.72
Basic weighted average common shares outstanding							559	563
Diluted weighted average common shares outstanding							561	565

	Exchanges Segment		Fixed Income and Data Services Segment		Mortgage Technology Segment		Consolidated	
	Three Months Ended September 30,							
	2022	2021	2022	2021	2022	2021	2022	2021
Total revenues, less transaction-based expenses	\$ 1,001	\$ 959	\$ 534	\$ 477	\$ 276	\$ 366	\$ 1,811	\$ 1,802
Operating expenses	301	330	337	338	260	256	898	924
Less: Amortization of acquisition-related intangibles	17	19	44	45	91	92	152	156
Less: Transaction and integration costs	—	2	—	—	19	11	19	13
Adjusted operating expenses	\$ 284	\$ 309	\$ 293	\$ 293	\$ 150	\$ 153	\$ 727	\$ 755
Operating income	\$ 700	\$ 629	\$ 197	\$ 139	\$ 16	\$ 110	\$ 913	\$ 878
Adjusted operating income	\$ 717	\$ 650	\$ 241	\$ 184	\$ 126	\$ 213	\$ 1,084	\$ 1,047
Operating margin	70 %	66 %	37 %	29 %	6 %	30 %	50 %	49 %
Adjusted operating margin	72 %	68 %	45 %	39 %	46 %	58 %	60 %	58 %
Net income/(loss) attributable to ICE common stockholders							\$ (191)	\$ 633
Add: Amortization of acquisition-related intangibles							152	156
Add: Transaction and integration costs							19	13
Add: Accrual relating to legal settlement							—	16
Add: Net interest expense on pre-acquisition-related debt							31	—
Add: Accelerated unamortized costs related to the early payoff of the June 2023 floating rate senior notes							—	4
Add: Impairment on Bakkt equity method investment							40	—
Less: Gain on sale and fair value adjustment of Euroclear equity investment and dividends received							—	(64)
Add/(Less): Net losses/(income) from unconsolidated investees							1,055	(8)
Less: Income tax effect for the above items							(355)	(38)
Less: Deferred tax adjustments on acquisition-related intangibles							(18)	(1)
Adjusted net income attributable to ICE common stockholders							\$ 733	\$ 711
Basic earnings/(loss) per share attributable to ICE common stockholders							\$ (0.34)	\$ 1.12
Diluted earnings/(loss) per share attributable to ICE common stockholders							\$ (0.34)	\$ 1.12
Adjusted basic earnings per share attributable to ICE common stockholders							\$ 1.31	\$ 1.26
Adjusted diluted earnings per share attributable to ICE common stockholders							\$ 1.31	\$ 1.26
Basic weighted average common shares outstanding							558	563
Diluted weighted average common shares outstanding							560	566

Amortization of acquisition-related intangibles are included in non-GAAP adjustments as excluding these non-cash expenses provides greater clarity regarding our financial strength and stability of cash operating results.

Transaction and integration costs are included as part of our core business expenses, except for those that are directly related to the announcement, closing, financing, or termination of a transaction. However, we adjust for the acquisition-related transaction and integration costs for acquisitions such as Ellie Mae given the magnitude of the \$11.4 billion purchase price of the acquisition. We also adjusted for the acquisition-related transaction costs related to the merger between Bakkt and VIH, and for our pending acquisition of Black Knight, due to the significance of these transactions.

During the nine months ended September 30, 2022, we included an accrual related to a legal settlement as a non-GAAP adjustment, and during the nine months ended September 30, 2021, we adjust for a gain related to the settlement of an acquisition-related indemnification claim from a prior acquisition. We do not consider events of this type to be reflective of our core business operations.

During the nine months ended September 30, 2022, we exclude the \$30 million of costs associated with the May and June 2022 extinguishment of four series of senior notes that would have matured in 2022 and 2023 using proceeds from our May 2022 issuance of new senior notes as a non-GAAP adjustment. We also exclude \$79 million of interest expense on pre-acquisition-related debt from our May 2022 debt refinancing related to the pending Black Knight acquisition. This adjustment is net of \$30 million of interest income earned on investments from the pre-acquisition debt proceeds. We do not consider these to be part of our normal operations.

The acceleration of unamortized costs related to the early payoff of the June 2023 floating rate senior notes is included as a non-GAAP adjustment during the three months ended September 30, 2021 as it relates to the early redemption of our June 2023 floating rate senior notes. We do not consider these adjustments to be reflective of our normal operations.

As of September 30, 2022 the carrying value of our Bakkt investment was \$439 million and exceeded its fair value by \$40 million. Due to the decline in fair value, we were required to determine if this decline was other than temporary. Based on

our review, we determined that the decline in fair value was other than temporary in nature. During the three months ended September 30, 2022, we recorded an impairment charge on our investment in Bakkt of \$40 million as other expense. This investment is not considered to be a part of our core business operations. See Note 4 of our consolidated financial statements in this Quarterly Report where discussed further.

During the nine months ended September 30, 2022, we exclude the \$41 million gain on the sale of our Euroclear investment and the related historical dividends received of \$60 million and \$30 million during the nine and three months ended September 30, 2021, respectively. During the nine and three months ended September 30, 2021, we also exclude the \$34 million fair value gain on our Euroclear equity investment. During the nine months ended September 30, 2021, we exclude the gain on the sale of our Coinbase investment. Such transactions are not considered a part of our core business operations.

We exclude net income/(losses) from our unconsolidated equity method investees for purposes of calculating non-GAAP measures. As of September 30, 2022, this adjustment includes our share of profits or losses from OCC, Bakkt, BondLink and our other equity method investments, and as of September 30, 2021, it included only OCC and BondLink. During the three months ended September 30, 2022, Bakkt reported an impairment of goodwill and intangible assets of approximately \$1.5 billion, of which \$1.0 billion is included in our share of estimated net losses from our unconsolidated equity method investees. This is consistent with how we treat changes in the fair value of our equity investments. We believe these adjustments provide greater clarity of our performance, given that equity and equity method investments are non-cash and not a part of our core operations.

The income tax effects relating to all non-GAAP adjustments above are included as non-GAAP adjustments. We also include non-GAAP adjustments for deferred tax adjustments on acquisition-related intangibles. The deferred tax adjustments of \$3 million and \$18 million for the nine and three months ended September 30, 2022, respectively, relate primarily to U.S. state apportionment changes. The deferred tax adjustment of \$196 million for the nine months ended September 30, 2021 relates primarily to the 2021 U.K. corporate income tax rate increase from 19% to 25% effective April 1, 2023.

For additional information on these items, refer to our consolidated financial statements included in this Quarterly Report and “—Consolidated Operating Expenses”, above.

Contractual Obligations and Commercial Commitments

During the nine months ended September 30, 2022, there were no significant changes to our contractual obligations and commercial commitments from those disclosed in the section “Management’s Discussion and Analysis of Financial Condition and Results of Operations” in our 2021 Form 10-K.

Off-Balance Sheet Arrangements

As described in Note 12 to our consolidated financial statements, which are included elsewhere in this Quarterly Report, certain clearing house collateral is reported off-balance sheet. We do not have any relationships with unconsolidated entities or financial partnerships, often referred to as structured finance or special purpose entities.

New and Recently Adopted Accounting Pronouncements

During the nine months ended September 30, 2022, there were no significant changes to the new and recently adopted accounting pronouncements applicable to us from those disclosed in Note 2 of our 2021 Form 10-K.

Critical Accounting Policies

During the nine months ended September 30, 2022, there were no significant changes to our critical accounting policies and estimates from those disclosed in the section “Management’s Discussion and Analysis of Financial Condition and Results of Operations” in our 2021 Form 10-K.

ITEM 3. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

As a result of our operating and financing activities, we are exposed to market risks such as interest rate risk, foreign currency exchange rate risk and credit risk. We have implemented policies and procedures designed to measure, manage, monitor and report risk exposures, which are regularly reviewed by the appropriate management and supervisory bodies.

Interest Rate Risk

We have exposure to market risk for changes in interest rates relating to our cash and cash equivalents, short-term and long-term restricted cash and cash equivalents, short-term and long-term investments and indebtedness. As of September 30, 2022 and December 31, 2021, our cash and cash equivalents and short-term and long-term restricted cash and cash equivalents were \$7.6 billion and \$2.0 billion, respectively, of which \$313 million and \$276 million, respectively, were denominated in pounds sterling, euros or Canadian dollars, and the remaining amounts are denominated in U.S. dollars. We do not use our investment portfolio for trading or other speculative purposes. A hypothetical 50% decrease in short-term interest rates would decrease our annual pre-tax earnings by \$12 million as of September 30, 2022, assuming no change in the amount or composition of our cash and cash equivalents and short-term and long-term restricted cash and cash equivalents.

As of September 30, 2022, we had \$18.1 billion in outstanding debt, all of which relates to our senior notes. In addition, we have \$7 million outstanding under lines of credit at our ICE India subsidiaries. See Part I, Item 2 "Management's Discussion and Analysis of Financial Condition and Results of Operations - Debt" and Note 8 to our consolidated financial statements included in this Quarterly Report.

The interest rates on our Commercial Paper Program are currently evaluated based upon current maturities and market conditions. The weighted average interest rate on our Commercial Paper Program was 0.33% as of December 31, 2021. The effective interest rate of commercial paper issuances will continue to fluctuate based on the movement in short-term interest rates along with shifts in supply and demand within the commercial paper market.

Foreign Currency Exchange Rate Risk

As an international business, we are subject to foreign currency exchange rate risk. We may experience gains or losses from foreign currency transactions in the future given that a significant part of our assets and liabilities are recorded in pounds sterling, Canadian dollars or euros, and a significant portion of our revenues and expenses are recorded in pounds sterling or euros. Certain assets, liabilities, revenues and expenses of foreign subsidiaries are denominated in the local functional currency of such subsidiaries. Our exposure to foreign denominated earnings for the nine and three months ended September 30, 2022 and 2021 is presented by primary foreign currency in the following table (dollars in millions, except exchange rates):

	Nine Months Ended September 30, 2022		Three Months Ended September 30, 2022		Nine Months Ended September 30, 2021		Three Months Ended September 30, 2021	
	Pound Sterling	Euro	Pound Sterling	Euro	Pound Sterling	Euro	Pound Sterling	Euro
Average exchange rate to the U.S. dollar in the current year period	1.2589	1.0649	1.1774	1.0071	1.3854	1.1967	1.3784	1.1788
Average exchange rate to the U.S. dollar in the same period in the prior year	1.3854	1.1967	1.3784	1.1788	1.2708	1.1242	1.2916	1.1691
Average exchange rate increase/(decrease)	(9)%	(11)%	(15)%	(15)%	9 %	6 %	7 %	1 %
Foreign denominated percentage of:								
Revenues, less transaction-based expenses	8 %	6 %	7 %	6 %	7 %	6 %	7 %	6 %
Operating expenses	8 %	2 %	7 %	2 %	8 %	2 %	8 %	2 %
Operating income	7 %	11 %	8 %	11 %	5 %	10 %	5 %	11 %
Impact of the currency fluctuations ⁽¹⁾ on:								
Revenues, less transaction-based expenses	\$ (42)	\$ (44)	\$ (22)	\$ (20)	\$ 29	\$ 18	\$ 8	\$ 1
Operating expenses	\$ (21)	\$ (6)	\$ (11)	\$ (2)	\$ 18	\$ 3	\$ 5	\$ —
Operating income	\$ (21)	\$ (38)	\$ (11)	\$ (18)	\$ 11	\$ 15	\$ 3	\$ 1

(1) Represents the impact of currency fluctuation for the nine and three months ended September 30, 2022 and 2021 compared to the same periods in the prior year.

We have a significant part of our assets, liabilities, revenues and expenses recorded in pounds sterling or euros. During the nine and three months ended September 30, 2022, 14% and 13%, respectively, of our consolidated revenues, less transaction-based expenses were denominated in pounds sterling or euros and for the nine and three months ended September 30, 2022, 10% and 9%, respectively, of our consolidated operating expenses were denominated in pounds

sterling or euros. As the pound sterling or euro exchange rate changes, the U.S. equivalent of revenues and expenses denominated in foreign currencies changes accordingly.

Foreign currency transaction risk related to the settlement of foreign currency denominated assets, liabilities and payables occurs through our operations, which are received in or paid in pounds sterling, Canadian dollars, or euros, due to the increase or decrease in the foreign currency exchange rates between periods. We incurred foreign currency transaction losses of \$9 million and \$11 million for the nine months ended September 30, 2022 and 2021, respectively, and \$2 million for the three months ended September 30, 2021 inclusive of the impact of foreign currency hedging transactions. We did not incur any significant foreign currency transaction gains/(losses) during the three months ended September 30, 2022. The foreign currency transaction losses were primarily attributable to the fluctuations of the pound sterling and euro relative to the U.S. dollar. A 10% adverse change in the underlying foreign currency exchange rates as of September 30, 2022, assuming no change in the composition of the foreign currency denominated assets, liabilities and payables and assuming no hedging activity, would result in a foreign currency loss of \$14 million.

We entered into foreign currency hedging transactions during the nine and three months ended September 30, 2022 and 2021 as economic hedges to help mitigate a portion of our foreign exchange risk exposure and may enter into additional hedging transactions in the future to help mitigate our foreign exchange risk exposure. Although we may enter into additional hedging transactions in the future, these hedging arrangements may not be effective, particularly in the event of imprecise forecasts of the levels of our non-U.S. denominated assets and liabilities.

We have foreign currency translation risk equal to our net investment in our foreign subsidiaries. The financial statements of these subsidiaries are translated into U.S. dollars using a current rate of exchange, with gains or losses included in the cumulative translation adjustment account, a component of equity. Our exposure to the net investment in foreign currencies is presented by primary foreign currencies in the table below (in millions):

	As of September 30, 2022		
	Position in pounds sterling	Position in Canadian dollars	Position in euros
Assets	£ 729	\$ 3,475	€ 171
of which goodwill represents	565	395	92
Liabilities	117	3,021	57
Net currency position	£ 612	\$ 454	€ 114
Net currency position, in \$USD	\$ 683	\$ 328	\$ 112
Negative impact on consolidated equity of a 10% decrease in foreign currency exchange rates	\$ 68	\$ 33	\$ 11

Foreign currency translation adjustments are included as a component of accumulated other comprehensive income/(loss) within our balance sheet. See the table below for the portion of equity attributable to foreign currency translation adjustments as well as the activity for the nine and three months ended September 30, 2022 included within our statement of other comprehensive income. The impact of the foreign currency exchange rate differences in the tables below were primarily driven by fluctuations of the pound sterling as compared to the U.S. dollar which were 1.1165, 1.2178 and 1.3524 as of September 30, 2022, June 30, 2022 and December 31, 2021, respectively, and by fluctuations of the euro as compared to the U.S. dollar which were 0.9802, 1.0483 and 1.1372 as of September 30, 2022, June 30, 2022 and December 31, 2021, respectively.

	Changes in Accumulated Other Comprehensive Loss from Foreign Currency Translation Adjustments (in millions)
Balance, as of December 31, 2021	\$ (150)
Other comprehensive income/(loss)	(208)
Income tax benefit/(expense)	1
Net current period other comprehensive loss	(207)
Balance, as of September 30, 2022	\$ (357)

	Changes in Accumulated Other Comprehensive Loss from Foreign Currency Translation Adjustments (in millions)
Balance, as of June 30, 2022	\$ (259)
Other comprehensive income/(loss)	(98)
Income tax benefit/(expense)	—
Net current period other comprehensive loss	(98)
Balance, as of September 30, 2022	\$ (357)

The future impact on our business relating to the U.K. leaving the EU and the corresponding regulatory changes are uncertain at this time, including future impacts on currency exchange rates.

Credit Risk

We are exposed to credit risk in our operations in the event of a counterparty default. We limit our exposure to credit risk by rigorously selecting the counterparties with which we make our investments, monitoring them on an ongoing basis and executing agreements to protect our interests.

Clearing House Cash Deposit Risks

The ICE Clearing Houses hold material amounts of clearing member margin deposits which are held or invested primarily to provide security of capital while minimizing credit, market and liquidity risks. Refer to Note 12 to our consolidated financial statements for more information on the ICE Clearing Houses' cash and cash equivalent margin deposits and guaranty funds, invested deposits, delivery contracts receivable and unsettled variation margin which were \$164.7 billion as of September 30, 2022. While we seek to achieve a reasonable rate of return which may generate interest income for our clearing members, we are primarily concerned with preservation of capital and managing the risks associated with these deposits. As the ICE Clearing Houses may pass on interest revenues (minus costs) to the clearing members, this could include negative or reduced yield due to market conditions. For a summary of the risks associated with these deposits and how these risks are mitigated, see Part II, Item 7(A) "Quantitative and Qualitative Disclosures About Market Risk" in our 2021 Form 10-K.

ITEM 4. CONTROLS AND PROCEDURES

(a) *Evaluation of Disclosure Controls and Procedures.* As of the end of the period covered by this report, an evaluation was carried out by our management, with the participation of our Chief Executive Officer and Chief Financial Officer, of the effectiveness of our disclosure controls and procedures (as defined in Rule 13a-15(e) under the Securities Exchange Act of 1934). Based upon that evaluation, our Chief Executive Officer and Chief Financial Officer concluded that these disclosure controls and procedures were effective as of the end of the period covered by this report.

(b) *Changes in Internal Controls over Financial Reporting.* There were no changes in our internal controls over financial reporting that occurred during our most recent fiscal quarter that have materially affected, or are reasonably likely to materially affect, our internal controls over financial reporting. As a result, no corrective actions were taken.

PART II. Other Information

ITEM 1. LEGAL PROCEEDINGS

See Note 13 to the consolidated financial statements and related notes, which are incorporated by reference herein.

ITEM 1(A). RISK FACTORS

During the nine months ended September 30, 2022, there were no significant new risk factors from those disclosed in Part I, Item 1A, "Risk Factors" in our 2021 Form 10-K, other than those set forth below under the heading "Risks Relating to the Black Knight Acquisition". In addition to the other information set forth in this Quarterly Report, including the information in the "- Regulation" section of Item 2 - Management's Discussion and Analysis of Financial Condition and Results of Operations and the risk factors set forth below, you should carefully consider the factors discussed under "Risk

Factors” and the regulation discussion under “Business - Regulation” in our 2021 Form 10-K. These risks could materially and adversely affect our business, financial condition and results of operations. The risks and uncertainties in our 2021 Form 10-K and those described below are not the only ones facing us. Additional risks and uncertainties not presently known to us, or that we currently believe to be immaterial, may also adversely affect our business.

Risks Relating to the Black Knight Acquisition

The merger will not be completed unless important conditions are satisfied or waived, including regulatory approvals.

Specified conditions set forth in the merger agreement must be satisfied or waived to complete the merger, including the expiration or early termination of the applicable waiting period under the Hart-Scott-Rodino Antitrust Improvements Act of 1976, as amended, or the HSR Act. If the conditions are not satisfied or, to the extent permitted by law, waived, the merger will not occur or will be delayed, and we may lose some or all of the intended benefits of the merger. If the merger is not completed, our ongoing businesses, financial condition, financial results and stock price may be materially and adversely affected and, without realizing any of the benefits of having completed the merger, we will be subject to a number of risks, including (i) the market price of our common stock could decline to the extent the current market price reflects an assumption that the merger will be completed; (ii) we could owe a termination fee to Black Knight under certain circumstances; and (iii) we will be required to pay costs relating to the merger, such as legal, accounting, financial advisory, financing (including the redemption by us of \$5 billion of bonds at 101% of par value) and printing fees.

Regulatory approvals may not be received, may take longer than expected or may impose conditions that are not presently anticipated, that could have an adverse effect on ICE following the merger or that are otherwise unacceptable to ICE.

Completion of the merger is conditioned on, among other things, the expiration or early termination of the waiting period applicable to the consummation of the merger under the HSR Act. There can be no assurance that this condition will be satisfied on a timely basis or at all, or that, if regulatory approvals are granted, they will not result in the imposition of conditions, limitations, obligations or restrictions that have the effect of preventing the completion of the merger, imposing additional material costs on or materially limiting our revenues following the merger or otherwise reducing the anticipated benefits of the merger, or result in the delay or abandonment of the merger. Under the merger agreement, we are not obligated to agree to any structural or behavioral remedy that any government entity may seek to impose.

In addition, at any time before or after the completion of the merger, and notwithstanding the termination of applicable waiting periods, the applicable U.S. antitrust authorities or any state attorney general could take such action under the antitrust laws as any such party deems necessary or desirable in the public interest. Such action could include, among other things, seeking to enjoin the completion of the merger or seeking divestiture of substantial assets of the parties. In addition, in some circumstances, a third party could initiate a private action under antitrust laws challenging, seeking to enjoin, or seeking to impose conditions on the merger. Although we are not obligated under the merger agreement to litigate to defeat such efforts to enjoin the merger under U.S. antitrust laws, we and Black Knight, at our option, may engage in litigation to obtain the clearances, consents, approvals and waivers under U.S. antitrust laws so as to enable the parties to close the merger. We may not prevail and may incur significant costs in defending or settling any such action.

We may be unable to successfully integrate Black Knight's business and realize the anticipated benefits of the merger, and we will incur significant costs in connection with the merger and the integration of Black Knight.

The success of the merger will depend on, among other things, our ability to successfully integrate the business of Black Knight into the ICE Mortgage Technology business in a manner that facilitates growth opportunities, realizes anticipated synergies, and achieves the projected cost savings, revenue growth and profitability targets of the combined businesses without adversely affecting current revenues and investments in future growth. There is a significant degree of difficulty and management distraction inherent in the process of integrating an acquisition, which may involve delays or additional and unforeseen expenses. The integration and other disruptions from the merger may also disrupt our ongoing businesses. If we are not able to successfully achieve our objectives from the merger within the anticipated time frame, or at all, the anticipated benefits may not be realized fully or at all, may take longer to realize than expected, and our business or stock price may be adversely affected.

In connection with the merger and the integration of Black Knight's business, we have incurred and expect to continue to incur significant costs. We may also incur additional costs or suffer loss of business under third-party contracts of Black Knight that are terminated or that contain change in control or other provisions that may be triggered by the completion of the merger, and/or losses of, or decreases in orders by, customers, and may also incur costs to maintain employee morale and to retain certain key management personnel and employees. These incremental transaction-related costs may exceed the savings and efficiencies we expect to achieve from the integration of the businesses.

As a result of the merger, we will be subject to risks relating to the business conducted by Black Knight.

Following the consummation of the merger, we will be subject to a variety of risks relating to the business conducted by Black Knight, many of which we, and more specifically, ICE Mortgage Technology, already face in our business, as described in Item 1(A) "Risk Factors" included in our 2021 Form 10-K. Some of the specific risks facing Black Knight include risks relating to the mortgage lending industry, including general conditions in the industry; changes in inflation rates and interest rates; changes in current or new regulations and legislation and potential structural changes in the mortgage lending industry; technology risks, including cyber security and data privacy risks relating to Black Knight's services; risks relating to intellectual property held or used by Black Knight; the ability of Black Knight to adequately compete with products or other companies, including through attracting new customers and retaining or selling additional service offerings to existing customers; risks relating to Black Knight's use of international third-party service providers and Black Knight's international operations; risks relating to Black Knight's indebtedness; risks relating to Black Knight's investment in Dun & Bradstreet Holdings, Inc.; and risks relating to current and future legal proceedings or disputes involving Black Knight. In connection with any such legal proceeding or other dispute, we could incur significant expenses. An adverse resolution of any such proceeding or dispute may require us to pay substantial damages or impose restrictions on how we conduct business, either of which could adversely affect our business, financial condition and operating results.

In addition, Black Knight is and will continue to be subject to business uncertainties and contractual restrictions while the merger is pending. The occurrence of any of such risks could have a material adverse impact on the financial condition, business or results of operations of Black Knight, which could impair or eliminate our ability to achieve the expected cost savings and synergies from the merger on a timely basis, if ever, or could impair our ability to achieve such cost savings and synergies without adversely affecting our current revenues or investments in future growth. Additionally, such risks could impair our ability to integrate the business of Black Knight with our businesses in an efficient and timely manner, if at all.

After the completion of the merger, we will be more leveraged than we currently are, and the financing arrangements that we will enter into will contain restrictions and limitations that could, under certain circumstances, impact our ability to operate our business.

In connection with the merger, we have incurred approximately \$5 billion of additional indebtedness in the form of senior notes and, assuming that the closing of the merger occurs in the first half of 2023, we intend to incur approximately \$4 billion to \$5 billion in additional indebtedness consisting of commercial paper and term loans, in order to finance a portion of the cash consideration in the merger and related transactions including, without limitation, the refinancing of certain existing indebtedness of Black Knight, and we will assume approximately \$1 billion of Black Knight's outstanding senior notes. After the completion of the merger, we expect to have consolidated indebtedness of approximately \$24 billion. Our increased indebtedness after the completion of the merger may have the effect, among other things, of reducing our flexibility to respond to changing business and economic conditions, requiring us to use increased amounts of cash flow to service indebtedness and increasing our borrowing costs.

In addition, our credit ratings impact the cost and availability of future borrowings, and, as a result, our cost of capital. Each of the ratings organizations reviews our ratings periodically, and there can be no assurance that our current ratings will be maintained in the future. Downgrades in our credit ratings could adversely affect our businesses, cash flows, financial condition and operating results. Furthermore, if the merger is completed and Black Knight's senior notes are downgraded and rated below investment grade by two out of three of Moody's, S&P and Fitch within 60 days following the change of control or the announcement thereof, this may constitute a change of control triggering event under the indenture governing those notes. Upon the occurrence of a change of control triggering event, Black Knight (or we on its behalf) would be required to offer to repurchase those notes at 101% of the principal amount thereof plus accrued and unpaid interest. However, it is possible that Black Knight or we would not have sufficient funds at the time of the change of control triggering event to make the required repurchase of notes or that restrictions in other debt instruments would not allow such repurchase.

We also expect that the agreements governing the indebtedness that we will incur will contain covenants that may, under certain circumstances, place limitations on certain actions that we could seek to undertake, which could impact our ability to operate our business.

ITEM 2. UNREGISTERED SALES OF EQUITY SECURITIES AND USE OF PROCEEDS

Stock Repurchases

The table below sets forth the information with respect to purchases made by or on behalf of ICE or any “affiliated purchaser” (as defined in Rule 10b-18(a)(3) under the Exchange Act) of our common stock during the nine months ended September 30, 2022.

Period (2022)	Total number of shares purchased (in thousands)	Average price paid per share	Total number of shares purchased as part of publicly announced plans or programs (in thousands) (1)	Approximate dollar value of shares that may yet be purchased under the plans or programs (in millions) (1)
July 1 - July 31	—	N/A	4,955	\$2,518
August 1 - August 31	—	N/A	4,955	\$2,518
September 1 - September 30	—	N/A	4,955	\$2,518
Total	—	N/A	4,955	\$2,518

Three months ended	Total number of shares purchased (in thousands) (1)	Average price paid per share	Amount of repurchases (in millions)	Total cumulative year-to-date shares purchased as part of publicly announced plans or programs (in thousands)	Approximate dollar value of shares that may yet be purchased under the plans or programs (in millions)
March 31, 2022	3,674	\$129.30	\$475	3,674	\$2,675
June 30, 2022	1,281	\$122.54	\$157	4,955	\$2,518
September 30, 2022	—	N/A	\$—	4,955	\$2,518
Total (1)	4,955	\$127.56	\$632		

(1) Includes 4.6 million shares at a cost of \$582 million under our Rule 10b5-1 trading plan and 0.4 million shares at a cost of \$50 million on the open market during an open trading period.

Refer to Note 10 to our consolidated financial statements and related notes, which are included elsewhere in this Quarterly Report for details on our stock repurchase plans.

ITEM 3. DEFAULTS UPON SENIOR SECURITIES

None.

ITEM 4. MINE SAFETY DISCLOSURES

Not applicable.

ITEM 5. OTHER INFORMATION

Not applicable.

ITEM 6. EXHIBITS

Exhibit Number		Description of Document
3.1	—	<u>Sixth Amended and Restated Certificate of Incorporation of Intercontinental Exchange, Inc., effective August 22, 2022 (incorporated by reference to Exhibit 3.1 to Intercontinental Exchange, Inc.'s Current Report on Form 8-K filed with the SEC on August 22, 2022, File No. 001-36198).</u>
3.2	—	<u>Ninth Amended and Restated Bylaws of Intercontinental Exchange, Inc., effective August 22, 2022 (incorporated by reference to Exhibit 3.2 to Intercontinental Exchange, Inc.'s Current Report on Form 8-K filed with the SEC on August 22, 2022, File No. 001-36198).</u>
31.1	—	<u>Rule 13a-14(a)/15d-14(a) Certification of Chief Executive Officer.</u>
31.2	—	<u>Rule 13a-14(a)/15d-14(a) Certification of Chief Financial Officer.</u>
32.1	—	<u>Section 1350 Certification of Chief Executive Officer.</u>
32.2	—	<u>Section 1350 Certification of Chief Financial Officer.</u>
0.101	—	The following materials from Intercontinental Exchange, Inc.'s Quarterly Report on Form 10-Q for the quarter ended September 30, 2022, formatted in Inline XBRL (Extensible Business Reporting Language): (i) the Consolidated Balance Sheets, (ii) the Consolidated Statements of Income, (iii) the Consolidated Statements of Comprehensive Income, (iv) the Consolidated Statements of Changes in Equity and Redeemable Non-Controlling Interest (v) Consolidated Statements of Cash Flows and (vi) Notes to Consolidated Financial Statements, tagged as blocks of text.
104	—	The cover page from Intercontinental Exchange, Inc.'s Quarterly Report on Form 10-Q for the quarter ended September 30, 2022 formatted in Inline XBRL.

* Certain exhibits and similar attachments to this agreement have been omitted in accordance with Item 601(b)(2) of Regulation S-K. A copy of any omitted exhibit or other attachment will be furnished supplementally to the SEC upon request.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the Registrant has duly caused this Report to be signed on its behalf by the undersigned, thereunto duly authorized.

Intercontinental Exchange, Inc.
(Registrant)

Date: November 3, 2022

By: /s/ A. Warren Gardiner

A. Warren Gardiner
Chief Financial Officer
(Principal Financial Officer)

CERTIFICATION

I, Jeffrey C. Sprecher, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Intercontinental Exchange, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - (c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - (d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
 - (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: November 3, 2022

/s/ Jeffrey C. Sprecher

Jeffrey C. Sprecher
Chair of the Board and
Chief Executive Officer

CERTIFICATION

I, A. Warren Gardiner, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Intercontinental Exchange, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - (c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - (d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
 - (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: November 3, 2022

/s/ A. Warren Gardiner

A. Warren Gardiner
Chief Financial Officer

**CERTIFICATION PURSUANT TO
18 U.S.C. SECTION 1350,
AS ADOPTED PURSUANT TO
SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002**

In connection with the Quarterly Report of Intercontinental Exchange, Inc. (the “Company”) on Form 10-Q for the period ended September 30, 2022 as filed with the Securities and Exchange Commission on the date hereof (the “Report”), I, Jeffrey C. Sprecher, Chair of the Board and Chief Executive Officer of the Company, certify, pursuant to 18 U.S.C. § 1350, as adopted pursuant to § 906 of the Sarbanes-Oxley Act of 2002, that, to the best of my knowledge:

- (1) The Report fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934; and
- (2) The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company.

Date: November 3, 2022

/s/ Jeffrey C. Sprecher

Jeffrey C. Sprecher
Chair of the Board and
Chief Executive Officer

**CERTIFICATION PURSUANT TO
18 U.S.C. SECTION 1350,
AS ADOPTED PURSUANT TO
SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002**

In connection with the Quarterly Report of Intercontinental Exchange, Inc. (the “Company”) on Form 10-Q for the period ended September 30, 2022 as filed with the Securities and Exchange Commission on the date hereof (the “Report”), I, A. Warren Gardiner, Chief Financial Officer of the Company, certify, pursuant to 18 U.S.C. § 1350, as adopted pursuant to § 906 of the Sarbanes-Oxley Act of 2002, that, to the best of my knowledge:

- (1) The Report fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934; and
- (2) The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company.

Date: November 3, 2022

/s/ A. Warren Gardiner

A. Warren Gardiner
Chief Financial Officer